NIH National Institute on Aging

Sample Application for Research Training and Career Development Funding

Through the F99/K00 Transition to Aging Research for Predoctoral Students Awards, NIA supports exceptional graduate students in finishing their dissertation research on any topic and continuing to a postdoctoral position conducting research on aging. Each award has two phases: the F99 phase supporting predoctoral training and the K00 phase supporting postdoctoral training.

Please note that these applications do not reflect the new Fellowship review criteria, which are effective January 25, 2025, as well as other updates to application instructions. Be sure to refer to the latest Fellowship Instructions and the current Notice of Funding Opportunity when preparing an F99/K00 application.

Find more NIA sample applications and information about training and career development funding:
https://www.nia.nih.gov/research/training/sample-applications

PI: **Yu, Kexin**	Title: Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology	
Received: 10/22/2020	Opportunity: RFA-AG-21-022	Council: 05/2021
Competition ID: FORMS-F	FOA Title: Transition to Aging Research for Predoctoral Students (F99/K00)	
1F99AG068492-01A1	Dual:	Accession Number: 4511212
IPF: 7636101	Organization: UNIVERSITY OF SOUTHERN CALIFORNIA	
Former Number: 1F99AG068492-01	Department: Social Work	
IRG/SRG: ZAG1 ZIJ-D (M1)	AIDS: N	Expedited: N
Subtotal Direct Costs (excludes consortium F&A) Year 1: 0	Animals: N Humans: Y Clinical Trial: N Current HS Code: E7 HESC: N HFT: N	New Investigator: Early Stage Investigator:
Senior/Key Personnel:	*Organization:*	*Role Category:*
Kexin Yu	University of Southern California	PD/PI
Lawrence Palinkas	University of Southern California	Other (Specify)-Co-Sponsor
Shinyi Wu	University of Southern California	Other (Specify)-Co-Sponsor
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
Lisa Silbert	Oregon Health & Science University	Other (Specify)-Co-Sponsor
Iris Chi	University of Southern California	Other (Specify)-Sponsor

Reference Letters

[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]

APPLICATION FOR FEDERAL ASSISTANCE
SF 424 (R&R)

3. DATE RECEIVED BY STATE	**State Application Identifier**
1. TYPE OF SUBMISSION*	**4.a. Federal Identifier** [redacted]
❍ Pre-application ❍ Application ● Changed/Corrected Application	**b. Agency Routing Number**
2. DATE SUBMITTED 2020-10-22 / **Application Identifier**	**c. Previous Grants.gov Tracking Number** GRANT13232560

5. APPLICANT INFORMATION **Organizational DUNS*:** [redacted]

Legal Name*: University of Southern California
Department:
Division:
Street1*: [redacted]
Street2: [redacted]
City*: [redacted]
County: [redacted]
State*: [redacted]
Province:
Country*: [redacted]
ZIP / Postal Code*: [redacted]

Person to be contacted on matters involving this application
Prefix: Mr. First Name*: Steven Middle Name: L. Last Name*: Misuraca Suffix:
Position/Title: Senior Contract and Grant Officer
Street1*: [redacted]
Street2:
City*: [redacted]
County: [redacted]
State*: [redacted]
Province:
Country*: [redacted]
ZIP / Postal Code*: [redacted]
Phone Number*: [redacted] Fax Number [redacted] Email: [redacted]

6. EMPLOYER IDENTIFICATION NUMBER *(EIN) or (TIN)** [redacted]

7. TYPE OF APPLICANT* O: Private Institution of Higher Education

Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

8. TYPE OF APPLICATION*
❍ New ● Resubmission
❍ Renewal ❍ Continuation ❍ Revision

If Revision, mark appropriate box(es).
❍ A. Increase Award ❍ B. Decrease Award ❍ C. Increase Duration
❍ D. Decrease Duration ❍ E. Other *(specify)* :

Is this application being submitted to other agencies?* ❍Yes ●No What other Agencies?

9. NAME OF FEDERAL AGENCY*
National Institutes of Health

10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER
TITLE:

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT*
Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology

12. PROPOSED PROJECT
Start Date* 07/01/2021
Ending Date* 06/30/2026

13. CONGRESSIONAL DISTRICTS OF APPLICANT
CA-037

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE — Page 2

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix: First Name*: Kexin Middle Name: Last Name*: Yu Suffix:
Position/Title: Graduate Student
Organization Name*: University of Southern California
Department: Social Work
Division:
Street1*: [redacted]
Street2:
City*: [redacted]
County: [redacted]
State*: [redacted]
Province:
Country*: [redacted]
ZIP / Postal Code*: [redacted]
Phone Number*: [redacted] Fax Number: Email*: [redacted]

15. ESTIMATED PROJECT FUNDING

[redacted]
b. Total Non-Federal Funds* [redacted]
c. Total Federal & Non-Federal Funds* [redacted]
d. Estimated Program Income* [redacted]

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

a. YES ❍ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:

DATE:

b. NO ● PROGRAM IS NOT COVERED BY E.O. 12372; OR

❍ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

● I agree*

** The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.*

18. SFLLL or OTHER EXPLANATORY DOCUMENTATION File Name:

19. AUTHORIZED REPRESENTATIVE

Prefix: Ms. First Name*: Aimee Middle Name: C Last Name*: Barnard Suffix:
Position/Title*: Contracts and Grants Officer
Organization Name*: University of Southern California
Department: Vice President For Research
Division: Provost's Office
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Street2: [redacted]
City*: [redacted]
County: [redacted]
State*: [redacted]
Province:
Country*: [redacted]
ZIP / Postal Code*: [redacted]
Phone Number*: [redacted] Fax Number: [redacted]

Signature of Authorized Representative*	Date Signed*
Aimee C Barnard	10/22/2020

20. PRE-APPLICATION File Name:

21. COVER LETTER ATTACHMENT File Name:2__Cover_Letter_10_20_20201019212508.pdf

424 R&R and PHS-398 Specific
Table Of Contents

OMB Number: 4040-0010
Expiration Date: 12/31/2022

Project/Performance Site Location(s)

Project/Performance Site Primary Location

❍ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name:	University of Southern California
Duns Number:	[redacted]
Street1*:	[redacted]
Street2:	
City*:	[redacted]
County:	[redacted]
State*:	[redacted]
Province:	
Country*:	[redacted]
Zip / Postal Code*:	[redacted]
Project/Performance Site Congressional District*:	CA-037

Additional Location(s) File Name:

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED Other Project Information

1. Are Human Subjects Involved?* ● **Yes** ○ **No**

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ● Yes ○ No

If YES, check appropriate exemption number: _ 1 ✓ 2 _ 3 _ 4 _ 5 _ 6 _ 7 ✓ 8

If NO, is the IRB review Pending? ○ **Yes** ○ **No**

IRB Approval Date:

Human Subject Assurance Number 00007099

2. Are Vertebrate Animals Used?* ○ **Yes** ● **No**

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ○ Yes ○ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. Is proprietary/privileged information included in the application?* ○ **Yes** ● **No**

4.a. Does this project have an actual or potential impact - positive or negative - on the environment?* ○ **Yes** ● **No**

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ○ Yes ○ No

4.d. If yes, please explain:

5. Is the research performance site designated, or eligible to be designated, as a historic place?* ○ **Yes** ● **No**

5.a. If yes, please explain:

6. Does this project involve activities outside the United States or partnership with international collaborators?* ○ **Yes** ● **No**

6.a. If yes, identify countries:

6.b. Optional Explanation:

	Filename
7. Project Summary/Abstract*	04__Project_Summary1019212449.pdf
8. Project Narrative*	05__Project_Narrative1019212514.pdf
9. Bibliography & References Cited	06__BIBLIOGRAPHY1019212451.pdf
10.Facilities & Other Resources	07__Facilities_and_Other_Resource1019212453.pdf
11.Equipment	No_Major_Equipment_Doc1019212452.pdf
12. Other Attachments	Certification_Ltr_Yu__Kexin_10_21_201019212567.pdf

PROJECT SUMMARY

Social wellbeing and cognitive health are two pillars of successful aging, and they are interrelated. Three decades of research have consistently shown that social isolation is associated with declines in cognitive functioning. Although existing literature has explored the individual-level risk and protective factors, researchers' attention has been less paid to the extent to which physical and social characteristics of the living environment affect social isolation and cognitive health. The Ecological Theory of Aging (ETA) posits environmental factors have significant influences on health and social wellbeing. Negative physical characteristics of the environment were hypothesized to be related to increased social isolation and cognitive decline longitudinally. An increasing amount of research investigated the association between ICT use and social isolation, yet whether ICT use would interact with the effect of environmental factors on the risk of social isolation has not yet been empirically tested. Besides, the influence of ICT use on cognitive health has not been well understood. Promoting social interaction using technology-based approaches have the potential to alleviate social isolation among older people and prevent cognitive decline. The proposed dissertation work (F99) will examine the effect of physical and social environments of living on social isolation and cognitive health. The effects of ICT use will be investigated within living contexts. Multi-level modeling and structural equation modeling methods will be employed to conduct secondary data analysis using data from the National Health & Aging Trend Study (NHATS) wave 5 to 9. The postdoctoral stage (K00) is guided by the overarching aim of building knowledge on the behavioral and contextual pathways that could explain the association between social isolation and cognitive decline and developing a community-based behavioral health intervention. At the K00 phase, the PI will increase understanding of the social isolation and cognitive health research landscape by conducting a systematic review. Longitudinal data from the ongoing Internet-based Conversational Engagement Clinical Trial (I-CONECT) will be analyzed to examine the association between social isolation and change in neuropsychological outcomes. The PI will utilize the longitudinal data collected with the Collaborative Aging Research Using Technology (CART) platform to investigate the relationships between daily activities, social engagement, and the cognitive health of older participants. Knowledge to be built at F99 and K00 stages will be synthesized to inform the participatory co-design of a technology-facilitated intervention program for social isolation and cognitive decline. The proposed research will produce knowledge on the influence of environmental factors on social isolation and cognitive health, and achieve the PI’s long-term professional goal of developing behavioral health intervention to address social isolation and prevent the onset of Alzheimer's Disease and Related Dementias.

PROJECT NARRATIVE

The proposed dissertation work will contribute to the fundamental knowledge with longitudinal empirical evidence on the influence of physical, social environmental factors on social isolation and cognitive health, and contextualized the role of technology use. The postdoctoral research will build knowledge on the behavioral and contextual pathways that could explain the association between social isolation and cognitive decline in later life. Knowledge to be built in the predoctoral and postdoctoral stages will be synthesized to inform the development of a technology-facilitated community-based intervention program to reduce social isolation and prevent cognitive decline among older adults.

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FACILITIES & OTHER RESOURCES

The University of Southern California (USC) will serve as the primary site for Ms. Yu's predoctoral training and research. The Oregon Health and Science University (OHSU) will support the proposed postdoctoral work. Through the F99 award period, Ms. Yu will have easy access to the various facilities and other resources at the Suzanne Dworak-Peck School of Social Work (SSW) and the university at large. She has built collaborative relationships with scholars in the USC Alzheimer Disease Research Center (ADRC). Ms. Yu has access to the training opportunities, support, and resources at the USC ADRC Research and Education Core (REC). In both F99 and K00 phases, Ms. Yu will be able to access resources and support from OHSU School of Medicine and Oregon Layton Aging and Alzheimer's Disease Center (OADC). The three OHSU based cosponsors all serve leadership roles in OADC (Dr. Jeffery Kaye, director of OADC; Dr. Hiroko Dodge, director of the OADC data core; Dr. Lisa Silbert, director of the OADC neuroimaging core). They will direct Ms. Yu to different resources of OADC as her research projects require. Access to facilities and resources at both USC and OHSU will ensure the successful completion of the proposed research and training plan.

University of Southern California (Predoctoral Research and Training Site)

USC Suzanne Dworak-Peck School of Social Work

The USC SSW is strategically placed at the crux of critical societal problems in urban Los Angeles, a vastly diverse setting that reflects, in many ways, all modern complex urban environments. The School is home to the **Hamovitch Center for Science in the Human Services (HRC),** which was first endowed in 1988; the **Edward R. Roybal Institute on Aging**, which conducts translational research on aging in minority populations; the Center for Innovation and Research on Veterans & Military Families, dedicated to advancing evidence-based interventions and workforce development for service members, veterans, and their families; the USC Center for Artificial Intelligence in Society which conducts research to develop, test, iterate, and demonstrate how AI can be used to tackle the most difficult societal problems; the Children's Data Network that maintains a network for a California-wide integrated child services big data system specifically designed for epidemiological research on neighborhood-level effects on child development and the implementation of interventions that will provide therapeutic alternatives to the present juvenile justice system; the National Center for Excellence in Homeless Services' Research Core that conducts research on homelessness and housing focusing on enhancing the health and well-being of persons experiencing homelessness and on solutions to homelessness through permanent housing and supportive services; the Center for LGBT Health Equity that leads scientific inquiry into the physical, emotional and social health of LGBT youth, adults and family, and the California Social Welfare Archives, which creates and houses the most complete documentation of social welfare history in California. Ms. Yu's proposed F99/K00 research and training plans will be situated in the SSW's doctoral program. She has been actively engaged in the research and professional development opportunitiesprovided at the **Edward R. Roybal Institute on Aging**. Ms. Yu has been supported with summer research awards by the **Hamovitch Center for Science in the Human Services (HRC)** and will receive further training and administrative support in writing and submitting Extramural grant applications.

IT service. The School provides more than 20,000 square feet of dedicated research space in two locations (University Park Center and the City Center) for the investigators, research project staff, and administration; secure data storage facilities; symposia spaces that include state-of-the-art video conferencing equipment; and an information technology (IT) network that contains computer stations and digital information and training laboratories offering academic and technological resources and services for students and the faculty. A dedicated IT department maintains computers and local network services. The IT department provides technical assistance to investigators and is responsible for access and security on computers, laptops, and mobile devices. Maintained by the IT department, the offices are equipped with Dell Optiplex 990 desktop computers, featuring Microsoft Office Suite, SAS 9.2 for advanced statistical analysis, NVivo for qualitative data analysis, Bluejeans teleconferencing software, and a camera and microphone. The IT department provides daily backups of each computer. These backups and the data stored on these computers are secured by the university's large-data cyber security system with appropriate firewalls. The software and IT services will provide the necessary support for Ms. Yu to access data analysis software and safeguard online communication with research collaborators during and beyond the COVID-19 pandemic.

Administrative support. The School maintains a strong infrastructure of staff experts in research administration, data analysis and database management. Under the direction of the Research Council, an elected faculty governance body, the Associate Dean of Research and the Director of Research Administration, research administration staff, featuring two nationally Certified Research Administrators (CRAs), provides investigators assistance with proposal development and submission, pre-award responses and budget development, post-award project and budget management, and research compliance. With the assistance of the CRA-led staff, PIs have access to Cayuse and TARA, USC's electronic research administration and information portals. These portals assist researchers and the research administration staff with various aspects of managing sponsored research projects, including pre-award and post-award activities. In addition, investigators have access to iStar, the USC online institutional review board (IRB) application system used to submit, track, and review all research studies involving human subjects. The IRB staff or IRB designee conducts an initial review of all research applications involving human subjects. The IRB staff conducts a thorough pre-review of the application to verify the correct level of review and evaluate the protocol and supporting documents (e.g., consent form, recruitment materials). Any significant changes to the approved study must be submitted and reviewed by the IRB prior to implementation of changes. For studies designated as expedited or full board, IRB review is required from a designated reviewer or the full board. Access to other USC systems relevant to the research mission include disClose for conflict of interest disclosure and management and Sophia for managing inventions, intellectual property, and patents. The administrative team at USC will help to manage the proposed F99/K00 grant project and facilitate the transition from the F to K stage.

Data analysis support. A team of biostatistician, bioengineering statistician and a database manager assists investigators and doctoral students in creating databases and datasets, developing design and analytic approaches, estimating sampling size requirements for specific research questions, planning and performing interim and final analyses, and writing reports to summarize results for manuscript preparation. Additionally, the School has access to senior biostatisticians and methodologists with expertise in specialized areas through its Southern California Statistical Support Network. PIs with big datasets and complex computational models also have access to the USC High Performance Computing Cluster's (HPCC) extensive computer facilities. HPCC features a diverse mix of computing and data resources. Two Linux clusters constitute the principal computing resource. In addition, HPCC has a central facility that provides more than 1.2 petabytes of combined disk storage and potential access to nearly a petabyte of tape storage. The HPCC computing resource consists of two shared-head nodes and various compute nodes. All PIs can obtain free accounts to supplement the HRC database management and data analysis resources to enable successful performance of statistical analyses of longitudinal "big data." Ms. Yu will receive support from biostatisticians for any analytical questions she might have for the proposed F99/K00 research.

SSW doctoral program. Dr. Michael Hurlburt is the director of the SSW's distinguished PhD program. The SSW has one of the most comprehensive support networks for doctoral student researchers. In particular, the SSW doctoral program regularly hosts seminars and training workshops on relevant topics, including proposal development and writing, funding opportunities and strategies, research preparation and training, and research talks and discussions. The overarching goal of the doctoral program is to enhance the success of doctoral student research by providing support and building community. The SSW doctoral program has its own administrative coordinator who works with Dr. Hurlburt, providing additional resources for travel to conferences and stipends. Ms. Yu will have access to informative research workshops organized by the USC Center for Excellence in Research specially designed for early stage investigators and focused on research skills needed for development into an independent scientist.

USC Institute for Creative Technologies

The USC Institute for Creative Technologies is an interdisciplinary laboratory devoted to the advancement of virtual reality training applications, and provides a unique and supportive environment for the advancement of virtual human technology. With two hundred faculty, researchers, graduate students and support staff, no other laboratory in the world has such ready access to excellent talent across the range of disciplines that impact virtual human design: computer graphics, computer animation, computer games, audio rendering, artificial intelligence, natural language processing, and entertainment industry professional content producers. Institute for Creative Technologies has long enjoyed a close relationship with the entertainment industry through its School of Cinema and Television, as well as its Entertainment Technology Center. In 1999, the University expanded its relationship with the entertainment industry when it entered into an agreement with the US Army Research Office to create a University Affiliated Research. Institute for Creative Technologies now represents a unique collaboration between the Hollywood entertainment community, the US Army, and academia, with its mission to create a world-class research facility to develop training and simulation applications, and a successful track record of transitioning technology.

Institute for Creative Technologies has three stories of dedicated space for administration, investigators, and research project staff members and students; secure data storage facilities; two symposia spaces with projector equipment as well as various smaller conference rooms. Institute for Creative Technologies has a full time IT staff that maintains computers and local network services, and a central facility that provides encrypted data storage. The IT department provides technical assistance to investigators and is responsible for access and security on computers, laptops, and mobile devices. All Institute for Creative Technologies investigators on this grant have access to these support services, as well as both desktop and laptop computers with all required software (e.g., Microsoft Office Suite, analytic software such as SPSS or SAS to programing software such as Python, software for physiological data collection such as AcqKnowledge, etc.) as well as sufficient hardware (e.g., high-end graphics cards).

Institute for Creative Technologies also has substantial facilities for conducting research with human subjects, including a 20-participant computer lab and multiple rooms for running participants in individual settings. In addition, Institute for Creative Technologies has access to on-campus facilities for conducting research with human subjects at USC. These laboratories include single-, double- and group-occupancy configurations, and allow for easier recruitment of specialized subject populations (e.g., subjects with clinical training). Ms. Yu could explore the possibility of using facilities at the USC Institute for Creative Technologies to design intervention programs delivered using advanced technology tools.

Scientific Environment

The environment provided by SSW and the different centers contributes to the probability of success of the proposed application through excellent staffing and a climate of encouragement for scientific research. Evidence-based interventions and translational science are core components of the scientific agenda and intellectual foundation of USC as a university and School. SSW has pioneered the science of social work based on a foundation of an interdisciplinary faculty that along with PhD-level social workers also features sociologists, psychologists, anthropologists, nurses, engineers, and a developmental pediatrician. The School continues to have a solid record of success in receiving competitive federal grants by virtue of its strong research environment. These interdisciplinary faculty members work with other schools at USC to produce research that has an international impact and tackles the most challenging problems of society. Through these collaborative institutional networks, the School is well situated to share the benefits of the university's impressive scientific environment.

USC is qualified as one of a few premier research institutions on which our country depends for a steady stream of new knowledge, art, and technology. USC has more than $700 million in annual research expenditures and has ranked among the top 10 private universities in federally supported research activities. This federal support has created a rich array of core facilities and other resources that are directed by the university's internal funding programs to be directed toward establishing a culture and infrastructure for building NIH and NSF research centers. USC is the oldest private research university in the West and one of the nation's most highly selective institutions in terms of the student body it attracts. With 17 professional schools in addition to the Dornsife College of Letters, Arts and Sciences, USC is ranked 11th among U.S. private universities in dollar volume of federal research support. The accredited IRB, libraries, and other resources of the university provide exceptional support of research and scholarship. Federal research funding has doubled in the past 10 years. USC graduates have an invaluable worldwide support network of more than 180,000 living alumni in the Trojan Family. Alumni maintain a close association with the university, mentoring students and opening career doors for graduates. USC libraries actively support the research and scholarship of USC's faculty, students, and staff. The USC library system has 23 libraries, information centers, and the USC Digital Library.

The USC ADRC serves as another excellent source of institutional support for Ms. Yu's proposed training goals. The USC Memory and Aging Center comprises the USC NIH Alzheimer's Disease Research Center (USC ADRC) directed by Dr. Chui and the state-funded California Alzheimer Disease Center (USC CADC) directed by Dr. Schneider. These entities provide clinical care and continuity of research care for people with memory complaints and for non-cognitively-impaired older people who have agreed to long-term follow-up and brain autopsy. The studies conducted at USC ADRC include observational studies that follow participants over time while examining changes that may occur with age, and intervention studies such as exercise training, medications and vaccines. Ms. Yu will participate in training at the USC ADRC Research and Education Core, including topics on types of AD clinical studies and research design, conducting pragmatic trial studies, as well as grant planning and writing. Ms. Yu will also gain clinical experience related to ADRD by engaging in the clinical practice at the USC memory disorder clinic.

Oregon Health & Science University (Postdoctoral Research and Training Site)

OHSU is one of over 100 academic health centers in the nation and the only one in Oregon. The institution comprises 4 primary schools—Schools of Medicine, Dentistry, Nursing, and the newly established School of Public Health. These schools, in combination with its research centers and institutes, create one of the leading academic health and research centers in the nation. OHSU has undergone rapid expansion in recent years. OHSU receives more than $350 million each year in outside funding for sponsored projects. Most of these funds are from the National Institutes of Health. In Fiscal Year 2016, OHSU's funding expenditures totaled $389,602,823, a sum that includes $234,320,764 overall from the NIH. In Fiscal Year 2015, the Knight Challenge, a program sponsored by $500 million from Nike co-founder Phil Knight and his wife, Penny, $200 million in state money, and more matched by private donations—for a total of $1 billion—is allowing OHSU to expand both infrastructure and human resources in the fight against cancer. Construction on the new 320,000 square foot, $160 million Knight Cancer Institute research building began in July 2016 and is slated to be completed in July 2018. Emblematic of the expansion in research and collaboration at OHSU is the LEED Platinum Collaborative Life Sciences Building that brings together Oregon Health & Science University, Oregon State University, and Portland State University in one location. The $295 million building is the first on this scale to combine the resources of multiple universities to offer the best education and research opportunities for students.

Ms. Yu's professional goal of promoting cognitive health through improving older adults' living environment and using technological tools are in line with the research priorities and resources available at the Layton Center and ORCATECH. The facilities and resources at OHSU will facilitate the successful completion of the proposed project.

Layton Aging & Alzheimer's Disease Center: The Layton Aging & Alzheimer's Disease Center is a comprehensive research, clinical care and education center at OHSU focused on identifying the causes of Alzheimer's disease and related disorders, as well as understanding how to ensure healthy brain aging throughout life. The Center has provided the best in clinical diagnosis and care for thousands of patients and families in the northwest for over two decades. Ms. Yu's postdoctoral research will benefit from the wide-ranging training and pilot funding opportunities available at the Layton Center.

Layton Center is dedicated to conducting innovative, forward thinking research, and is recognized as one of the best venues for this work as part of a network of national centers funded by the National Institutes of Health, National Institute on Aging. Since 1990, the overarching goal of the Oregon Alzheimer's Disease Center (OADC) is twofold: (1) to focus on early detection of change in order to define mechanisms supporting healthy brain aging and the transition to MCI and early dementia, and (2) to facilitate developing effective therapies to prevent or limit these transitions. To achieve these objectives, seven cores (Administrative, Clinical & Satellite, Data, Neuropathology, Outreach Recruitment & Outreach, Biomarkers and Neuroimaging) are organized to catalyze and sustain innovative research and discovery in Alzheimer Disease (AD) and related disorders through an organizational infrastructure that supports a rich collaborative environment composed of leading edge technologies and methods, multi-disciplinary scientists, empowered research volunteers, and an inspired community of lay and professional stakeholders. The OADC provides the necessary materials to support research through well-characterized research participants, biological specimens, brain tissue, data provision and analytics. The OADC focuses on specific areas of emphasis: (1) Preclinical dementia and activity of disease emphasizing the oldest old; (2) Markers of meaningful change captured through studies of peripheral biomarkers, neuroimaging and continuous in-home behavioral monitoring; (3) Neuropathology of brain aging and late life dementia; (4) Novel testing of novel treatments; and (5) Improving education and sharing knowledge about dementia.

ORCATECH: ORCATECH (Oregon Roybal Center for Aging & Technology) is an OHSU research center dedicated to the translation of research advances in the basic behavioral and social sciences into new approaches for measuring key aspects of health and well-being. Supported in part by NIH NIA as an Edward R. Roybal Centers for Translation Research in the Behavioral and Social Sciences of Aging, ORCATECH is recognized as a national and global leader in its field. The motivation for ORCATECH's research is the fundamental need to change the way we acquire central outcome measures and to bring these novel methodologies into wider use in research and practice. The impetus for this change is the growing realization that current standard paradigms for capturing core data to inform decisions about health and well-being are limiting, failing to deliver the optimal meaningful information needed on which to base key decisions either for professionals or the lay community. These limitations stem from the nature of conventional approaches that rely largely on brief episodic encounters dependent on personal recall or data that imprecisely incorporate the time domain and are prone to be ecologically invalid.

Ongoing developments (pervasive and ubiquitous computing, in-home and community based embedded sensing, wireless communication and “data mining” technologies) provide a path to challenging the prevalent paradigm so as to provide real-time, real-world data that are objective, continuous and ecologically valid. Integrating these technologies allows one to go from piecing together activity and events to directly capturing behavior itself and then automatically relating one behavior to another. Beginning in 2004, the Oregon Roybal Center for Aging & Technology (ORCATECH) was funded to begin development of these new technology-assisted methods for translation into research and wider practice. The Center has focused on creating necessary research procedures, a unique Life Laboratory of community volunteers and researchers, basic functional algorithms of human health-related activity and distinctive collaborations to create new knowledge and facilitate dissemination of this new research approach. ORCATECH provides a unique organizational infrastructure that facilitates the process of developing, translating and disseminating innovative measurement and interpretation of health and well-being data. Taking advantage of advances in pervasive computing, in-home and community based embedded sensing, wireless communication and "data mining" technologies, the Center improves the conduct of timely, real world measurement of health and wellbeing.

Research Resources: The Layton Center and ORCATECH are committed to sharing data and facilitating collaborations to advance aging research. They work closely to make the data and specimens that comprehensively characterize longitudinal cohorts of research volunteers available to the wider research community. In order to enact this aspect of our shared mission, we collaborate extensively with OHSU investigators and researchers at other institutions to analyze and publish findings of professional interest in aging and dementia research. Such collaboration facilitates efficient and productive use of the data, as well as rapid development and dissemination of knowledge. The Layton Center and ORCATECH collect several different kinds of data and specimens, including but not limited to:

- clinical data,
- brain imaging data,
- activity data, video data,
- audio data,
- recruitment data,
- biomarker and genetic specimens,
- pathology specimens, and
- data generated from biomarker, genetic and pathologic specimens.

As of December 2017, our resources broadly include data and specimens from over 1,200 volunteers total, with 352 currently active. For details about management, contents and sources, request a copy of our Research Repository Protocol from Tracy Zitzelberger at [redacted]

Laboratory

The OADC Clinical Core uses the CLIA certified laboratories of the OHSU for patient assessments. The OHSU Department of Radiology supports one 1.5 T GE, and one 3 T Siemens MRI system utilized for research for the OADC. In addition, there are extensive resources for research imaging afforded by OHSU’s Advanced Imaging Research Center (see below).

Neuropathology Core: The OADC Neuropathology Core consists of 3 rooms containing a total of 775 square feet of dedicated laboratory space. This space houses a full histology area with appropriate venting (room 1352), a biochemical workspace with biosafety cabinet (room 1358), and a workspace (room 1364) for NP Core staff that includes 4 desks with computers as well as laboratory bench space. Fixed and frozen brain tissues are dissected in a dedicated 150 square foot alcove of the Basic Science Morgue (room 2310-C). Brain autopsies are performed in the fully equipped OHSU Department of Pathology Richard Jones Hall Morgue, room 2310. Dedicated storage space for tissues is found in room 2310 and in 150 square feet of room 1329 Richard Jones Hall. See also Facilities & Other Resources for the Neuropathology Core.

Biomarkers Core: The OADC Biomarkers Core is assigned a 400 square foot laboratory on the 4th floor of the Basic Research Building, adjacent to the building housing OADC staff offices. Blood is collected in the clinical research unit and CSF and plasma divided and snap frozen before transfer to the Biomarkers lab for long-term storage. DNA is extracted from whole blood in the Biomarkers Core laboratory. The lab is equipped with centrifuge, pipetters, bench space, hood space, and a NanoDrop 2000 spectrophotometer (NanoDrop Products, Wilmington, DE) for DNA extraction. The lab is also equipped with four -80 freezers for sample storage. The freezers are all alarmed to permit rescue of the samples in the event of a power or equipment failure. See also Facilities & Other Resources for the Biomarkers Core.

Neuroimaging Core: The OADC Neuroimaging Core is assigned offices on the 13th floor of the Hatfield Research Building, contiguous to OADC staff offices. Neuroimaging research has an over 20-year history of substantive longitudinal neuroimaging research, with a focus on healthy aging, mild cognitive impairment (MCI) and MRI markers of early dementia in the elderly, and includes an extensive longitudinal database of over 3700 MRI's from over 1000 subjects. In collaboration with the OHSU Advanced Imaging Research Center (AIRC), the OADC neuroimaging lab has recently implemented advanced imaging sequences and analysis techniques to further our understanding of MRI markers associated with age-related changes in cognitive and motor function, including healthy aging, early cognitive decline, and imaging markers of vascular disease and white matter integrity. These advances, have allowed the OADC neuroimaging lab to collaborate and foster MRI research in brain aging both within and outside of OHSU. The OADC neuroimaging lab supports in vivo 3T MRI acquisition and analysis of AD and related disorders, with an additional focus on vascular cognitive impairment and dementia (VCID). In addition to advanced in vivo MRI protocols, we have developed a 7T post-mortem imaging protocol of human brain tissue for the targeted sampling of MRI-defined regions of interest for histopathology. The recent addition of the OHSU Radionuclear Research Center (RRC) has enabled continued development efforts aimed at providing of novel neuroimaging technologies promoting AD-related research to the OHSU community, including tau-PET imaging. The Neuroimaging Core works to consolidate and substantially build upon these advances, using both in vivo and post-mortem MRI methods to detect early brain changes associated with cognitive decline in elderly at risk for dementia.

Advanced Imaging Research Center: OHSU's Advanced Imaging Research Center (AIRC) is located in the new 11 story Biomedical Research Building (BRB) on the main Marquam Hill campus. In 16,000 ft^2, the AIRC houses 3 research-dedicated MR instruments: 1) a whole-body Siemens TIM Trio 3T (with audiovisual fMRI presentation), 2) a whole-body Siemens 7T (with audiovisual fMRI presentation), and 3) a small animal Bruker BioSpin 12T [the 7T and 12T systems are in basement laboratories on bedrock]. The AIRC BRB space includes administrative, reception, and subject preparation areas, faculty and staff offices, sub-divided research associate office areas, a seminar/conference room, an image processing lab, a chemistry lab, an animal surgery, an electronics/machine shop, and a home-built mock scanner for fMRI training/acclimatization. The 12T small laboratory includes a custom-designed mouse holder that positions the head RF transceiver coil (M2M) and inserts into the magnet bore. It interfaces with the integrated animal monitoring (respiration and temperature) and temperature control system (SA Instruments, Inc.), and a Model SAR 830 small animal ventilator (Cwe, Inc.). The AIRC houses its own satellite pharmacy, with a DEA licensee, for controlled substances (e.g., anesthetics) used in animal research. A 2500 ft^2 AIRC satellite facility housing a whole-body Siemens Trio 3T, and animal preparation and recovery rooms, is located at the Oregon National Primate Research Center (OHSU West Campus).

Computing

Information Technology Group: ITG develops, implements, and maintains technology-based solutions enabling us to effectively manage information. This includes maintaining the OHSU network and web system, computer and software upgrades, e-mail, computerized systems for patient records and data, a campus computer and software purchasing and installation program, computer technical help including advice and setup of major new systems for OHSU's core research facilities and training of researchers in implementation of new computing systems.

Advanced Computing Center: The Advanced Computing Center at OHSU (OHSU ACC) is charged with meeting the advanced computing needs of the OHSU research, academic, and administrative community. The ACC is a revenue based service center, partially subsidized by the University, and architected to provide a scalable set of advanced computing services that augment and supplement no-cost core services provided to the OHSU Enterprise by the Information Technology Group. The OHSU ACC is currently collaborating with customers across 23 departments on more than 100 projects, and has the infrastructure, resources, capacity and expertise to accommodate the computing resources to support a full-scale operation of CART Phase II.

The OHSU ACC provides services that include consulting, research storage solutions, secure local and remote collaboration options, virtual server environments for Windows and Unix, equipment hosting for stand-alone research computing systems, application hosting and license management/compliance, and co-location and support for research computing systems. The facility presently provides access to more than 100 open-source research applications compiled for 64-bit platforms as well as a number of statistical and simulation packages. User accounts and access to core functionality such as commonly used open source software are available at nominal cost, while more complex individual needs are accommodated through a modest fee structure. The Advanced Computing Center, through its own and other collaborative research centers, has the following computing and research facilities available:

Facilities and support are available for hosting physical and virtual servers on customer owned equipment or on ACC provided hardware. The OHSU ACC currently runs Solaris, Mac OS X, RedHat and Ubuntu LTS Linux, and Windows but has been designed to accommodate any operating system required by the research community. The ACC maintains a VMWare cluster and supports hosted customer compute clusters. Through a groundbreaking collaboration with Intel, the Advanced Computing Center is actively involved in cloud cluster development and operations.

The OHSU ACC's network infrastructure consists of four high-speed switches delivering Gigabit Ethernet to all hosted resources as well as a limited number of 10 Gigabit connections. This exclusively switched architecture was designed to eliminate the network bandwidth contention common in research environments, while allowing reliable high and low bandwidth connections between OHSU researchers, the Internet, and OHSU's 10 Gigabit connection to Internet2.

The OHSU ACC's tape backup system consists of a dedicated high-end server with multiple Gigabit network, SCSI and fiber channel interfaces for high-speed backups and restores. It uses commercial-grade backup software to control a tape library with an overall capacity of approximately 120 terabytes. The system currently performs a full back-up every week and daily incremental backups, with schedules tuned for customer needs. In addition, an off-site tape rotation to a secure location keeps mission-critical data safe while maintaining compliance with OHSU data retention policies.

OHSU ACC customers authenticate against the OHSU Active Directory whenever possible. For collaboration with users outside OHSU, ACC staff work with the OHSU Security Engineering Team (SECE) to create an OHSU account for remote access, or an ACC-specific account when necessary. All ACC users have read, agreed to, and signed OHSU's Confidentiality Statement and Network Services Usage Agreement as per OHSU policy. All ACC users are required to use encryption in accordance with OHSU's security policies and directives, as well as HIPAA guidelines. ACC facilitates the use of encryption with open source and commercial solutions, and a Secure Remote Access Portal for remote access. In addition to following all OHSU policies & guidelines, ACC also employs a secondary firewall within the OHSU network. The Advanced Computing Center's architecture has been reviewed by the OHSU Office of Integrity, and undergoes periodic internal security audits.

Core staffing for the Center is provided by six full-time Research System Engineers, two part-time System Analysts, and two part-time Database Analysts. The facility is housed in the recently completed state-of-the-art data center on the OHSU West Campus. The 18,000 square foot Data Center West provides redundant systems for power, innovative air handling, physical security, and ample room for growth. Both data centers operated by the OHSU ACC require a check-in at a guard station and data center employee escort, and all employees must display their badge while on the premises. Once inside the data centers, server racks can only be accessed with both a key and keycard. In addition, these facilities have security cameras monitoring them 24/7, as well as the presence of OHSU Public Safety officers.

Work Space Computing: All personnel are provided personal computers. The OADC's Data Management and Statistics Core (DMSC) database server resides on an iMac with a 3.06 GHz Intel Core 2 Duo processor, a 500 GB hard drive and 8 GB of RAM. Incremental database backups to a password protected computer are performed each workday and are also copied to the OHSU Advanced Computing Center (ACC). The ACC's tape backup system consists of a dedicated high-end server with multiple Gigabit network, SCSI and fiber channel interfaces for high-speed backups and restores. In addition, the ACC maintains daily incremental tape backups of our 800 GB secure drive with a one-month retention including a weekly tape copy stored in an offsite secure location. See also Facilities & Other Resources for the DMSC.

Space

Office Space: Research space is allocated at OHSU through the department of each core leader and investigator. About 2,000 square feet of modern contiguous dedicated research space is allotted for the OADC in the Hatfield Research Building. This space houses most of the Center (Administrative, Clinical, Data Management and Statistics, Biomarkers, ORE, and Imaging Cores) and includes a 20-person conference room dedicated for OADC use where all relevant meetings such as OADC journal clubs, clinical-pathological conferences, reliability training sessions, and individual core and Executive Committee meetings are held. There is an additional 500 square feet of cubicle space in the Center for Health & Healing 1 which houses several ORCATECH staff. The Neuropathology Core offices and laboratory are located in the Basic Science Building, a 5-minute walk from the Hatfield Research Building.

Clinical Space: Aging & Alzheimer's Clinic patients are examined in the Platinum LEED certified, Center for Health and Healing (CHH), which is easily reached via a short aerial tram ride from the Hatfield Research Center. Non-cognitively impaired subjects from the community-based cohorts are assessed in the Clinical and Translational Research Center of the Oregon Clinical and Translation Research Institutes (OCTRI) in outpatient units staffed by research assistants and RNs containing 11 exam rooms, procedure rooms, and phlebotomy space in the Hatfield Research Center (10^{th} floor of the Hatfield Research Center, 3 floors below the OADC offices). Participants are also seen in their home in the community.

Offsite Space for Community-Based Studies: Clinical evaluations for the ORCATECH site in Portland, Oregon are performed at continuous care retirement centers, retirement homes, and subsidized housing venues, in addition to community centers and participants' homes across the Portland Metro area. Participating retirement communities' facilities may provide research staff with conference room for examination, testing, and interviewing. Most research visits are completed as individual home visits in the volunteers' homes.

Other: ORCATECH

ORCATECH Life Laboratory: The ORCATECH Life Laboratory (OLL) is a community-based resource of volunteer seniors who have agreed to allow emerging technologies to be deployed and tested in their homes, as well as be available to participate in studies of health and well-being outcomes using pervasive computing methodologies. As of December 2017, the OLL has 128 active participants in the Portland Metro area, residing in eight continuing care retirement communities, apartments, as well as single-family homes (Female: 74%; mean age mean age 86.8 years SD 7.8; 11% non-white). These volunteers undergo semi-annual cognitive and neurological batteries, and report weekly on changes to their health, medications and life events. Technologies permanently deployed in most OLL homes (see **Figure 1**) include activity sensors, contact (door) sensors, phone sensors, home computers, medication tracking devices, a bio-impedance scale and temperature and CO_2 sensors. These technologies, which are intentionally for research purposes, provide a cross-section of basic activity data that can be used as a basis for comparison to modified platforms that are be developed and evolve over time with technological advances. The OLL cohort size changes relative to research community needs and resources engaged from independently funded investigators that use the OLL; currently, the OLL includes about 110 homes. ORCATECH maintains a longitudinal database of the continuous activity data collected in these homes including weekly on-line health and life event reports sent by subjects to the Center. The OLL is an extremely important resource for testing new devices, system configurations and algorithms in parallel with standard systems in the community before any new methodologies are deployed to larger cohorts.

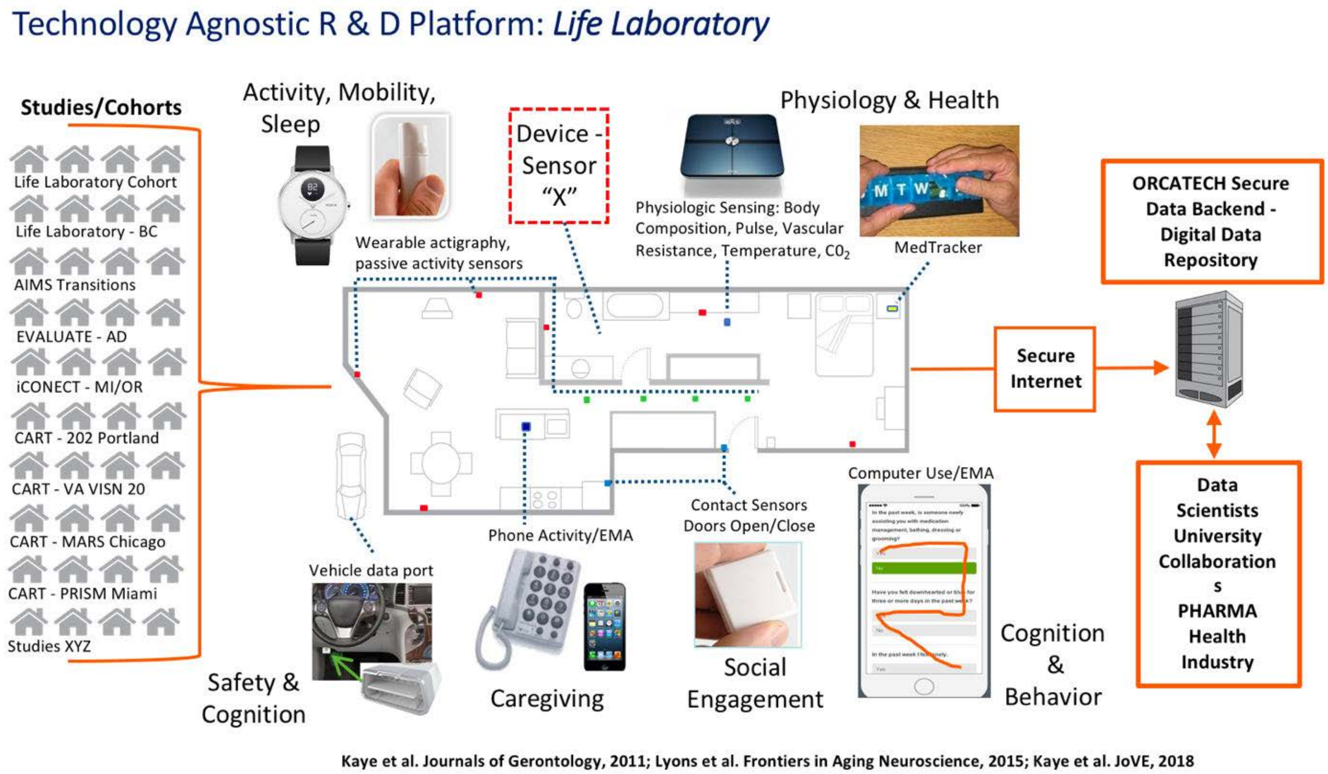

Figure 1. Oregon Center for Aging & Technology Life Laboratory Pervasive Home-based Computing System. The technology-agnostic, continuous-assessment platform can incorporate any "Device/Sensor X" using established protocols and provides a wide array of data capture capabilities: e.g., mobility assessment with passive or wearable sensors, automated physiological monitoring, automated medication adherence tracking, online queries, EMA, or cognitive testing (via laptop, tablet, smartphone), social engagement indicators, driving assessment (via vehicle data port monitoring), and other relevant outputs. Data is sent securely to a central server cluster for system monitoring, data annotation, curation, analytics, and storage. Representative cohorts or studies are listed on the left indicates that any number of studies or projects can use the basic system.

ORCATECH Information System: The ORCATECH Information System schematically shown in **Figure 2** is a sophisticated big data repository and configuration manager for acquiring data from remote Life Lab homes and the RITE Online Cohort, storing the data within a repository, and enabling clinicians to easily install and monitor the installed homes. Data is stored both within relational and non-relational databases on 1 and 10 Gbe high speed networks as well as within raw binary repositories. The back-end server farm includes a web server for delivering web content to residents of the homes and also to ORCATECH technicians ("Installation/Monitoring Unit") who install and configure the homes, and remotely monitoring ongoing system performance and pushing software updates. The sensor system within volunteer homes includes custom and off-the-shelf sensors for measuring movement trends (e.g., ambulation, room transitions) using Zigbee-based sensors (or other communication protocols), physiologic measures (body weight, pulse), atmosphere (temperature, carbon dioxide) and medication compliance (MedTracker) as well as social activity (e.g., computer and phone use). The 10 Gb network is an isolated network protected from outside non-authorized users and consists of a 26 TB SAN and an iSCSI tape backup. The 1 Gb network is designed to communicate directly with the Life Lab homes and/or the ROI cohort and consists of a cluster of 5 Apache Kafka nodes (SuperMicro 6019-MTR – VIPSYS-SR8 16-core 2.1 GHz processors 32 Gb RAM) and a cluster of 5 Xen servers (Dell R620 8-core 2.8 GHz processors with 128 GB of RAM) that host 16 VMs which handle web requests, DNS, VPN, email, NIS, builds and version control for new software releases, backup, hardware and software management, key capture software, graph database and Apache Spark interfaces. An isolated 1 Gb network consists of 7 Apache Cassandra/Spark nodes (SuperMicro SDR-1411-T04 16-core 2.4 GHz processors with 62 Gb of RAM and 3Tb HDD) for long term storage and analysis of sensor and subject generated data. The backend MySQL database runs on a 24-core 3.3 GHz machine with 64 GB of RAM for storage and retrieval of project, study, inventory, subject and home information. At its peak, this system has handled the remote management and the receipt and storage of 3 MB of data per home daily for up to 300 homes and is linearly scalable through the purchase of additional nodes.

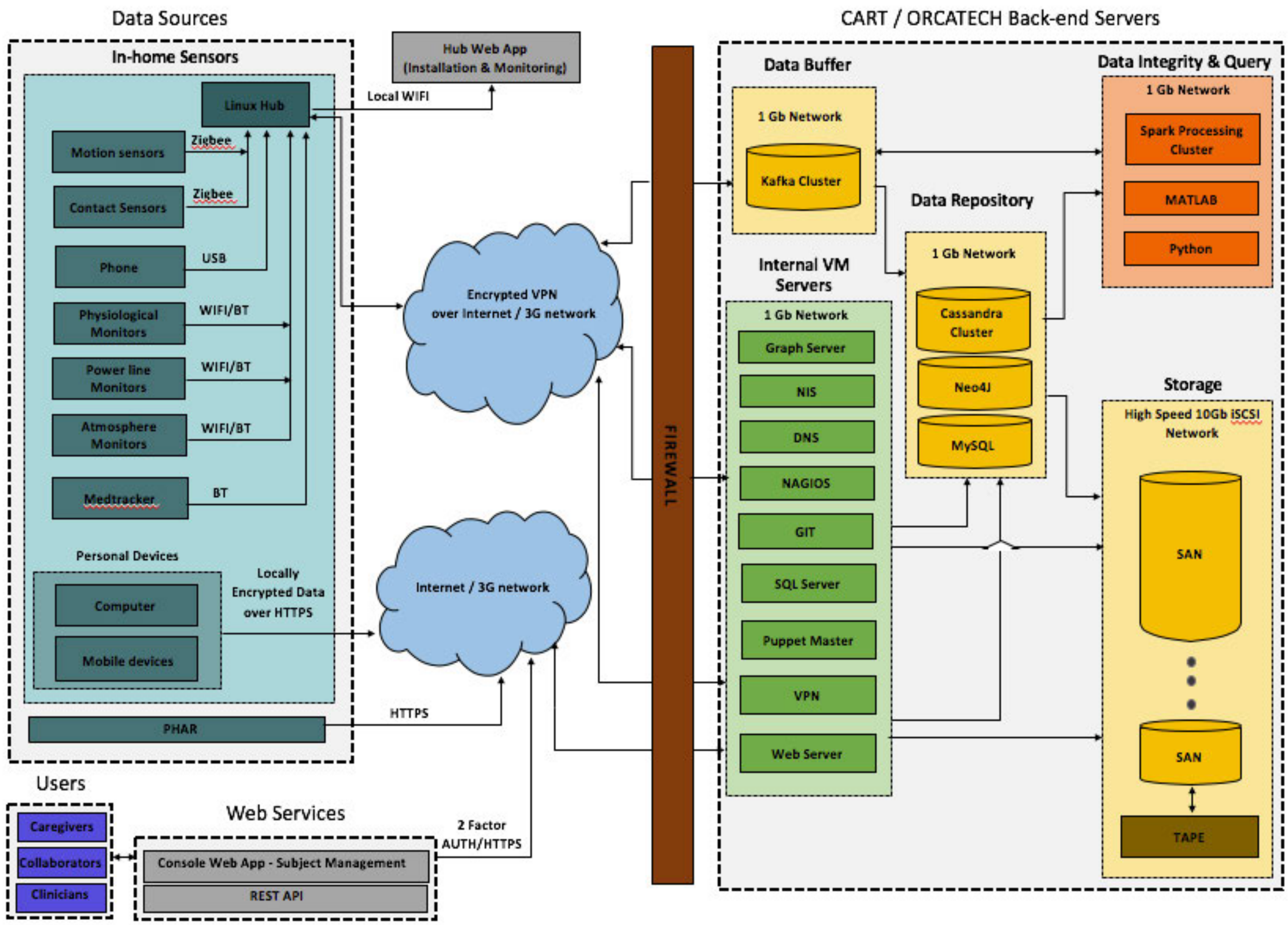

Figure 2. ORCATECH Information System schematic.

ORCATECH Management Console: The ORCATECH Management Console is an online database interface used to manage all ORCATECH cohorts, including over 500 subjects that have been consecutively involved in various projects at multiple sites. This tool allows secure remote management of the cohort (e.g., tracking adherence to completion of questionnaires, including weekly questionnaires, tracking upcoming in-person clinical evaluations, tracking health issues), as well as the in-home systems themselves (e.g., tracking inventory, sensor “health”, and data reliability, as well as supporting remote fixes of problems. The in home hub uses this environment to install the necessary sensor software. The console is continually updated and tailored to meet project needs.

Equipment Development Lab: We have a 200 square-foot room at the Center for Health & Healing 1 for storage and staging of ORCATECH equipment. It is well equipped with electronic test equipment and a stock of components for prototype design. The equipment includes digital and analog oscilloscopes, signal generators, digital counters and voltmeters, spectrum analyzer, power supplies, engineering work stations for circuit board layout, PSpice circuit simulation and PIC microcontroller firmware development. The lab has optical breadboards, mounts and components; is stocked with electronic components and hand tools for prototype fabrication and small production runs.

EQUIPMENT

No major items of equipment will be used in the activities proposed in this application.

USC Suzanne Dwoark-Peck School of Social Work

Ph.D. Program

Michael Hurlburt, Ph.D.
Associate Professor
Director of Doctoral Programs and Chair of the PhD Program
Suzanne Dworak-Peck School of Social Work
University of Southern California (USC)

October 2020

Review committee
Transition to Aging Research Award for Predoctoral Students (F99/K00) National Institute for Aging

Dear Sir/Madam:

RE: **KEXIN YU, PHD CANDIDATE AT THE UNIVERSITY OF SOUTHERN CALIFORNIA (USC) - SUZANNE DWORAK-PECK SCHOOL OF SOCIAL WORK**

I am pleased to submit this letter of certification for Ms. Kexin Yu's F99/K00 (RFA-AG-21-022) proposal, under the primary sponsorship of Drs. Iris Chi, and confirm her eligibility for this award. Ms. Yu is in excellent academic standing with respect to the timeline set forth by the USC School of Social Work Ph.D. program. The F99/K00 serves as a timely opportunity for her training at the dissertation research stage and beyond.

As the Chair of the Ph.D. program at USC Suzanne Dworak-Peck School of Social Work, I certify the following:

1. [redacted]

2. Ms. Yu has passed her qualifying exam and advanced to candidacy. She has completed all requirements for the doctoral degree except for the dissertation.

3. I certify that the USC Suzanne Dworak-Peck School of Social Work's facilities and environment will provide the necessary support for the completion of the proposed research. Ms. Yu is engaged in a multi-disciplinary scientific environment and will be mentored and supported by a highly experienced group of senior faculty members. USC provides an extensive system of library resources, computer labs, qualitative and quantitative analytic software, and IT support to Social Work Ph.D. Candidates.

Obtaining the F99/K00 award will support this outstanding young scientist to receive exceptional training and build collaborative relationships with leaders with similar research interests. The transition award will facilitate Ms. Yu toward becoming an independent investigator, achieving her long-term career goals of making significant contributions to aging research and practice.

Sincerely,

[redacted]

Michael Hurlburt, Ph.D.
Director of Doctoral Programs and Chair of the PhD Program
USC Suzanne Dworak-Peck School of Social Work
University of Southern California
[redacted]
Cell: [redacted]

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: First Name*: Kexin Middle Name Last Name*: Yu Suffix:

Position/Title*: Graduate Student
Organization Name*: University of Southern California
Department: Social Work
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: PD/PI Other Project Role Category:

Degree Type: MSW Degree Year: 2018

Attach Biographical Sketch*: File Name: Applicant_s_biosketch_Kexin_Yu .pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Professor First Name*: Iris Middle Name Last Name*: Chi Suffix:

Position/Title*: Professor
Organization Name*: University of Southern California
Department: Social Work
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: Other (Specify) Other Project Role Category: Sponsor

Degree Type: DSW Degree Year: 1985

Attach Biographical Sketch*: File Name: Biosketch_Chi .pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Shinyi Middle Name Last Name*: Wu Suffix:

Position/Title*: Associate Professor
Organization Name*: University of Southern California
Department: Social Work
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: Other (Specify) Other Project Role Category: Co-Sponsor

Degree Type: Ph.D. Degree Year: 2000

Attach Biographical Sketch*: File Name: Biosketch_Wu_10_09 pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Lawrence Middle Name A. Last Name*: Palinkas Suffix:

Position/Title*: Professor
Organization Name*: University of Southern California
Department: Social Work
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: Other (Specify) Other Project Role Category: Co-Sponsor

Degree Type: Ph.D. Degree Year: 1981

Attach Biographical Sketch*: File Name: Palinkas_NIH_biosketch .pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: First Name*: Middle Name Last Name*: Suffix:

Position/Title*:
Organization Name*:
Department:
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: Other (Specify) Other Project Role Category: Co-Sponsor

Degree Type: Ph.D. Degree Year:

Attach Biographical Sketch*: File Name:

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: First Name*: Lisa Middle Name C. Last Name*: Silbert Suffix:

Position/Title*: Associate Professor
Organization Name*: Oregon Health & Science University
Department: Neurology
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: Other (Specify) Other Project Role Category: Co-Sponsor

Degree Type: MD Degree Year: 1996

Attach Biographical Sketch*: File Name: Biosketch_Silbert_10_12 df

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Jeffrey Middle Name A. Last Name*: Kaye Suffix:

Position/Title*: Professor
Organization Name*: Oregon Health & Science University
Department: Neurology & Biomedical Enginee
Division:
Street1*:
Street2:
City*:
County:
State*:
Province:
Country*:
Zip / Postal Code*:

Phone Number*: Fax Number:

E-Mail*:

Credential, e.g., agency login:

Project Role*: Other (Specify) Other Project Role Category: Co-Sponsor

Degree Type: MD Degree Year: 1980

Attach Biographical Sketch*: File Name: Kaye_Biosketch_OCT_ .pdf

Attach Current & Pending Support: File Name:

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Kexin Yu

eRA COMMONS USER NAME (credential, e.g., agency login): [redacted]

POSITION TITLE: Graduate Student

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
Zhejiang University, Hangzhou, China	B.S.	08/2012	06/2016	Psychology
University of Southern California, Los Angeles, CA	M.S.W.	08/2016	05/2018	Social Work
University of Southern California, Los Angeles, CA	Ph.D.	08/2016	Present	Social Work

A. Personal Statement

My long-term research interests involve examining underlying pathways for the association between social isolation, loneliness, and cognitive decline and developing behavioral health intervention to prevent the onset of Alzheimer's Disease and Related Dementias (ADRD). I aspire to promote positive aging experiences by modifying the physical and social living environments of older adults and employing technology-assisted intervention tools.

My research to date centers on the social connectedness and interpersonal relationships of older adults. I have published two first-authored manuscripts examined the longitudinal relationships between internet use, health conditions, and loneliness. I worked as a research assistant in an intervention study employing mobile health to facilitate self-management among Type 2 Diabetes (T2D) patients. I led the design and data collection of a pilot study to understand the unmet T2D care needs of Chinese and Hispanic immigrants in Los Angeles. I organized focus group discussions with older adult residents and staff at affordable retirement communities to elicit community perspectives on the influence of living environments on health. I served as a co-investigator to co-design a mobile app with Chinese immigrant caregivers to increase self-care intervention accessibility. My research experience and academic training to date have laid the foundation for me to investigate the relationships between Technology use, social isolation experiences, and cognitive health of older adults.

The fieldwork at affordable senior housing communities made me realize the research gap on contextual risk factors impacting older adults' cognitive health and social well-being. Therefore, I devoted my doctoral dissertation to examining the effects of physical and social environments on older adults' social isolation and cognitive health. Social isolation is a known risk factor for developing ADRD. With the support of the F99/K00 award, I will receive interdisciplinary training and develop behavioral interventions that promote cognitive health by addressing social isolation. I will bring the "person in environment" perspective and emphasis on social justice from my social work background to my future ADRD behavioral health research. I believe my choices of sponsors, proposed research and training plans will prepare me to develop into an independent ADRD behavioral intervention researcher with unique sets of skills.

B. Positions and Honors

Professional positions:

06/13 – 08/13 Research Assistant, Human Factors & Applied Cognition Lab, North Carolina State University, Raleigh, NC

01/14 – 12/15 Research Assistant, Lab of Aging and Cognition, Zhejiang University, Hangzhou, China

02/15 – 07/15	Social Work Intern, Hangzhou Lenjoy Community & Charity Service, Hangzhou, Zhejiang Province, China
01/16 – 04/16	Research Assistant, Translating Research in Elder Care Program, University of Alberta, Edmonton, Alberta, Canada
09/16 – 05/17	Social Work Intern, ABC Adult Day Health Care Center, Los Angeles, CA, US
08/18 – 12/18	Teaching Assistant, University of Southern California, Los Angeles, CA
09/16 – 05/20	Research Assistant, Intergenerational Mobile Technology Opportunity Program (IMTOP), University of Southern California, Los Angeles, CA
02/18 –	Co-investigator, Co-Designing a Mobile App with Chinese Immigrant Caregivers for Self-Management of Their Health, University of Southern California, Los Angeles, CA
02/19 –	Co-investigator, Intergenerational mHealth Program for Affordable Housing Communities (ImPAHC) Feasibility Study
01/20 – 05/20	Co-instructor, University of Southern California, Los Angeles, CA

Academic and Professional Honors:

2013 – 2016	Morningside Scholar, Zhejiang University
2014	Zhejiang Province Undergraduate Innovation and Entrepreneurship Fellowship
2014 – 2015	Excellent Undergraduate Student First-class Scholarship, Zhejiang University
2015	Kefa Cen First-class Scholarship, Zhejiang University
2016	University of Alberta Research Assistant Award
2016 – present	University of Southern California, Suzanne Dworak-Peck School of Social Work Fellowship
2017-2020	USC Research Council Summer PhD Research Award
2018	Adult Mental Health and Wellbeing (AMHW) Department Pilot Award
2018	Clinical and Translational Science Institute (CTSI) pilot funding
2018	Nobuo Maeda International Research Award by American Public Health Association
2019	Adult Mental Health and Wellbeing (AMHW) Department Pilot Award
2019	Stanford Longevity Center Forming Science-Industry Research Collaboration Workshop Award
2020	GSA Mentoring and Career Development Technical Assistance Workshop Junior Investigator Diversity Fellow Award.

Service in Professional Societies:

2019 - Present	Reviewer, International Journal of Geriatric Psychiatry
2019 - Present	Reviewer, The Gerontologist
2019 - Present	Reviewer, Journal of Gerontology: Series B
2020 - Present	Reviewer, Ethnicity & Health
2020	Awards selection committee member, Nobuo Maeda International Research Award

C. Contributions to Science

<u>I. Social Isolation and Loneliness.</u> The primary outcome of my research as a graduate student has been social isolation and loneliness among older people. I have examined the longitudinal relationship between internet use on loneliness and changes in big-five personality and loneliness. To bridge my interests in health research and social isolation, I examined the time-sequence and directionality of the associated between geriatric conditions and loneliness. These papers contribute to science with longitudinal empirical evidence of predictors of loneliness. My dissertation work employs an ecological perspective of aging and contributes to the literature by studying the environmental influences on social isolation in later life.

Publications

[redacted]

2. **Yu, K.,** Wu, S., Chi, I. (2020). Internet use and loneliness of older adults over time: The mediating effect of social contact. *Journal of Gerontology: Series B*, gbaa004. https://doi.org/10.1093/geronb/gbaa004

[redacted]

II. Mobile Health (mHealth) Interventions. I have worked as a graduate research assistant on mHealth intervention needs assessment, study design, participant recruitment, and program evaluation. Using trial data collected in Taiwan, we examined the longitudinal effects of a mobile health intervention for T2D management in a rural setting. We conducted a pilot study to understand whether the intervention suits the needs of Chinese and Hispanic immigrants with T2D in the greater Los Angeles. With the results of my undergraduate research, I published a paper on the validity of an online self-administered PhQ-9 for depression screening. This internet screening tool contributed to the development of a mHealth intervention for depression. My research has advanced the field by expanding the mHealth intervention to diverse populations, including older adults living in rural areas and first-generation immigrants.

Publications

1. **Yu, K.,** Wu, S., Lee, P., Wu, D., Hsiao, H., Tseng, Y., Wang, Y., Cheng, C., Wang, Y., Lee, S., Chi, I. (2020) Longitudinal effects of an intergenerational mHealth program for older Type 2 Diabetes patients in a rural area. *The Diabetes Educator*, 46(2), 206–216. https://doi.org/10.1177/0145721720907301

[redacted]

4. Du, N., **Yu, K**., Ye, Y., & Chen, S. (2016). Validity study of Patient Health Questionnaire-9 items for Internet screening in depression among Chinese university students. *Asia-Pacific Psychiatry*, 9(3), e12266 https://doi.org/10.1111/appy.12266

III. Health Service at Retirement Housing Communities. The role of living environments in health service and its influence on health outcomes are not well understood. One of the following listed manuscripts examined the environments of retirement housing communities on T2D management and explored the feasibility of mobile health intervention in this context. The other contributed to the field with synthesized knowledge of oral health care in nursing homes.

Publications

1. Hoben, M., Clarke, A., Huynh, K. T., Kobagi, N., Kent, A., Hu, H., Pereira, R.A., Xiong, T., **Yu, K.**, Xiang, H. and Yoon, M.N., (2017). Barriers and facilitators in providing oral care to nursing home residents, from the perspective of care aides: A systematic review and meta-analysis. *International journal of nursing studies*, 73, 34-51. https://doi.org/10.1016/j.ijnurstu.2017.05.003

[redacted]

D. Research Support and Scholastic Performance

Ongoing Research Support
None

Completed Research Support

[redacted]

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors. **DO NOT EXCEED FIVE PAGES.**

NAME: Iris Chi, MSW, DSW

eRA COMMONS USER NAME (credential, e.g., agency login): [redacted]

POSITION TITLE: Professor and Golden Age Association Frances Wu Chair for the Chinese Elderly

EDUCATION/TRAINING *(Begin with baccalaureate or initial professional education, include postdoctoral and residency training.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Chinese University of Hong Kong	BS	06/1978	Social Science
California State University, San Diego	MSW	06/1981	Social Work
University of California, Los Angeles	DSW	05/1985	Social Welfare
The Gerontological Society of America & The Center for Asian American Study, University of California, Los Angeles	Postdoctoral Fellow	12/1987	Applied Gerontology and Asian American Studies

A. Personal Statement

The overall goal of F99/K00 application is to examine environmental factors related to social isolation and cognitive decline among older adults and develop technology-facilitated behavioral health interventions. As a mentor, I will leverage my expertise to oversee this training program through completion. I will work closely with the applicant, Kexin Yu by providing input on study design, variables construction, data analysis, data interpretation, manuscript writing, grant writing and scholarship development. I will leverage my expertise to guide this program through completion.

I have studied health and mental health issues among older Chinese populations and their caregivers over 30 years. I have the expertise in ethnographic, survey research, longitudinal study, secondary data analysis, and intervention research on behavioral health, chronic disease prevention and self-management studies. I have been involved in seven cross-national comparative studies in this field and most of them have pointed to the fact that responsiveness to cultural needs is very important in terms of enhancing the overall quality of life. My current scholarship has been devoted to improve the health and mental health of the low-income older Chinese immigrants to promote health equity and wellbeing. In addition, my research also examines the role of interpersonal relations and technology in moderating the impact of stress on wellbeing of older Asians and their caregivers. Since 2000, I developed a set of training materials including text, case studies and videos to support caregivers and the materials have been widely used among the Chinese societies around the world. Using these materials in a self-directed learning study has demonstrated effectiveness in strengthening caregivers' capacity and knowledge of caregiving. I am an honorary professor, fellow, consultant and advisor to more than 30 local and overseas universities and research institutes. I have an excellent track record of successful mentorship of mentees to become independent scholars. Many of them are holding prestigious academic positions and contributing to behavioral health and intervention sciences. Between 2001 and 2008, I served as mentor for the China-Rochester Suicide Research Center at the University of Rochester. Since 2015, I have been the mentor for the Chinese Health, Aging, and Policy Program (CHAPP) at Rush University and Asian RCMAR at Rutgers University. I received the Awards for Excellence in Mentoring from the Mellon Academic Mentoring Support Program. Finally, I have established very good relationships with the community health and social service providers who work closely with Asian communities in Southern California. Working with these partners, I am familiar with the barriers in immigrants' access to health care services in U.S.. We will tackle these barriers together by designing the most appropriate interventions to help the older immigrants and their caregivers to maintaining and improving their health.

1. Tang, F., Chi, I., Zhang, W. & Dong, X. (2018). Activity engagement and cognitive function: Findings from a community-dwelling U.S. Chinese Aging Population Study. *Gerontology and Geriatric Medicine*, 4, doi: 2333721418778180
2. Tang, F., Chi, I., & Dong, X. (2019). Gender differences in the prevalence and incidence of cognitive impairment: Does immigration matter? *Journal of the American Geriatrics Society*, 67, S513–S518.
3. Kwan, R., Kwan, C.W., Bai, Y. & Chi, I. (2020). Cachexia and cognitive function in the community-dwelling older adults: Mediation effects of oral health. *Journal of Nutrition, Health & Aging*, 24(2), 230-236. https://doi.org/10.1007/s12603-019-1303-x.
4. Wu, J., Liu, M., Ouyang, Y. & Chi, I. (2020). Beyond Just Giving Care: Exploring the Role of Culture in Chinese American Personal Care Aides' Work. *Journal of Cross-Cultural Gerontology*, *35*(3), 255–272. https://doi.org/10.1007/s10823-020-09404-w.

B. Positions and Honors

Positions

1980 – 1983	Executive Director, Chinese Social Service Centre, San Diego, California, USA
1987 – 1998	Assistant/Associate Professor, Department of Social Work & Social Administration, The University of Hong Kong (HKU), Hong Kong
1998 – 2003	Professor, Department of Social Work & Social Administration, HKU, Hong Kong
1998 – 2004	Director, Sau Po Center on Aging, HKU, Hong Kong
2001 – 2008	Mentor, China-Rochester Suicide Research Centre, University of Rochester
2004 –	Chinese American Golden Age Association Frances Wu Chair for the Chinese Elderly, USC Suzanne Dworak-Peck School of Social Work
2011 – 2014	External Research Assessor, College of Liberal Art and Social Science, City University of Hong Kong, Hong Kong and Department of Social Work, National Taiwan University, Taiwan
2010 -	Senior Scientist, Global and Community Health Program Lead, USC Edward R. Roybal Institute on Aging

Honors

1999 –	Honorary Professor and Associated Health Researcher, Center on Aging, University of Victoria, British Columbia, Canada
2000 –	Fellow, interRAI
2003 –	International Consultant, HKU Jockey Club Centre for Suicide Research and Prevention
2003	Award for Research, International Psychogeriatric Association
2004	Bronze Bauhinia Star, the Government of Hong Kong Special Administrative Region, China
2005 –	Honorary Professor of Gerontology, USC Leonard Davis School of Gerontology
2006 –	Advisor & Research Fellow, Center of Research on Disability and Development, China
2006	Awards for Excellence in Mentoring, Mellon Academic Mentoring Support Program
2009	Distinguished Faculty, S.C. Franklin Award, University of Southern California
2010 -	Fellow, Gerontological Society of America (GSA)
2010	Gary Andrews Fellow, Australian Association of Gerontology, Australia
2011 –	Honorary Advisor, Sau Po Centre on Aging, The University of Hong Kong, Hong Kong
2012 –	International Advisor, Center for Aging and Health Studies, School of Public Policy and Administration, Xi'an Jiaotong University, China
2012 – 2017	Honorary Professor, Nanjing University, Nanjing, China
2013 – 2018	Fellow, Risk Society and Policy Research Center, College of Social Sciences, National Taiwan University, Taiwan
2014 –	Fellow, American Academy of Social Welfare and Social Work
2015 – 2018	Expert and Resource Person, Asian Development Bank (ABD)
2017 –	Honorary Professor, Master of Social Work Education Center, Beijing Normal University, Beijing, China
2017 – 2022	Honorary Consultant, Evidence-based Social Work Research Center, Jiangsu Civic Affair Bureau, Nanjing, China
2018	Nabuo Maeda International Research Award, American Public Health Association
2019	Congressional Women of the Year Award, 27th Congressional District

Journal Reviewer: (over 30 ad hoc reviewer for journals)

Editorial Board and guest editor: (16 journals) JGMS, International Journal of Social Welfare, Geriatrics and Gerontology International, Australasian Journal on Ageing, Journal of Social Work Practice and Evaluation

Associate Editor: (2 journals) BMC Geriatrics & International Journal of Social Welfare

C. Contribution to Science

1. A major part of my publications addressed health and mental health issues in older immigrant populations in the United States. I have been collecting both survey and clinical data since 2004 and I lead a multidisciplinary research team to analyze the data and publish manuscripts. Below are the selected recent publications related to the studies of Chinese Immigrants:
   a. Liu, J., Guo, M., Xu, L., Mao, W. & Chi, I. (2020). Loss of friends and psychological well-being in later life: Evidence from older Chinese immigrants in the United States. *Aging and Mental Health.* https://doi.org/10.1080/13607863.2019.1693967.
   b. Tang, F., Zhang, W., Chi, I., Li, M. & Dong, X. (2020). Importance of activity engagement and neighbourhood to cognitive function among older Chinese Americans. *Research on Aging, 42*(7-8), 226–235. https://doi.org/10.1177/0164027520917064.
   c. Liu, J., Guo, M. & Chi, I. (eds.) (2019). Special issue on Intergenerational relationships and well-being of East Asian older adults in migrant families. *Journal of Ethnic & Cultural Diversity in Social Work*. Routledge: Taylor & Francis Group.
   d. Tang, F., Chi, I., & Dong, X. (2019). Gender differences in the prevalence and incidence of cognitive impairment: Does immigration matter? *Journal of the American Geriatrics Society*, 67, S513–S518.

2. Another contribution area is on the studies of Asian Caregivers:
   a. Wu, J., Liu, M., Ouyang, Y. & Chi, I. (2020). Beyond Just Giving Care: Exploring the Role of Culture in Chinese American Personal Care Aides' Work. *Journal of Cross-Cultural Gerontology*, *35*(3), 255–272. https://doi.org/10.1007/s10823-020-09404-w.
   b. Chi, I. (2020). Caregiver self- management training manual. USC Edward R. Roybal Institute on Aging.
   c. [redacted]
   d. Li, M., Mao, W., Chi, I., & Lou, V. W. (2019). Geographical proximity and depressive symptoms among adult child caregivers: social support as a moderator. *Aging & Mental Health*, 23(2), 205-213.

3. Another contribution is conduct diabetes self-management intervention program, and diabetes health literacy related issues. Below are the selected publications in Diabetes and Technology:
   a. Yu, K., Wu, S., Lee, P., Wu, D., Hsiao, H., Tseng, Y., Wang, Y., Cheng, C., Wang, Y.H., Lee, S. & Chi, I. (2020). Longitudinal effects of a mHealth intergenerational program for Type 2 Diabetes among older adults in rural Taiwan. *The Diabetes Educator, 46(2),* 206–216. https://doi.org/10.1177/0145721720907301.
   b. Yu, K., Wu, S. & Chi, I. (2020). Internet use and loneliness of older adults over time: The mediating effect of social contact. *Journal of Gerontology Series B: Social Sciences*. https://doi.org/10.1093/geronb/gbaa004.
   c. Hsiao, H., Chen, Y., Lu, P.L., Tseng, J., Chi, I. & Wu, S. (2020). Health awareness, attitudes, and behaviors among young-adult tutors in the Intergenerational Mobile Technology Opportunities Program in Taiwan. *Journal of Society for Social Work and Research*, 11(1), 83-102. https://doi.org/10.1086/705416.
   d. Li, M. F., Hagedon, A., Pan, L. C., Hsiao, H., Chi, I., & Wu, S. (2018, July). Obstacles of Utilizing a Self-management App for Taiwanese Type II Diabetes Patients. In International Conference on Human Aspects of IT for the Aged Population (pp. 74-88). Springer, Cham.

4. Beneath are the selected publications related to Cognitive Function:
   a. Kwan, R., Kwan, C.W., Bai, Y. & Chi, I. (2020). Cachexia and cognitive function in the community-dwelling older adults: Mediation effects of oral health. *Journal of Nutrition, Health & Aging, 24*(2), 230-236. https://doi.org/10.1007/s12603-019-1303-x.
   b. Tang, F., Chi, I., & Dong, X. (2019). Gender differences in the prevalence and incidence of cognitive impairment: Does immigration matter? *Journal of the American Geriatrics Society*, 67, S513–S518.

c. Tang, F., Chi, I., Zhang, W. & Dong, X. (2018). Activity engagement and cognitive function: Findings from a community-dwelling U.S. Chinese Aging Population Study. *Gerontology and Geriatric Medicine, 4, doi: 2333721418778180*
d. Lee, Y., Chi, I. & Palinkas, L. (2018). Retirement, leisure activity engagement, and cognition. *Journal of Aging and Health*. doi: 0898264318767030

5. I work immensely in social support for older adults in particularly with intergenerational support in older Chinese. Below are the selected publications related to Family and Social Support:

a. Mao, W.Y., Silverstein, M., Prindle, J.J. & Chi, I. (2020). The reciprocal relationship between instrumental support from children and self-rated health among older adults over time in rural China. *Journal of Aging and Health.* https://doi.org/10.1177/0898264320943759.
b. Tang, F., Zhang, W., Chi, I., Li, M. & Dong, X. (2020). Importance of activity engagement and neighbourhood to cognitive function among older Chinese Americans. *Research on Aging, 42*(7-8), 226–235*.* https://doi.org/10.1177/0164027520917064*.*
c. Mao, W., Chi, I., & Wu, S. (2019). Multidimensional intergenerational instrumental support and self-rated health among older adults in rural China: Trajectories and correlated change over 11 years. *Research on Aging*, *41(2),* 115-138.
d. Xu, L. & Chi, I. (2018). Change patterns of instrumental support from adult children and grandchildren among rural Chinese grandparents. *Journal of Family Studies*, 1-14. 10.1080/13229400.2018.1442240

Complete List of Published Work in My Bibliography:
https://www.ncbi.nlm.nih.gov/sites/myncbi/iris.chi.1/bibliography/51376360/public/?sort=date&direction=ascending

D. Additional Information: Research Support and/or Scholastic Performance
Ongoing Research Support

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

NIH, R01 Fang (PI) 2020 -
Preventive Effects of Activity Engagement in the Context of Chinese Immigrant Experience and Neighborhoods
Role: Consultant

Completed Research Support

[redacted]

[redacted]

NIH, R21, Wang (PI) 2016 – 2018
A Study of a Self-administered Acupressure Intervention Improving Fatigue and Physical Functioning of Chinese Breast Cancer Survivors
Role: Consultant
Via a randomized controlled trial to develop and pilot test the efficacy of a home-based, self-administered acupressure intervention in improving cancer-related fatigue (proximal) and physical functioning and other quality of life outcomes (distal outcomes) of Chinese immigrant breast cancer survivors (versus usual care control group). Using Los Angeles (LA) cancer registry cases, we will enroll 124 first-generation, Chinese-speaking immigrant women with moderate to severe level of fatigue, diagnosed with stage 0-III breast cancer, aged 21-69, and in 1-5 years post-primary treatment.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Shinyi Wu

eRA COMMONS USER NAME (credential, e.g., agency login): [redacted]

POSITION TITLE: Associate Professor of Social Work and Industrial and Systems Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Chung-Yuan Christian University, Taiwan	B.S.	06/1992	Industrial Engineering
University of Wisconsin – Madison	M.S.	12/1993	Industrial Engineering
University of Wisconsin – Madison	Ph.D.	05/2000	Industrial Engineering

A. Personal Statement

I will serve as a co-sponsor for Kexin Yu's proposed F99/K00 application titled "Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology". I have the research expertise and mentoring experiences relevant to the proposed study to support Kexin in successfully carrying out the aims of the study. I have a multidisciplinary background in industrial and systems engineering (ISE), health services and policy research, health economics, and social work. My research has focused on enhancing quality of health services and reducing health disparities through innovative design and implementation of cost-effective, patient-centered approaches including using information and digital health technology. I have successfully led a large-scale study called Diabetes-Depression Care-Management Adoption Trial (DCAT) for low-income, midlife, primarily Latino diabetic populations. The DCAT tested a home-grown integrated telehealth technological system in 8 safety-net clinics to implement evidence-based comorbid depression care management. The trial results showed the intervention's effects on improving health outcomes, reducing costs, and more cost-effective than usual care. Recognizing disadvantaged population needs greater self-management support, I co-investigated employing promotora-assisted depression and self-care management among predominantly Latinos with concurrent chronic illness. The result shows the improvement is equivalent to usual care quality improvement, leading my investigation towards testing individualized technology-assisted intervention. I then led a digital health study to design and test a text-messaging program for exercise prompting for low-income, limited English proficiency Latino diabetes patients. I also assisted in developing and evaluating a mHealth technology for team-based care management to support secondary stroke prevention. More recently I led a trial to develop and test an intergenerational self-management program that teaches skills of using tablet and internet technology and self-monitoring with a home-grown mobile app among older Chinese adults with diabetes. We found longitudinal effects of the program to improve diabetes self-care and health outcomes. I believe my previous experiences and expertise in technology, health services research, and in leading trans-disciplinary, digital health-related clinical trials to improve health outcomes for older adults makes me well qualified to serve as the co-sponsor and nicely complements the expertise of the rest of the sponsor team.

a. **Wu**, **S**., Vidyanti, I., Liu, P., Hawkins, C., Ramirez, M., Guterman, J., Gross-Schulman, S., Sklaroff, L., & Ell, K. (2014). Patient-centered technological assessment and monitoring of depression for low-income patients. *Journal of Ambulatory Care Management*, *37*(2), 138–147. PMID: 24525531. doi:10.1097/JAC.0000000000000027.
b. Ramirez, M.*, **Wu, S.**, & Beale, E. (2016). Designing a text messaging intervention to improve physical activity behavior among low-income Latino patients with diabetes: A discrete-choice experiment, Los Angeles, 2014–2015. *Preventing Chronic Disease*, *13*(160035). http://dx.doi.org/10.5888/pcd13.160035

c. Ramirez, M., **Wu**, **S**., Ryan, G., Towfighi, A., Vickrey, B. (2017). Using Beta-version mHealth technology for team-based care management to support stroke prevention: An assessment of utility and challenges. *JMIR Research Protocols,* 6(5), e94. PMCID: PMC5461415
d. Yu, K. *, **Wu, S.**, Lee, P., Wu, D., Hsiao, H., Tseng, Y., Wang, Y., Cheng, C., Wang, Y., Lee, S., Chi, I. (2020) Longitudinal effects of an intergenerational mHealth program for older Type 2 Diabetes patients in a rural area. *The Diabetes Educator*, 46(2), 206–216. https://doi.org/10.1177/0145721720907301

B. Positions and Honors

Positions and Employment

1999-2006	Associate Engineer, Health Program, RAND, Santa Monica, CA
2006-2007	Engineer, Health Program, RAND, Santa Monica, CA
2006-2009	Associate Director, Roybal Center for Health Policy Simulation, RAND
2008-2014	Assistant Professor, Daniel Epstein Department of Industrial and Systems Engineering, University of Southern California, Los Angeles, CA
2008-2012	Affiliated Adjunct Engineer, Technology & Applied Science Group and Health Program, RAND
2014-present	Associate Professor, School of Social Work and Daniel J. Epstein Department of Industrial and Systems Engineering, University of Southern California, Los Angeles, CA
2015-2016	Associate Director of Social and Health Services Research, Edward R. Roybal Institute on Aging, University of Southern California, Los Angeles, CA
2016-2017	Associate Director, Center for Artificial Intelligence in Society, University of Southern California, Los Angeles, CA

Other Experience

2008-2011	National Science Foundation (NSF) Peer Review Committee: Service Enterprise Systems
2011-2014	Committee Member, Quality Committee of the Hospital Governing Boards of USC University Hospital and USC Norris Cancer Hospital
2012-2015	Technical Consultant to US Department of Veterans Affairs Health Services Research & Development: Center of Excellence (COE) and Center for the Study of Healthcare Innovation, Implementation & Policy (CSHIIP)
2013	Expert Testimony at Informational Hearing on Improving Outcomes through the Patient Centered Medical Home, Assembly Committee on Health, Assembly California Legislature.
2013-2014	Primary Care Practice Facilitation Technical Expert Panel to the Agency for Healthcare Research and Quality (AHRQ)

Honors

2004	Most Outstanding Abstract, AcademyHealth 21th Annual Research Meeting, San Diego, CA
2007	RAND Merit Bonus Award for Improve Policy and Decisionmaking through Research and Analysis, Santa Monica, CA
2007	Honor of Service for Quality, Chinese Society for Quality, Taipei, Taiwan
2008	Faculty Development Award, Chinese American Faculty Association of Southern California, Los Angeles, CA
2014	Best Poster Paper Award, International Conference on Big Data and Analytics in Health Care
2014	Honorable Mention, Preventing Chronic Disease Annual Student Paper Contest, National Center for Chronic Disease Prevention and Health Promotion, U.S. Centers for Disease Control and Prevention (With Doctoral student Brian Wu et al.)
2015	Outstanding Teacher of the Year, Daniel J. Epstein Department of Industrial and Systems Engineering, University of Southern California
2015	Sterling C. Franklin Award for Distinguished Faculty, School of Social Work, University of Southern California
2015	Honorable Mention, Preventing Chronic Disease Annual Student Paper Contest, National Center for Chronic Disease Prevention and Health Promotion, U.S. Centers for Disease Control and Prevention (With Doctoral student Haomiao Jin et al.)
2015	Top Rated Abstract, Society for Medical Decision Making 37th Annual North American Meeting, St. Louis, MO
2018	Award for Excellence in Mentoring in the category of Faculty Mentoring Graduate Students, University of Southern California
2018	Nobu Maeda International Research Award, American Public Health Association
2019	Finalist, Fannie Mae Sustainable Communities Innovation Challenge

C. Contributions to Science

1. My research contributes to implementation science to reduce health disparities in safety net chronic illness patients. In the context of a national evaluation of *Hablamos Juntos*, an innovative language access services for Spanish-speaking patients with limited English proficiency, I led the analysis of implementation costs and lessons learned. The key lessons learned are that there are a variety of creative models that can be employed to develop, enhance, and sustain language access services, but there are considerable operating costs and many internal and external system changes required for their successful implementation and sustainable adoption. To address health disparities in low-income, minority population, in partnership with the LA County Department of Health Services (DHS), I led a team to design the DCAT cutting-edge automated telephonic assessment system and integrated the telecommunications with the clinical registry system to prompt providers with tasks and alerts. This solution automates individualized, regular assessment of a patient's condition (based on patient medical history and preferences such as language), thus identifying their unmet needs in a timelier manner and strengthening emotional support. To test this technology, I conducted a quasi-experimental 3-group trial with 1,406 low-income, predominantly Hispanic/Latino diabetes patients served by DHS safety-net clinics. Sample *Hablamos Juntos* and DCAT publications not repeating the above include:

   a. **Wu**, **S**., Ridgely, M. S., Escarce, J. J., & Morales, L. S. (2007). Language access services for Latinos with limited English proficiency: Lessons learned from *Hablamos Juntos*. *Journal of General Internal Medicine*, *22*(Suppl 2), 350–355. PMCID: PMC2078542
   b. **Wu**, **S**., Duan, N., Wisdom, J. P., Kravita, R. L., Owen, R. R., Sullivan, G., Wu, A.W., Di Capua, P., & Hoagwood, K. E. (2014). Integrating science and engineering to implement evidence-based practices in health care settings. *Administration and Policy in Mental Health and Mental Health Services Research*, *42*(5), 588-592. PMCID: PMC4363001
   c. **Wu**, **S**., Ell, K., Jin, H., Vidyanti, I., Chou, C-P., Lee, P-J., Gross-Schulman, S., Sklaroff, L., Belson, D., Nezu, A.M., Hay, J., Wang, C-J., Scheib, G., Di Capua, P., Hawkins, C., Liu, P., Ramirez, M., Wu, B.W., Richman, M., Myers, C., Gonzalez, J., Agustines, D., Dasher, R., Kopelowicz, A., Allevato, J., Roybal, M., Ipp, E., Haider, U., Lakshaman, R., Graham, S. Mahabadi, V., & Guterman, J. (2018). Comparative effectiveness of a technology-facilitated depression care management model in safety-net primary care patients with type 2 diabetes: 6-month outcomes of a large clinical trial. *Journal of Medical Internet Research, 20*(4), e147. PMCID: PMC5938593
   d. Evanson, O.* & **Wu**, **S**. (2019). Comparison of satisfaction with comorbid depression care models among low-income patients with diabetes. *Journal of Patient Experience*. ePub on https://doi.org/10.1177/2374373519884177.

2. My cost-effectiveness analysis work aims to facilitate implementation of evidence-based health interventions and programs. In many preventive and health service areas, cost-effectiveness analyses can provide critical information to help policy-makers and community planning organizations devise strategies that maximize the use of scarce resources to promote population health and social justice. To advance that goal, I have conducted cost-effectiveness analysis and analytic modeling in many health-related areas, including HIV prevention, falls prevention, physical activity promotions, and technology-facilitated care-management.

   a. **Wu**, **S**., Keeler, E. B., Rubenstein, L., Maglione, M., & Shekelle, P. (2010). A cost-effectiveness analysis of a proposed national falls prevention program. *Clinics in Geriatric Medicine*, *26*(4), 751-766. PMID: 20934620
   b. **Wu**, **S**., Shi, Y., Cohen, D. A., Pearson, M. L., & Sturm, R. (2011). Economic analysis of physical activity interventions. *American Journal of Preventive Medicine*, *40*(2), 149–158. PMCID: PMC3085087
   c. Babey, S., **Wu**, **S**., & Cohen, D. A. (2014). How can schools help youth increase their physical activity? An Economic analysis comparing school-based programs. *Preventive Medicine*, *69*, S55–S60. PMID: 25456799
   d. Hay, J., Lee, P. J., Jim, H., Guterman, J. J., Gross-Schulman, S., Ell, K., & **Wu, S.** (2017). Cost-effectiveness of a technology-facilitated depression care management adoption model in safety-net primary care patients with type 2 diabetes. *Values in Health, 21*(5), 5616-568. PMCID: PMC5953558.

3. My recent efforts in finding technological solutions to provide population-based chronic illness care involve assessment of engagement, acceptance, and preferences among patients and providers in technology-

facilitated interventions and testing of intervention effectiveness. I served as the PI of the following studies and advised junior investigators in these publications as their dissertation advisor or clinical scholar mentor.

a. Vidyanti, I., Wu, B., & **Wu**, **S**. (2015). Low-income minority patient engagement with automated telephonic depression assessment and impact on health outcomes. *Quality of Life Research*, *24*(5), 1119-1129, PMCID: PMC4412647
b. Di Capua, P., Wu, B., Sednew, R., Ryan, G., & **Wu, S**. (2016). Complexity in redesigning depression care: Comparing intention vs. implementation of an automated depression screening and monitoring program. *Population Health Management, 19*(5), 349-56. PMID: 27028043
c. Ramirez, M. & **Wu, S**. (2017). Phone messaging to prompt physical activity and social support among low-income Latino patients with type 2 diabetes: A randomized pilot study. *JMIR Diabetes* 2(1), e8. PMID: 30291094
d. Jin, H. & **Wu**, **S**. (2020). Screening depression and related conditions via text messaging vs. interview assessment for underserved, predominantly minority safety net primary care patients: A Validity Study. *Journal of Medical Internet Research*, *22*(3):e17282. doi: 10.2196/17282.

4. My research over the last two decades has examined systems approaches to improve quality of health service delivery. My work included development and evaluation of supportive approaches, such as toolkit development and coaching, for healthcare organizations' implementation of the Chronic Care Model (CCM) for improving care for people with chronic illnesses. I also led a study to evaluate the impact of implementing Lean principles to facilitate safety-net clinics to improve care and outcomes for people with chronic illnesses. I served as the principal investigator for those studies. Recently, together with my doctoral students, I am using predictive modeling and agent-based simulation to understand delivery method and system leverages to improve quality of care.

a. Coleman, C., Pearson, M., & **Wu**, **S**. (2011). Integrating chronic care and business strategies in the safety net: A practice coaching manual. In R. Simons (Ed.), Operations Management: A Modern Approach (pp. 267–296). Canada: Apple Academic Press, Inc., 2011. ISBN: 978-1-926692-90-6.
b. **Wu**, **S**., Liu, P., & Belson, D. (2013). Multiple-hospital lean initiative to improve congestive heart failure care: A mixed-methods evaluation. *Journal of the Society of Healthcare Improvement Professionals*, *3*.
c. Jin, H., **Wu**, **S**., & Di Capua, P. (2015). Development of a clinical forecasting model to predict comorbid depression among diabetes patients and an application in depression screening policy making. *Preventing Chronic Disease*, *12*, E142. PMCID: PMC4561536 (Selected as Honorable Mention in the 2015 Preventing Chronic Disease Annual Student Paper Contest.)
d. Alibraham, A., & **Wu**, **S**. (2018). An agent-based simulation model of patient choice of health care providers in accountable care organizations. *Health Care Management Science*, *21*(1), 131-143. PMID: 27704322

Complete List of Published Work in My Bibliography:
http://www.ncbi.nlm.nih.gov/sites/myncbi/shinyi.wu.1/bibliography/50282292/public/?sort=date&direction=ascending

D. Additional Information: Research Support and/or Scholastic Performance
Ongoing Research Support

[redacted]

[redacted]

Completed Research Support

[redacted]

CBET-1548502 Mataric (PI) 09/01/2015 – 08/31/2019
National Science Foundation
EAGER: Studying the Dynamics of In Home Adoption of Socially Assistive Robot Companions for Elderly
There is a clear need for ethical, age-appropriate, and personalized companion technologies for the elderly and their caregivers, and studies already show a user preference for embodied (robot) companions over screen-based ones. This project will iteratively develop and probe the effects of introducing an in-home socially assistive robot at multiple levels (with the older adult user, with an in-home family caregiver, and with other family members) and in different interaction contexts (user-caregiver and user-visitors).
Role: Co-Principal Investigator

[redacted]

U54NS081764 Vickrey (PI) 09/01/2012 – 08/31/2018
National Institute of Neurological Disorders and Stroke, NIH
Los Angeles Stroke Prevention/Intervention Research Program in Health Disparities
This program is a multi-partner research center that will conduct two randomized, controlled, community-based trials of stroke-prevention interventions. My research is to design a mobile health technology application and an evaluation to measure perceptions of usefulness, ease of use, and willingness to adopt the mobile health technology from multiple stakeholders' perspectives.
Role: Subaward Co-Principal investigator

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Lawrence A. Palinkas

eRA COMMONS USER NAME (credential, e.g., agency login): [redacted]

POSITION TITLE: Albert G. and Frances Lomas Feldman Professor of Social Policy and Health

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of Chicago, IL	B.A.	06/1974	Anthropology
University of California, San Diego	M.A.	06/1975	Anthropology
University of California, San Diego	Ph.D.	06/1981	Anthropology

1. Personal Statement

My extensive content expertise in mental health services and implementation science and my methodological expertise in mixed method designs position me as an ideal participant in the proposed F99/K00 project. In my own research program, I am particularly interested in the role of social networks and the use of research evidence in the implementation of evidence-based practices in child welfare, child mental health and juvenile justice systems; psychological trauma and disaster mental health; and prevention of HIV and drug use in vulnerable populations. I have held positions of leadership in studies that expanded my understanding of the several aspects of the current proposal. I conducted research on the implementation of a psychoeducational intervention designed to prevent HIV and STD incidence in female sex workers in Mexico. I was the Principal Investigator of a NIDA-funded study designed to develop a system for measuring sustainment of SAMHSA-funded substance use and suicide prevention programs. I was the Principal Investigator of a large NIDA-funded randomized controlled trial of social skills training, social network restructuring and case management on treatment and prevention of drug and alcohol use in female adolescents. I have been engaged in developing new types of mixed method designs that target implementation of evidence-based practices and addressing the behavioral health needs of youth in child welfare, mental health, and juvenile justice settings. Finally, my own interest and expertise in the areas discussed above gives me a profound appreciation of the research and community impact of the proposed grant, such that I am deeply committed to its execution and successful completion.

1. **Palinkas LA**, Aarons GA, Horwitz SM, Chamberlain P, Hurlburt, M, Landsverk J. Mixed method designs in implementation research. *Admin Policy Ment Health;* 2011; 38:44-53. PMID 20967495
2. **Palinkas LA**, Horwitz SM, Green CA, Wisdom JP, Duan N, Hoagwood KE. Purposeful sampling for qualitative data collection and analysis in mixed method implementation research. *Admin Policy Ment Health,* 2015; 42:533-544. doi: 10.1007/s10488-013-0528-y. PMID:24193818
3. **Palinkas LA**, Zatzick D. Rapid assessment procedure informed clinical ethnography (RAPICE) in Pragmatic clinical trials of mental health services implementation: Methods and applied case study. *Admin Policy Ment Health*; 2018, https://doi.org/10.1007/s10488-018-0909-3 PMID:30488143
4. **Palinkas LA**, Spear SW, Mendon SJ, Villamar J, Reynolds C, Green CD, Olson C, Adade A, and Brown CH. Conceptualizing and measuring sustainability of prevention programs, policies and practices. *Translat Behav Med*; 2019 Nov 25. pii: ibz170. doi: 10.1093/tbm/ibz170. PMID:31764968

B. Positions and Honors

1981-1982 Statistician, Naval Health Research Center (NHRC), San Diego, CA

1981-1984	Visiting Lecturer, University of California, San Diego (UCSD)
1982-1984	National Research Council Postdoctoral Research Associate, NHRC, San Diego, CA
1984-1989	Research Psychologist, Naval Health Research Center, San Diego, CA
1986-1989	Assistant to Associate Adjunct Professor, Department of Community and Family Medicine, UCSD
1993-2005	Associate to Professor, Department of Family and Preventive Medicine, UCSD
1996-2005	Director, Immigrant/Refugee Health Studies, UCSD
1999-2005	Professor, UCSD/San Diego State University Joint Doctoral Program in Clinical Psychology
1998-2005	Professor, UCSD/San Diego State University Joint Doctoral Program in Public Health
2005-2014	Adjunct Professor, Dept of Family and Preventive Medicine, UCSD
2005-present	Professor, School of Social Work and Department of Anthropology, University of Southern California (USC), Los Angeles
2007-present	Professor, Department of Preventive Medicine, USC
2008-present	Albert G and Frances Lomas Feldman Professor of Social Policy and Health, USC
2009-2014	Adjunct Professor, Department of Medicine, UCSD
2014-2019	Chair, Department of Children, Youth and Families, School of Social Work, USC

Other Experience and Professional Memberships

1992	Elected Fellow, American Anthropological Association
1996-present	United States Representative to the Human Biology and Medicine Working Group, Scientific Committee on Antarctic Research
1997-2000	Member, Committee on Space Biology and Medicine, Space Studies Board, National Research Council/ National Academy of Sciences
2003-2005	Chair, External Advisory Council, National Space Biomedical Research Institute
2004-2005	Member, Committee on Reviewing NASA's Biomedical Critical Path Roadmap, Institute of Medicine
2006-2010	Member, Services Review Non-Specialty IRG, National Institute of Mental Health
2009-2010	Member, Decadal Survey on Physical and Biological Sciences in Space: Human Behavior and Mental Health Panel, Space Studies Board, National Research Council/National Academy of Sciences
2010-11	Member, Committee to Review the Federal Response to the Health Effects Associated with the Gulf of Mexico Oil Spill, Institute of Medicine.
2013	Member, Committee on Ethics Principles and Guidelines for Health Standards for Long Duration and Exploration Spaceflights, Institute of Medicine
2013-16	Member, Forum on Child Cognitive, Affective and Behavioral Development, Institute of Medicine
2014-	Member, Committee on Aerospace Medicine and Medicine of Extreme Environments, Institute of Medicine

Honors

1989	Antarctic Service Medal (National Science Foundation/U.S. Navy)
1998	Visiting Professor, Peking Union Medical College, Beijing, China
2002	Deputy Chief Officer, Life Sciences Standing Scientific Committee, Scientific Committee on Antarctic Research
2004-2006	Visiting Professor, University of Oulu, Finland
2010	Sterling C. Franklin Award for Outstanding School of Social Work Faculty, USC
2011-12	Visiting Professor, School of Public Health, University of California, Berkeley
2015	Excellence in Leadership and Creativity Award, School of Social Work, USC
2015	Elected Fellow, American Academy of Social Welfare and Social Work
2017	Elected Fellow, Society for Social Work and Research
2019	Elsevier Atlas Award for research that could significantly impact people's lives around the world

C. Contributions to Science

1. I have devoted much of my career to the challenges of developing innovative approaches to integrating quantitative and qualitative methods in mixed methods designs for health services research. This work has led to development of guidelines for using qualitative and mixed methods in NIH and PCORI-funded research and for sampling study participants in mixed method implementation research. It has also led to the creation of new methods for measuring processes and outcomes of implementation and sustainment of

evidence-based programs and practices, including a rapid assessment version of clinical ethnography, procedures for quantifying qualitative data, use of social network analysis to develop and tailor strategies to facilitate implementation, and assessment of sustainment of prevention programs and initiatives.

a. Valente T, **Palinkas LA**, Czaja S, Chu KH, Brown CH. Social network analysis for program implementation (SNAPI). *PLOS One*, 2015; 10(6):e0131712. doi: 10.1371/journal.pone.0131712 PMID:26110842
b. Zatzick DF, Russo J, Darnell D, Chambers DA, **Palinkas LA**, Van Eaton E, et al. An effectiveness-implementation hybrid trial study protocol targeting posttraumatic stress disorder and comorbidity. *Implement Sci*, 2016; 11(1):58. doi: 10.1186/s13012-016-0424-4. PMID:27130272
c. Brown CH, Curran G, **Palinkas LA,** Wells KB, Jones L, Collins LM, et al. An overview of research and evaluation designs for dissemination and implementation. *Ann Rev Public Health,* 2017; 38:1-22. doi:10.1146/annurev-publhealth-021816-044215. PMID:28384085
d. **Palinkas LA,** Mendon SJ, Hamilton AB. Innovations in mixed methods evaluations. *Ann Rev Public Health*, 2019; 40:20.1–20.20. https://doi.org/10.1146/annurev-publhealth-040218-044215

2. My current work is focused on the challenges involved in overcoming the wide gap in research to practice through understanding barriers to the implementation of evidence-based practices in mental health services and disease prevention, and developing and evaluating strategies designed to overcome these barriers. My contribution to this work has been in understanding the influence of social networks and use of research evidence in evidence-based practice implementation and in the application of novel mixed method designs in implementation research. This work has demonstrated that EBP implementation requires effective partnerships and collaborations that exchange information and resources through influence networks governed by the sharing of distributed understandings.
   a. Aarons GA, **Palinkas LA**. Implementation of evidence-based practice in child welfare: service provider perspectives. *Admin Policy Ment Health,* 2007; 34(4):411-419. PMID:17410420
   b. **Palinkas LA**, Holloway IW, Rice E, Fuentes D, Wu Q, Chamberlain, P. Social networks and implementation of evidence-based practices in public youth-serving systems: a mixed methods study. *Implementation Sci* 2011; 6:113. doi: 10.1186/1748-5908-6-113. PMID:21958674
   c. **Palinkas LA,** Spear SE, Mendon SJ, Villamar J, Valente T, Chou CP, Landsverk J, Kellam SG, Brown CH. Measuring sustainment of prevention programs and initiatives: A study protocol. *Implement Sci* 2016; 11:95 doi: 10.1186/s13012-016-0467-6 PMID:27422149
   d. **Palinkas LA**, Chou CP, Spear SE, Mendon SJ, Villamar J, Brown CH. Measurement of sustainment of prevention programs and initiatives: The Sustainment Measurement System Scale. *Implement Sci* 2020, 15:71. doi: 10.1186/s13012-020-01030-x. PMID: 32883352

3. Much of my research on evidence-based practice implementation has occurred in the context of hybrid effectiveness – implementation studies of child mental health services. The Clinical Treatment Project, funded by the MacArthur Foundation, found that modularized versions of evidence-based treatments of depression, anxiety, and conduct disorders in youth were more effective in reducing symptoms than standard manualized versions. The modular approach is also more likely to be sustained by practitioners upon conclusion of a clinical trial because it enables a form of cultural exchange between researchers and practitioners.
   a. **Palinkas LA**, Schoenwald SK, Hoagwood K, Landsverk J, Chorpita BF, Weisz JR, Research Network on Youth Mental Health. An ethnographic study of implementation of evidence-based treatment in child mental health: first steps. *Psychiatr Serv* 2008; 59:738-746. PMID:18586990
   b. Weisz JR, Chorpita BF, **Palinkas LA**, Schoenwald SK, Miranda J., Bearman SK, et al. Testing standard and modular designs for psychotherapy with youth depression, anxiety, and conduct problems: a randomized effectiveness trial. *Arch Gen Psychiatry* 2012; 69(3):274-282. PMID:22065252
   c. **Palinkas LA,** Weisz JR, Chorpita B, Levine B, Garland A, Hoagwood KE, Landsverk J. Use of evidence-based treatments for youth mental health subsequent to a randomized controlled effectiveness trial: A qualitative study. *Psychiatr Serv* 2013; 64:1110-1118. PMID:23904009
   d. Chorpita BF, Weisz JR, Daleiden EL, Schoenwald SK, **Palinkas LA,** Miranda J, et al. Long term outcomes for the Child STEPs randomized effectiveness trial: a comparison of modular and standard treatment designs with usual care. *J Consult Clin Psychol* 2013; 81(6):999-1009 PMID:23978169

4. I have also devoted considerable effort in understanding and reducing disparities in physical and mental health disorders associated with migration and acculturation. This includes studies of the association between acculturation and tobacco use in Hispanics living in California, tobacco use and HIV in southeast Asian refugees, and mental and behavioral health problems in Pilipino adolescents.
    a. **Palinkas LA**, Pierce J, Rosbrook B, Pickwell S, Johnson M, Bal D. Smoking beliefs and behavior among Hispanics in California. *Am J Prev Med* 1993; 9: 331-337. PMID:8311982
    b. Cabassa LJ, Hansen MC, **Palinkas LA**, Ell K. *Azúcar y Nervios*: Explanatory models and treatment experiences of Hispanics with diabetes and depression. *Soc Sci Med* 2008; 66:2413-2424. PMID:18339466
    c. Caprio S, Daniels S, Drewnowski A, Kaufman F, **Palinkas L**, Rosenbloom A, Schwimmer J. Influence of race, ethnicity, and culture on childhood obesity: Implications for prevention and treatment: a consensus statement of Shaping America's Health and the Obesity Society. *Diabetes Care* 2008; 31: 2211-2221. PMID:18955718
    d. Javier JR, Coffey DM, **Palinkas LA**, Kipke MD, Miranda J, Schrager S. Promoting enrollment in parenting programs among a Filipino population: A randomized trial. *Pediatrics* 2019; 143(2): e20180553 PMID:30679379

5. Much of my work on implementation of evidence-based prevention programs has occurred in low and middle-income countries, making my research relevant to the field of global health. For instance, I have contributed my expertise in qualitative and mixed methods in conducting studies of substance use and HIV prevention in Mexico. This work has highlighted the importance of understanding local context when implementing evidence-based programs and practices developed in the United States.
    a. **Palinkas LA**, Robertson AM, Syvertsen JL, Hernandez DO, Rangel MG, Martinez G, Strathdee SA. Client perspectives on design and implementation of a couples-based intervention to reduce sexual and drug risk behaviors among female sex workers and their noncommercial partners in Tijuana and Ciudad Juárez, México. *AIDS Behav* 2014, 18:583-594. doi: 10.1007/s10461-014-0715-1. PMID:24510364
    b. **Palinkas LA,** Chavarin CV, Rafful C, Um MY, Mendoza DV, Staines H, Aarons, GA, Patterson TL. Sustainability of evidence-based practices for HIV prevention among female sex workers in Mexico. *PLoS One* 2015; 10(10):e0141508. doi: 10.1371/journal.pone.0141508. eCollection 2015. PMID:26517265
    c. Pitpitian E, Aarons GA, **Palinkas LA**, Chavarin CV, Mendoza DA, Magis C, Staines H, Patterson T. Factors associated with program effectiveness in the implementation of a sexual risk reduction intervention for female sex workers across Mexico: Results from a randomized trial. *PLOS One* 2018; 13(9):e0201954. doi: 10.1371/journal.pone.0201954. PMID:30204761
    d. **Palinkas LA**, Um MY, Aarons GA, Rafful C, Chavarin CV, Mendoza DV, Staines H, Patterson TL. Implementing evidence-based HIV prevention for female sex workers in Mexico: Provider assessments of feasibility and acceptability. *Glob Soc Welfare* 2019, 6(2): 57-68. PMID:31632894

List of Published Work in my Bibliography (from over 285 peer-reviewed publications):
https://www.ncbi.nlm.nih.gov/sites/myncbi/lawrence.palinkas.1/bibliography/47624041/public/?sort=date&direction=ascending

D. Additional Information: Research Support and/or Scholastic Performance

Ongoing Research Support
Northwestern University (Sub), 60044679USC Brown (PI) 9/1/16 –6/30/21
NIH-NIDA P30 DA027828
Center for Prevention Implementation Methods for Drug Abuse and Sex Risk Behavior
To Integrate and extend system science methods to address critical research challenges of federally funded implementation research aimed at preventing drug use and sexual risk behaviors.
Role: Site Principal Investigator

New York University (Sub) Hoagwood (PI) 9/21/19 – 4/30/23
NIH-NIMH P50 MH113662-01A1 Accelerator Strategies for States to Improve System Transformations Affecting Children, Youth and Families
To develop tools to improve the uptake of child mental health (MH) research evidence by state policymakers, and test two practical accelerator strategies to improve D&I of evidence-based practices in state systems.
Role: Site Principal Investigator and Co-PI, Methods Core

University of Maryland Baltimore (Sub), F211896-2 Dubowitz (PI) 4/1/19 – 8/31/23
NIH-NICHD R01 HD092489-01A1.
Dissemination and implementation of the Safe Environment for Every Kid (SEEK) Model for Preventing Child Maltreatment.
To evaluate the impact of independent online (IND) versus in-depth structured (MOC) training and (2) use of SEEK*online* vs. the Traditional approach to guide fidelity of model delivery on adoption and delivery of SEEK in primary care settings, and subsequent prevention of CM.
Role: Site Principal Investigator

[redacted]

No Assigned # Springgate (PI) 7/01/20 – 9/30/20
Louisiana State University (Prime: NAS)
Community Resilience Learning Collaborative and Research Network
To conduct a rapid qualitative assessment of community priorities, strengths, and needs as a result of the COVID-19 pandemic.
Role: Service agreement Principal Investigator

Recently Completed
NIH-NIDA R34 DA 037516-01A1 Palinkas (PI) 4/15/15 – 3/30/19
Measuring Sustainment in Prevention Programs and Initiatives
To design and pilot test a measurement system for monitoring and providing feedback regarding sustainment within four of SAMHSA's prevention-related grant programs.
Role: Principal Investigator

Emory University (Sub), T811267 DiClemente (PI) 7/1/13 – 6/30/18
NIH-NIDA U01 DA036233
KiiDS: Knowing About Intervention Implementation in Detention Sites
To develop implementation strategies that enhance the adoption, fidelity, quality, reach, sustainability and cost-effectiveness of evidence-based substance use and HIV prevention interventions for youth in the juvenile justice system
Role: Site Principal Investigator

The Oregon Social Learning Center (Sub) 004054-00004, Chamberlain (PI) 1/1/14–4/30/17
NIH-NIDA P50A035763
Translational Research on the Prevention of Drug Abuse and HIV Risk
To understand underlying mechanisms and processes associated with exposure to high levels of early life adversity, and specific to risky decision-making in certain social contexts that are common for CWS youth during early adolescence; develop interventions to reduce high rates of drug use and engagement in HIV-risk behaviors in adolescent girls in the CWS; and identify methods for implementing evidence- based interventions into CWS real-world settings
Role: Site Principal Investigator

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Silbert, Lisa C.

eRA COMMONS USER NAME (credential, e.g., agency login): [redacted]

POSITION TITLE: Associate Professor, Department of Neurology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of California, Los Angeles, LA, CA	BS	06/92	Psychobiology
Indiana University School of Medicine, Indpls., IN	MD	06/96	Medicine
Indiana University School of Medicine, Indpls., IN	Internship	06/97	Internship
University of California, Los Angeles, LA, CA	Residency	06/00	Neurology Residency
Oregon Health & Science University, Portland, OR	Fellowship	06/02	Clinical Neurophysiology
Oregon Health & Science University, Portland, OR	Fellowship	06/03	Geriatric Neurology
Oregon Health & Science University, Portland, OR	Master	06/06	Clinical Research

A. Personal Statement

As Co-Associate Director of the Oregon Alzheimer's Disease Research Center (OADRC) and Director of the OADRC Neuroimaging Core, I am very familiar with biomarkers of cognitive function and neurodegenerative disease, and have experience in mentoring multi-level learners, including those in their pre and post-doctoral training. As Principal Investigator of a Paul B. Beeson/NIA K23 Career Development Award on Aging Research study investigating cortical excitability in elderly with subcortical white matter disease, and NIH/NIA R01 awards investigating in vivo and post-mortem magnetic resonance imaging characteristics of Alzheimer's disease and Related Disorders, I have expertise in investigating structural brain changes, and their molecular correlates, on cognitive processing in the elderly. I have been a Neurologist at the OADRC for 20 years, and have unique expertise in the neurological evaluation and cognitive assessment of older individuals with Alzheimer's Disease and Related Dementias (ADRD) In summary, I believe my extensive research and clinical experience in ADRD, has provided me with the expertise to contribute to the proposed F99/K00 application. I am highly enthusiastic to work with Ms. Yu to help familiarize her with clinical and neuroradiological aspects of ADRD research and to provide her with the necessary skills to succeed in her important research on social isolation and cognition and become an independent investigator in her field.

B. Positions and Honors

<u>Positions</u>

2000-2002 Senior Instructor of Neurology, Oregon Health & Science University, Portland, Oregon
May, 2000 Visiting Research Assistant: National Hospital for Neurology and Neurosurgery, Queen Square, London.
2000-present Staff Neurologist, Portland Veteran's Administration Hospital, Portland, Oregon.

2002-2012 Assistant Professor of Neurology, Oregon Health & Science University, Portland, Oregon;
2002-2006 Consulting Neurologist, University of California, San Diego-University of Guam Lytico-Bodig Research Consortium, Mangilao, Guam.
2008-present Consulting Staff, Shriners Hospital for Children, Portland, Oregon.
2010-2017 Director of Neuroimaging, NIA Layton Aging & Alzheimer's Disease Center, OHSU.
2012-2019 Associate Professor of Neurology, Oregon Health & Science University, Portland, Oregon.
2014-2020 Director, Dementia Clinic, Portland Veteran's Affairs Medical Center, Portland, Oregon
2015-present OHSU Site PI, Alzheimer's Disease Neuroimaging Initiative (ADNI)
2015-present UME Scholarly Projects Concentration Lead for Clinical Research, OHSU SOM.
2017-present Director of the Neuroimaging Core, NIA-Layton Oregon Alzheimer's Disease Center, OHSU
2019-present Professor of Neurology, Oregon Health & Science University, Portland, Oregon.
2019-present Co-Associate Director, NIA-Layton Oregon Alzheimer's Disease Center, OHSU

Honors

1987 Psi Chi, the National Honor Society in Psychology. University of California, Los Angeles.
1988 Alpha Lambda Delta; National freshman honors society; Association of College Honor Societies.
2004-2009 Paul B. Beeson/NIH K23 Career Development Award in Aging Research
2004 Clinical Research Training Fellowship – American Academy of Neurology
2011 OHSU Golden Rose Award: Recognizing outstanding service excellence.
2018-present Gibbs Family Endowed Professor for Research in Alzheimer's Disease and Related Disorders
2018 Invited reviewer for NIA Division of Neuroscience (DN) review by the Institute's National Advisory Council on Aging (NACA)
2020-present NIA/VA Workshop to increased Veteran participation in clinical research on Alzheimer's Disease and related Dementia, invited member.

Professional Memberships

1999-present Member, American Academy of Neurology
2001-present Member, American Academy of Neurology Geriatric Neurology section.
2001-present Member, American Academy of Neurology Clinical Neurophysiology section.
2008-present Member, International Society to Advance Alzheimer Research and Treatment
2012-present Member, The International Society of Behavioral and Cognitive Vascular Disorders
2013-present Fellow of the American Academy of Neurology (FAAN)
2018-present International Society for Cerebral Blood Flow & Metabolism

Recent Granting Agency Review Work

2013-2015 Alzheimer's Research UK, Senior Research Fellowship application, Reviewer
2015 Lawrence M. Brass, MD, Stroke Research Fellowship; American Brain Foundation and the American Heart Association/American Stroke Association, Reviewer
2015 NASA Directed Research Proposal Review of submission from HRP Human Health Countermeasures Element, Reviewer
2015 The Netherlands Organisation for Health Research and Development VCI grant, Reviewer
2015 The Netherlands Organisation for Scientific Research (NWO) Innovational Research Incentives Scheme grant application on microvascular function and MRI brain networks, Reviewer
2016 NIH/CSR Clinical Neuroscience and Neurodegeneration (CNN) Study Section, Center for Scientific Review, ad hoc reviewer
2016-2019 National Alzheimer's Coordination Center (NACC) Scientific Review Committee member
2017-2021 NIH/CSR Clinical Neuroscience and Neurodegeneration (CNN) Study Section, Center for Scientific Review, Section Member

C. Contributions to Science

1. While many cross-sectional studies had previously documented detrimental associations between cerebrovascular white matter hyperintensities (WMHs) and cognitive and motor function, I have led research showing the impact of such common radiographic findings as they progress over time. These publications directly assessed the longitudinal effects of age-related WMH progression on cognitive and motor function. They document the impact of WMH accumulation in relation to poorer gait performance

and increased risk of mild cognitive impairment. This body of work established the clinical importance of *accumulating* cerebrovascular disease burden in contributing to age-related cognitive impairment and dementia. By providing evidence of the detrimental effects of WMHs over time, these studies support further research into treatments aimed to halt WMH accretion. This work also showed age-related WMH to be associated with increased cortical excitability in older, dementia-free individuals

a. **Silbert LC**, K Nelson, BA Holman, R Eaton, MS Oken, JS Lou, and JA Kaye. Cortical excitability and age-related volumetric MRI changes. Clinical Neurophysiology. 2006; 117(5):1029-36. PMID: 16564739.
b. **Silbert LC**, C Nelson, DB Howieson, MM Moore, JA Kaye. Impact of white matter hyperintensity volume progression on rate of cognitive and motor decline. Neurology. 2008; 71(2):108-113. PMID: 18606964.
c. **Silbert LC**. Howieson DB, Dodge H, Kaye JA. Cognitive Impairment Risk: White Matter Hyperintensity Progression Matters. Neurology. 2009; 73:113-119. *PMICD: PMC2713187*
d. **Silbert LC**, Dodge HH, Perkins LG, Sherbakov L, Lahna D, Woltjer R, Shinto L, Kaye JA. Trajectory of White Matter Hyperintensity Burden Preceding Mild Cognitive Impairment. Neurology. 2012; 79:741-747. *PMCID: PMC3421153.*

2. This research was the first to use MRI to identify a white matter cerebral blood flow (CBF) penumbra, an approximately 1 cm area of normal appearing white matter surrounding WMH lesions. The WMH CBF penumbra is more extensive than the DTI structural penumbra, and independently associated with WMH expansion over time. This work established the WMH CBF penumbra as a potentially modifiable biomarker in future trials aimed at preventing vascular cognitive impairment in older individuals.

a. N. Promjunyakul, D. Lahna, J. Kaye, H Dodge, D Erten-Lyons, William Rooney, **Silbert LC**. Characterizing the White Matter Hyperintensity Penumbra: a pulsed arterial spin labeling study. Neuroimage: Clinical. 2015; 8:224-229. *PMCID: PMC4473817.*
b. Promjunyakul, D. Lahna, J. Kaye, H Dodge, D Erten-Lyons, William Rooney, **Silbert LC**. Comparisons of cerebral blood flow and structural penumbras in relation to white matter hyperintensities. J Cereb Blood Flow Metab. 2016; *Sept; 36(9):1528-36. PMID:PMC27270266*
c. Nutta-on Promjunyakul, Hiroko H. Dodge, David L. Lahna, Erin Boespflug, Jeffrey A. Kaye, Deniz Erten-Lyons, William D. Rooney, **Silbert LC**. Baseline normal appearing white matter structural integrity and cerebral blood flow predict white matter hyperintensity expansion over time. Neurology. 2018; 90(24): e1-e8.

3. Throughout my research career, I have maintained a strong interest in identifying neuroimaging markers of neurodegenerative disease. Accordingly, I have examined MRI biomarkers of cognitive impairment in older individuals. This body of worked contributed to elucidating neuroimaging biomarkers for detection of early cognitive impairment and neurodegeneration risk. Included in these studies were some of the first to show that greater rates of change in brain volume predicts increased Alzheimer's disease pathology, and that increased WMH progression is associated with both arteriolosclerosis *and* neurofibrillary tangles.

a. **Silbert LC**, Quinn JF., Moore MM., Corbridge E., Ball MJ., Murdoch G., Sexton G., Kaye JA. Changes in Premorbid Brain Volume Predict Alzheimer's disease Pathology. Neurology. 2003; 61(4): 487-492*. PMID: PMC12939422*
b. Deniz Erten-Lyons, Randall Woltjer, Jeffrey A. Kaye, Nora Mattek, Hiroko H. Dodge, Sarah C. Green, Huong Tran, Diane B. Howieson, Katherine Wild, **Silbert LC**. Neuropathologic Basis of White Matter Hyperintensity Accumulation with Advanced Age. Neurology. 2013; 81(11):977-83. *PMCID: PMC3888199*
c. **Silbert LC**, David Lahna, Hiroko Dodge, Nutta-on Promjunyakul, Nora Mattek, Deniz Erten-Lyons, Daniel Austin, Tamara Hayes, Jeffrey Kaye. Less daily computer use is related to smaller hippocampal volumes in cognitively intact elderly. Journal of Alzheimer's Disease. 2016; 52(2):713-7. *PMCID: PMC4866889.*
d. **Silbert LC,** David Lahna, Nutta-On Promjunyakul, Erin Boespflug, Yusuke Ohya, Yasushi Higashiuesato, Junko Nichihira, Takashi Tokashiki, Hiroko Dodge. Risk Factors Associated with Decreased Cortical Thickness and White Matter Hyperintensities in Dementia Free Okinawan Elderly.

Journal of Alzheimer's Disease. 2018; vol. 63, no. 1, pp. 365-372. PMCID: PMC5900560

4. Glial-lymphatic function has been identified as a potential mechanism by which toxic solutes are cleared from the brain, and may play a role in the accumulation of AD pathology. MR-visible perivascular spaces (PVS) have been associated with both Alzheimer's disease and Vascular Cognitive Impairment, and may reflect changes in vascular pulsatility leading to diminished fluid efflux from the perivascular space. This work represents novel methods and subsequent investigations of MR-visible PVS that characterize not just PVS count on one slice (as has been the gold standard), but rather whole brain PVS counts as well as PVS volume and morphological characteristics in young and old individuals.

   a. Boespflug EL, Schwartz DL, Lahna D, Rooney W, Pollock J, Iliff J, Kaye J, **Silbert LC**. MRI-based Mulitimodal Autoidentification of Perivascular Spaces (mMAPS): Automated morphological segmentation of perivascular spaces at clinical field strength. Radiology. 2018;286(2):632-642. PMID:28853674.
   b. Ryan A. Opel, Alison Christy, MD, PhD, Erin L. Boespflug, PhD, Brendan Case, MD, Jeffery Pollock, MD, **Lisa C. Silbert, MD, MCR,** Miranda M. Lim, MD, PhD. The effects of traumatic brain injury on sleep and the enlargement of perivascular spaces. Cerebral Blood Flow and Metabolism, 2018. PMID: 30092696
   c. Daniel L. Schwartz, Erin L. Boespflug, David L. Lahna, Jeffrey Pollock, Natalie E. Roese, **Lisa C. Silbert**. Autoidentification of Perivascular Spaces using Clinical Field Strength T1 and FLAIR MR Imaging. Neuroimage, Aug 2019, 202(15). PMID: 31461676, PMCID: PMC6819269
   d. Juan Piantino, Daniel Schwartz, Madison Luther, Angelica M. Morales, Amber Lin, Jeffrey Kaye, David Lahna, Ryan van Fossen**, Lisa C. Silbert**, Bonnie Nagel, Erin L Boespflug. Characterization of MRI-visible perivascular spaces in white matter of healthy adolescents at 3 Tesla. AJNR 2020, *in press.*

5. I have collaborated with others to investigate novel and potentially modifiable risk factors of MRI WMHs. These publications document lipid mediators of vascular cognitive impairment, and provide evidence in support of new treatment modalities to prevent white matter degeneration. Results from several of these studies resulted in an NIH-funded clinical trial investigating treatment aimed at decreasing WMH accumulation over time in an older population.

   a. Bowman GL, Dodge HH, Mattek N, Barbey AK, **Silbert LC**, Shinto L, Howieson DB, Kaye JA and Quinn JF. Plasma omega-3 PUFA and white matter mediated executive decline in older adults. Front. Aging Neurosci. 2013; 5:92. *PMCID: PMC3863786.*
   b. Jonathan W Nelson, PhD; Jennifer M Young, BS; Rohan N Borkar, BS; Randy L Woltjer,MD, PhD; Joseph F Quinn, MD; **Lisa C Silbert**, MD; Marjorie R Grafe, MD, PhD; Nabil J Alkayed, MD, PhD. Role of Soluble Epoxide Hydrolase in Age-Related Vascular Cognitive Decline. Prostaglandins & Other Lipid Mediators. 2014; 113-114; 30-37. *PMCID: PMC4254026.*
   c. Gene L. Bowman, **Lisa C. Silbert**, Hiroko H. Dodge, David Lahna, Kirsten Hagen, Charles F. Murchison, Diane Howieson, Jeffrey Kaye, Joseph F. Quinn, and Lynne Shinto. Randomized trial of n-3 polyunsaturated fatty acids for the prevention of cerebral small vessel disease and inflammation in aging (PUFA trial): Rationale, design and baseline results. Nutrients. *In Press*, 2019.
   d. Lynne Shinto ND, David Lahna BA, Hiroko Dodge PhD, Kirsten Hagen, Dennis Koop PhD Joseph Quinn, MD, Jeffrey Kaye MD, **Lisa Silbert MD**. Oxidized Products of Omega-6 and Omega-3 Long Chain Fatty Acids (PUFA) are Associated with White Matter Hyperintensities and Poorer Executive Function Performance in a Cohort of Cognitively Normal Hypertensive Older Adults. Journal of Alzheimer's Disease. Dec 2019, 74(1):65-77. PMID: 32176647.

Complete List of Published Work in My Bibliography:
https://www.ncbi.nlm.nih.gov/myncbi/collections/bibliography/40325675/

D. Additional Information: Research Support and/or Scholastic Performance
Ongoing Research Support

Department of Veterans Affairs, HSR&D COVID-19 Supplement (PI: Silbert) 06/01/20 – 02/28/21

"Impact of Social Distancing During the COVID-19 Pandemic on Cognitive, Physical, and Mental Health in Urban and Rural Veterans in the Pacific Northwest". The primary aims of this study are: 1) To examine whether there are changes in weekly reported social activities during the COVID-19 pandemic social distancing mandate and whether changes are associated with reports of greater physical and mental illness; 2) To determine which factors (i.e. rurality, marriage) work as buffers between social isolation and its negative effect of physical and psychological state; and 3) To determine which home-based digital biomarkers (i.e. sleep duration, daily step count) are associated with physical and psychological distress.

NIH/NIA U2C AG054397 (CART PI:Kaye; Silbert: VA-PI) 09/30/16 – 02/28/21
ORCATECH Collaborative Aging (in Place) Research Using Technology (CART)
The ORCATECH CART program develops and validates an infrastructure for rapid and effective conduct of research utilizing technology to facilitate aging in place (AiP).

NIH/NIA R01 AG056712-01 (PI: Silbert/Woltjer) 04/01/18 – 01/31/23
"*White Matter Hyperintensity-associated Astrocytopathy in Alzheimer's disease and Vascular Cognitive Impairment: A Targeted Histopathologic Study using Postmortem 7T MRI"*
The primary aims of this study are designed to identify which pathologic processes are most relevant to cognitive impairment in VCID, how WMHs function as biomarkers of impairment. Role: MPI; Silbert: Imaging/Vascular Dementia, Woltjer: Neuropathology

P30AG066518 Kaye (PI) 07/01/20-02/28/25
P30AG008017 Kaye (PI) 10/01/89-03/31/21
Oregon Alzheimer's Disease Research Center
The goal is to facilitate AD research by providing core resources for clinical and basic research, fostering interdisciplinary collaboration, sharing resources and expertise, mentoring new professionals, raising awareness and providing education. Cores: Administrative, Clinical, Data, Biomarkers, Neuropathology, Outreach Recruitment & Education, Neuroimaging, and Digital Technology; Research Education Component. Role: Co-Investigator; Clinical Core Neurologist (2002-present); Neuroimaging Core Lead since (2017-present); Co-Associate Director (2020).

NIH/NIA R01AG051628 (PI: Dodge) 07/01/16-06/30/21
Conversational Engagement as a Means to Delay Alzheimer's Disease Onset: Phase II
The goal of this grant is to examine conversational engagement as a means to improve cognitive functions among subjects aged 80 and older with Mild Cognitive Impairment and limited social interactions. Role: Co-Investigator: Co-I (see below description)

NIH/NIA R01AG056102 (PI: Dodge) 07/01/16/-06/30/21
Web-enabled social interaction to delay cognitive decline among seniors with MCI: Phase I
The goal of this project is to examine conversational engagement as a means to improve cognitive functions amount subjects aged 80 and older with Mild Cognitive Impairment and limited opportunities for social interactions. Role: Co-Investigator: Responsible for MRI protocol design, acquisition and MRI analysis

NIH/NIA U0179634019 (PI:Weiner) 09/15/2016 - 09/31/2021
Alzheimer's Disease Neuroimaging Initiative: The purpose of this observational study is to determine the relationships among clinical, cognitive, imaging, genetic, and biomarker characteristics of the entire spectrum of Alzheimer's disease as it progresses from a preclinical stage to very mild symptoms to mild cognitive impairment (MCI) to dementia. Participants sought include those with normal cognition, MCI, or mild Alzheimer's dementia. Role: Site PI

Completed Research Support

NIH/NIA R01AG043398 (PI:Bowman/Shinto) 09/01/2013 – 05/31/2020
Omega 3 PUFA for the vascular component of age–related cognitive decline
A phase II randomized and double masked, placebo controlled trial to examine the effects of a supplement on brain structure and function in non-demented older adults at high risk for dementia over 3 years
Role: Co- Investigator; Responsible for MRI protocol design, acquisition and analysis

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Kaye, Jeffrey A.

eRA COMMONS USER NAME (credential, e.g., agency login): [redacted]

POSITION TITLE: Professor of Neurology and Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Amherst College, Amherst, MA	BA	06/1976	Independent Scholar
New York Medical College, Valhalla, NY	MD	06/1980	Medicine
Hartford Hospital, Hartford, CT		1980-81	Internship
Boston University, Boston, MA		1981-84	Residency in Neurology

A. Personal Statement

I have been fortunate to be deeply involved in brain aging research for over 25 years. During this time, my research has focused on the question of why some individuals remain protected from functional decline and dementia with advancing age while others succumb at much earlier times. This work has relied on a number of approaches ranging across the fields of genetics, neuroimaging, physiology and continuous life activity monitoring. It has involved leading several major longitudinal studies on aging and clinical trials including the *Oregon Brain Aging Study*, the *Intelligent Systems for Detection of Aging Changes* (ISAAC), the *Ambient Independence Measures for Guiding Care Transitions,* and *the Collaborative Aging Research using Technology (CART)* studies. All this work has required and flourished in an environment of highly interactive team science, training and collaborative research found at the Oregon Center for Aging & Technology (ORCATECH) and the NIA-Layton Aging & Alzheimer's Disease Research Center which I direct. I find myself truly excited to facilitate and support this F99/K00 proposal focused on ultimately understanding socially isolated older adults at high risk of cognitive decline and the onset of Alzheimer's disease and related dementias. I look forward to my role as a member of the terrific mentor team assembled, especially providing my knowledge, experience and Center resources regarding high-frequency digital biomarker methodologies and data collection applied to objective assessment of environmental factors, social isolation, and cognitive health.

B. Positions and Honors

<u>Positions and Employment</u>

1983-84 Chief Resident in Neurology, Boston University Affiliated Hospitals, Boston, Massachusetts
1984-85 Fellow in Movement Disorders and Neuropharmacology, Boston University School of Medicine, Boston, Massachusetts
1984-85 Instructor in Neurology, Boston University School of Medicine, Boston, Massachusetts
1985-88 Medical Staff Fellow, Laboratory of Neurosciences, National Institute on Aging, National Institutes of Health, Bethesda, Maryland
1988-93 Assistant Professor of Neurology, Oregon Health & Science University, Portland, Oregon
1988- Director, Geriatric Neurology, Oregon Health & Science University and VA Medical Center, Portland, Oregon
1993-98 Associate Professor of Neurology, Oregon Health & Science University, Portland, Oregon

1996-	Director, Layton Aging and Alzheimer's Disease Center, Oregon Health & Science University, Portland, Oregon
1998-	Professor of Neurology, Oregon Health & Science University, Portland, Oregon
2003-	Professor of Biomedical Engineering, Oregon Health & Science University, Portland, Oregon
2004-	Director, Oregon Roybal Center for Translational Research on Aging
2005-	Director, Oregon Center for Aging and Technology (ORCATECH)

Other Experience and Professional Memberships

1999-03	Executive Committee, National Alzheimer's Centers Directors (Chair, 2003)
1999-03	Executive Committee, National Alzheimer's Coordinating Center
2000	Advisory Panel on Exceptional Longevity, National Institute on Aging
2001	Chair, Medical Research Advisory Group, Geriatrics, Department of Veteran's Affairs
2001	Co-chair, National Institute on Aging Panel on Characterization of Participants in Studies of Exceptional Survival in Humans
2002	Dementia Safety Workgroup, U.S. Dept. of Veterans Affairs, Office of the Inspector General
2005-07	Neuroscience of Aging Review Committee, NIA-N, National Institute on Aging
2006-12	Chair, Work Group on Technology, National Alzheimer's Association
2007-08	Co-Chair, Work Group, Guidelines for Clinical Care for Dementia, U.S. Dept. of Veterans Affairs
2007-	Commissioner, Center for Aging Services and Technologies
2011-14	Leadership Council, Network on Environments, Services and Technologies, American Society on Aging
2011-14	Vice-Chair, International Society to Advance Alzheimer's Research and Treatment
2014-	Chair, International Society to Advance Alzheimer's Research and Treatment

Honors

2001	Charles Dolan Hatfield Award for Alzheimer's Research
2003-	Best Doctors in America
2004-	Fellow, American Academy of Neurology
2011	Redifining Aging in Oregon Award
2011-	Layton Endowed Professor of Neurology
2012	Business Partnership Achievement Award, Oregon Health & Science University

C. Contributions to Science

It is critical that I first emphasize that all the major contributions I have successfully achieved have come about through team science and work with exceptional colleagues. Thus, although the NIH biosketch format requires 'your' major contributions to science, I must say that most of my achievements are truly shared - the result of long and productive collaborations with gifted scientists, assistants and not to be underestimated in importance, the research volunteers themselves. In fact I would say that I am particularly proud of the contributions to science I have made through working collaboratively with others, many of whom are noted coauthors in the publications that I cite below. With this in mind I outline those contributions which I consider most significant and for which I was the principal investigator or thought leader.

1. Brain Aging – Risks and Trajectory of Dementia in the Oldest Old. In 1989 I established the Oregon Brain Aging Study (OBAS) to address the question of why some people retained excellent function especially in cognition and mobility, two major domains that when they fail, threaten independence. Thus the OBAS cohort was established with entry criteria of individuals with no cognitive or medical illnesses at advanced ages (mean age = 89), a 'rare disease' in the oldest old. Subsequent enrollment waves and sister cohorts included comparison groups of more average healthy elders. These unique aging cohorts comprising over 900 subjects have yielded a rich fund of knowledge over the past 25 years. The most important insights that I have contributed have emanated from studies with these cohorts. These include the realization that even enjoying pristine health into advanced ages, is not protective of surviving cognitively intact; risk continues, the dementia-free survival curve is 'simply' shifted to the right to yet even later ages. Subsequent work described for the first time the more specific trajectory of change in cognitive decline revealing that major cognitive domains (episodic memory, fluency, visual-perceptive function) declined predictably, but with a more rapid acceleration of cognitive change *3-4 years prior* to the diagnosis of MCI. At the same time we also for the first time showed similar patterns of decline in motor function, first describing that there were motor changes predicting subsequent cognitive decline and then that there was a similar acceleration of motor decline measured as walking speed *12 years prior* to the diagnosis of MCI.

This work has influenced the field to begin to realize that there are clinical and behavioral or functional biomarkers that can be utilized to presymptomatically predict those most likely to develop MCI or dementia. These observations point to important opportunities for dementia prevention trials providing a basis for identifying sensitive behavioral biomarkers (see (**4**) below) to gauge the effect of new prevention treatments.

a. Gonzales McNeal M, Zareparsi S, Camicioli, R., Dame A, Howieson D, Quinn J, Ball M, Kaye J, Payami H. Predictors of Healthy Brain Aging. Journal of Gerontology: Biological Sciences, 56A:7, B294-B301, 2001
b. Marquis S, Moore M, Howieson D, Sexton G, Payami H, Kaye J, Camicioli R. Independent predictors of cognitive decline in healthy elderly persons. Archives of Neurology, 59:601-606, 2002
c. Camicioli RM., Howieson D., Oken B., Sexton G., Kaye JA. Motor slowing precedes cognitive impairment in the oldest old. Neurology, 50:1496-1498, 1998.
d. Buracchio T, Dodge H, Howieson D, Wasserman D, Kaye J. The trajectory of gait speed preceding MCI. Archives of Neurology, 67 (8): 980-986, 2010. PMC2921227

2. Brain Aging – Biomarkers of Brain Aging and Cognitive Decline. In parallel to the fundamental identification of the earliest clinical changes associated with healthy brain aging, and their trajectory of change other markers and techniques in the Oregon Aging Studies were deployed to better understand the biology underlying the clinical changes including post-mortem brain tissue histopathology, plasma biomarkers, genetics and imaging. Early studies were among the first to show that the non-demented OBAS oldest old population was enriched in apolipoprotein E2 suggesting a protective (against dementia) effect for the latter allele. The collection of DNA in OBAS cohorts proved particularly valuable as in recent years these well-characterized subjects (including post-mortem tissues) have been included in many of the major GWAS and genome sequencing efforts to identify risk genes for Alzheimer's disease. Other ground-breaking work has employed OBAS dietary studies and plasma samples along with neuroimaging biomarkers (further described below) to create novel nutritional biomarker panels associated with specific neuroimaging signatures of brain pathology (vascular disease and neurodegeneration). We have been among the few groups to utilize longitudinal clinical data combined with post-mortem data to identify presymptomatic rates of change in clinical (and neuroimaging) markers to show that the neuropathology accompanying healthy aging supports a continuum in which AD pathology is infrequent in the healthy, cognitively stable oldest old. The minimal abnormalities in cognitively stable individuals are consistent with the notion that preclinical pathologic AD *precedes* obvious cognitive impairment. Longitudinal cognitive testing showed an increased burden of neuropathologic changes in those who have cognitive decline but are not functionally impaired and do not meet criteria for the diagnosis of dementia. The strong relationship between cumulative pathologic changes and rate of cognitive decline has suggested that these lesions may have *clinical consequences at any age and are not likely to be benign senescent changes*. Further, these studies have shown that the classic pathologies of AD (especially neurofibrillary or 'tau' tangles), although contributing to the rate of cognitive decline are not the majority drivers. Vascular disease in these studies is demonstrated to be increasingly important in this equation. There are other yet to be uncovered pathologies (estimated as at least two-thirds of unexplained variance) that explain the cognitive decline and neurodegeneration of the aging.

a. Green S, Kaye J, Ball M. The Oregon Brain Aging Study: Neuropathology accompanying healthy aging in the oldest old. Neurology, 54:105-113, 2000.
b. Silbert LC., Quinn JF., Moore MM., Corbridge E., Ball MJ., Murdoch, G., Sexton, G., Kaye, JA. Changes in Premorbid Brain Volume Predict Alzheimer's disease Pathology. Neurology. 61(4):487-492, 2003.
c. Erten-Lyons D, Dodge H., Woltjer R, Silbert, L, Howieson D, Kramer P, Kaye J. Neuropathological Basis of Age-Associated Brain Atrophy. JAMA Neurology. May 1;70(5):616-22, 2013. PMC3898525
d. Erten-Lyons D, Woltjer R, Kaye J, Mattek N, Dodge H, Green S, Tran H, Howieson D, Wild K, Silbert L. Neuropathologic Basis of White Matter Hyperintensity Accumulation with Advanced Age. Neurology. 81:1-7, 2013. PMC3888199

3. Brain Aging – Neuroimaging Biomarkers of Brain Aging and Cognitive Decline. Neuroimaging biomarkers have from the beginning been integral in my research in the study of brain aging and identifying the risk and drivers of cognitive and functional decline. We were the first to show that volume loss in the hippocampus predicted those healthy elderly who were destined to develop dementia. We subsequently went on to show that rates of volume loss differed across regions of the brain and during specific phases of dementia from presymptomatic to clinically manifest AD. We were the first to show that the rates of brain loss were not linear, but exhibited regionally and pathologically differentiated accelerated degeneration with atrophy and CSF space expansion *accelerating 3-4 years prior* to MCI while white matter hyperintensity volume (indexing small vessel disease) accelerated *over a decade before onset* of MCI. This work corroborated our original observation that in unaffected elderly, these white matter changes indicative of vascular disease were the earliest and independent predictors of later cognitive decline and dementia. These observations were critical to the launching at our Center, the first phase II trial geared to prevention of cognitive decline based on vascular disease with the primary outcome biomarker being white matter hyperintensity volume.

a. Kaye JA, Swihart T, Howieson D, Dame A, Moore MM, Karnos T, Camicioli R, Ball M, Oken B, Sexton G. Volume loss of the hippocampus and temporal lobe in healthy elderly destined to develop dementia. Neurology, 48:1297-1134, 1997.
b. Kaye J, Moore M, Dame A, Quinn J, Camicioli R, Howieson D, Corbridge E, Care B, Nesbit G, Sexton G. Asynchronous regional brain volume losses in presymptomatic to moderate AD. Journal of Alzheimer's Disease, 8:51-56, 2005.
c. Adak S, Illouz K, Gorman W, Tandon R, Zimmerman E, Guariglia R, Moore M, Kaye J. Predicting the rate of cognitive decline in aging and early Alzheimer disease. Neurology , 63 (1):108-114, 2004.
d. Silbert LC, Dodge HH, Perkins LG, Sherbakov L, Lahna D, Erten-Lyons D, Woltjer R, Shinto L, Kaye JA. Trajectory of White Matter Hyperintensity Burden Preceding Mild Cognitive Impairment. Neurology, 79(8):741-747, 2012. PMC3421153

4. Pervasive Computing and Sensing for Objective Continuous Assessment and interventions. Years of conventional longitudinal follow-up in our cohort studies led to the realization that the in clinic or brief home based assessment methodologies employed to gauge change over time are critically limited by the brief annual assessments, self-reports and isolated biomarker determinations commonly employed. This latter approach inherently cannot provide the continuous time series data and trajectories of change necessary to pinpoint critical transition periods and their causes. Further, prevailing assessment methodology lacks ecological validity. This led me to collaboratively develop alternative methods of assessment with the basic principle being to bring assessment into the home using a variety of unobtrusive pervasive computing and sensing technologies. Thus we successfully created a unique in-home pervasive computing platform providing continuous monitoring of behavior and activity in everyday life. We have established a Life Laboratory deploying this platform in several hundred volunteer's homes and collected continuous activity and sensor data for now up to 8 years forming the largest known database of continuous annotated human activity and health data. This approach has allowed us to radically change the clinical assessment paradigm and afforded new understanding of clinical changes beginning decades prior to when MCI is diagnosed. Using the Life Lab and related cohorts combining the annual measures of the conventional clinical assessment with more continuous measures and on-line report, we have begun to demonstrate the potential capacity of this novel approach to transform our assessment paradigm. Thus, motor activity and gait speed, sleep patterns, medication taking behavior, socialization and computer use can be entirely passively and remotely assessed continuously for years at a time. Using this approach trajectories and patterns of motor activity, sleep and computer use are shown to distinguish cognitively normal from MCI cases *without formal psychometric tests*. Most importantly, the continuous data provide greater precision of trajectory estimates and because these changes are sensitive to change with MCI, they allow for drastically reduced sample sizes or time of follow-up needed to identify clinical trial end points. Many additional related automated assessment approaches are extremely promising such as analyzing digitized data (audio or video files) captured routinely from clinical or online encounters that increase the sensitivity of testing while allowing one to relate back to the legacy method's summary scores.

a. Kaye JA, Maxwell SA, Mattek N, Hayes TL, Dodge H, Pavel M, Jimison H, Wild K, Boise L, Zitzelberger TA. Intelligent Systems for Assessing Aging Changes: Home-Based, Unobtrusive and Continuous Assessment of Aging. Journal of Gerontology: Psychological Sciences, 66 (Suppl. 1): i180-i190, 2011. PMC3132763
b. Dodge HH, Mattek NC, Austin D, Hayes TL, Kaye JA. In-home walking speeds and variability trajectories associated with Mild Cognitive Impairment. Neurology 78:1946-1952, 2012. PMC3369505
c. Hayes TL, Riley T, Mattek N, Pavel M, Kaye JA. Sleep Habits in Mild Cognitive Impairment. Alzheimer Disease & Associated Disorders. Oct 17. [Epub ahead of print], 2013. PMC3990656
d. Kaye J, Mattek N, Dodge H, Campbell I, Hayes T, Austin D, Hatt W, Wild K; Jimison H. Unobtrusive measurement of daily computer use to detect mild cognitive impairment. Alzheimer's & Dementia, 10(1):10-17, 2014. PMC3872486

A list of my peer-reviewed publications is available in PubMed at:
http://www.ncbi.nlm.nih.gov/pubmed/?term=jeffrey+kaye

D. Additional Information: Research Support
Ongoing Research Support

[redacted]

P30 AG024978 Kaye (PI) 09/04 - 07/24
Oregon Roybal Center for Care Support Translational Research Advantaged by Integrating Technology The goals are to: 1) develop basic social, behavioral and biological knowledge about independent aging using technology/engineering; 2) Establish scalable methodologies for technology-based health monitoring and independent support, and; 3) Accelerate development and translation of new work in the field.
Role: PI

P30 AG008017 Kaye (PI) 07/90 - 03/25
Oregon Alzheimer Disease Research Center
The goal is to facilitate AD research by providing the core resources for clinical and basic research. Six cores (Administrative, Data, Clinical, Genetic, Neuropathology and Education) provide subjects, standardized data, and tissue and biological samples for use in a wide range of collaborations.
Role: PI

R01AG051628 Dodge (PI) 07/16 - 06/21
Conversational Engagement as a Means to Delay Alzheimer's Disease Onset: Phase II
The goal is to examine conversational engagement as a means to improve cognitive functions among subjects aged 80 and older with Mild Cognitive Impairment and limited opportunities of social interactions.
Role: Co-investigator

R01AG056102 Dodge (PI) 09/16/ - 08/22
NIH/NIA
Web-enabled social interaction to delay cognitive decline among seniors with MCI: Phase I
The goal is to examine conversational engagement as a means to improve cognitive functions amount subjects aged 80 and older with Mild Cognitive Impairment and limited opportunities for social interactions.
Role: Co-investigator

U01 AG10483 Feldman (PI) 09/91 - 06/21
Alzheimer's Disease Cooperative Study (UCSD subcontract)
The goals are to develop trials for promising agents designed to ameliorate behavioral symptoms, improve cognition, slow the rate of decline, or delay the appearance of AD.
Role: Co-investigator

U01 AG016976 Kukull (PI) 07/99 - 06/21
National Alzheimer's Coordinating Center (University of Washington subcontract)
The major goal of this project is to collect clinical and neuropathological data from all NIH Alzheimer's Centers.
Role: Co-investigator

Completed Research Support

[redacted]

R01 AG042191 Kaye (PI) 07/13 - 06/18
Ambient Independence Measures for Guiding Care Transitions
The goals are to develop systems that improve our ability to unobtrusively monitor important health changes due to chronic disease and aging and evaluate how such technologies may lead to improved care transitions.
Role: PI

P01 AG043362 Hofer/Kaye/Puccinin/Kuh (Co-PIs) 07/13 - 06/18
Integrative Analysis of Longitudinal Studies of Aging
The goal is to develop, maintain, and make use of a system for coordinated analysis of over 40 international long-term studies for advancing knowledge about aging-related change.
Role: Co-PI

PHS Fellowship Supplemental Form

OMB Number: 0925-0001
Expiration Date: 02/28/2023

Introduction

1. Introduction to Application (for Resubmission applications)	03__INTRODUCTION1019212467.pdf

Fellowship Applicant Section

2. Applicant's Background and Goals for Fellowship Training*	13__BACKGROUND_AND_GOALS1019212468.pdf

Research Training Plan Section

3. Specific Aims*	14_SPECIFIC_AIMS1019212456.pdf
4. Research Strategy*	15_RESEARCH_STRATEGY1019212457.pdf
5. Respective Contributions*	16__RESPECTIVE_CONTRIBUTIONS1019212458.pdf
6. Selection of Sponsor and Institution*	17__SELECTION_OF_SPONSOR___INST1019212459.pdf
7. Progress Report Publication List (for Renewal applications)	
8. Training in the Responsible Conduct of Research*	18__TRAINING_IN_RCR1019212460.pdf

Sponsor(s), Collaborator(s) and Consultant(s) Section

9. Sponsor and Co-Sponsor Statements	19_SPONSOR_AND_CO_SPONSOR_STMT1019212462.pdf
10. Letters of Support from Collaborators, Contributors and Consultants	Letters_of_Support1019212465.pdf

Institutional Environment and Commitment to Training Section

11. Description of Institutional Environment and Commitment to Training	Description_of_Insititutional_E_CT1019212491.pdf
12. Description of Candidate's Contribution to Program Goals	

Other Research Training Plan Section

Vertebrate Animals

The following item is taken from the Research & Related Other Project Information form and repeated here for your reference. Any change to this item must be made on the Research & Related Other Project Information form.

Are Vertebrate Animals Used? ☐ Yes ☑ No

13. Are vertebrate animals euthanized?
    If "Yes" to euthanasia
    Is method consistent with American Veterinary Medical Association (AVMA) guidelines?
    If "No" to AVMA guidelines, describe method and provide scientific justification

14. Vertebrate Animals

PHS Fellowship Supplemental Form

Other Research Training Plan Information

15. Select Agent Research

16. Resource Sharing Plan

17. Authentication of Key Biological and/or Chemical Resources — BiologicalChemicalResources1019212472.pdf

Additional Information Section

18. Human Embryonic Stem Cells

Does the proposed project involve human embryonic stem cells?* ☐ Yes ☑ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s), using the registry information provided within the agency instructions. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s):

19. Alternate Phone Number: [redacted]

20. Degree Sought During Proposed Award:

Degree:	If "other", indicate degree type:	Expected Completion Date (MM/YYYY):
PHD: Doctor of Philosophy		05/2021

21. Field of Training for Current Proposal*: 980 Social Work

22. Current or Prior Kirschstein-NRSA Support?* ☐ Yes ☑ No

If yes, identify current and prior Kirschstein-NRSA support below:

Level*	Type*	Start Date (if known)	End Date (if known)	Grant Number (if known)

23. Applications for Concurrent Support?* ☐ [redacted] [redacted]

If yes, describe in an attached file:

24. Citizenship*

U.S. Citizen U.S. Citizen or Non-Citizen National? ☐ [redacted] [redacted]

Non-U.S. Citizen ☐ [redacted]

[redacted]

If you are a non-U.S. citizen with a temporary visa applying for an award that requires permanent residency status, and expect to be granted a permanent resident visa by the start date of the award, check here: ☐

25. ☐ Change of Sponsoring Institution — Name of Former Institution:*

PHS Fellowship Supplemental Form

Budget Section

All Fellowship Applicants:

26. Tuition and Fees*:

☐ None Requested ☑ Funds Requested

Year 1	$16,000.00
Year 2	$16,000.00
Year 3	$16,000.00
Year 4	$16,000.00
Year 5	$16,000.00
Year 6 (when applicable)	
Total Funds Requested:	$80,000.00

Senior Fellowship Applicants Only:

27. Present Institutional Base Salary: Amount | Academic Period | Number of Months

28. Stipends/Salary During First Year of Proposed Fellowship:

a. Federal Stipend Requested: Amount | Number of Months

b. Supplementation from Other Sources: Amount | Number of Months

Type (e.g.,sabbatical leave, salary)

Source

Appendix

29. Appendix

INTRODUCTION

I appreciate the opportunity to strengthen the proposal and resubmit my F99/K00 application (1 F99 AG068492-01), which aimed to examine environmental factors related to social isolation and cognitive decline among older adults and develop a technology-facilitated behavioral health intervention. I sincerely thank the reviewers for providing constructive critiques. As a result of addressing the issues raised, I believe the proposal has significantly improved. The feedback received was synthesized into the following priority areas, and the steps taken to address each concern are outlined.

Sponsor Team Expertise. To address the concern on the absence of neurologists on the team, Drs. Lisa Silbert and Jeffrey Kaye from Oregon Health & Science University (OHSU) were invited and agreed to serve as co-sponsors. Dr. Lisa Silbert is a neuroimaging specialist and leads the MRI research core at Oregon Alzheimer's' Disease Research Center (OADC). Dr. Kaye specializes in employing technology to promote positive aging. A sponsor team coordination plan is described in the Sponsor & Co-Sponsor Statement.

Enriched the Training Plan. To address the concern that the training plan was underdeveloped, more training activities are included, and the training goals are further clarified (see Table 1. Training Activities and Timeline). I will work with Dr. Kaye to study the association between older adults' social engagement and their cognitive health using data collected with wearable devices and ambient technology. Co-sponsor Dr. Dodge will serve as the primary mentor in the K stage. I will move to Portland and join her research team as a postdoc fellow at OHSU. We agreed to have biweekly online meetings in the F99 stage and weekly in-person meetings in the K00 stage. I have identified more specific environmental and institutional resources that serve the proposal's aims, as documented in the tailored Facilities & Other Resources and revised Selection of Sponsor and Institution.

Neuroimaging and Statistical Methods. To address the concern that the neuroimaging and statistical training planned were inadequate, Dr. Silbert will provide mentored training to help me develop knowledge and statistical skills in employing and interpreting neuroimaging outcomes in intervention research. I will familiarize myself with examining neuroimaging outcomes in the F99 stage by working with collaborator Dr. Judy Pa (see letter of support) and drafting publishable manuscripts with her study datasets. I will further develop my capacity in advanced statistics collaborating with Dr. Dodge, an expert in biostatistics for intervention study and cognitive health research. To respond to the comment that the self-directed learning of R programming might have its limitations, I have added a formal course in R programming at USC to have a more engaged learning experience.

Clinical Experience. The reviewers raised the issue of lacking training in the clinical world of Alzheimer's Disease and Related Dementias (ADRD). In this resubmitted proposal, I propose to complete clinical internships in neurology clinics, mentored by collaborator Dr. John Ringman (see letter of support) in the F99 stage at USC and by co-sponsor Dr. Kaye in the K00 stage at OHSU. These clinical experiences will help me formulate practically informed research questions and develop a intervention program with significant clinical implications.

Grant Writing. The reviewers commented that teaching and mentoring should be less prioritized in my current career stage. In the resubmitted proposal, more time is planned for grant writing and scholarship development. I propose to develop an intervention program and to prepare a pilot study designed to test the feasibility and estimate the intervention effect size. I will seek both pilot funding at OHSU and extramural funding to conduct the pilot study. The pilot study will lead to future independent research awards (e.g., R21, R01) for testing the effectiveness of the intervention program.

Clinical Trial Methods. The reviewers raised the need to anticipate and make plans for the challenges in clinical research. I propose in training goal 4 to receive more specific training in clinical trial interventions. The sponsor team has extensive experience in clinical trial research. I will convene quarterly expert panel meetings with all sponsors to identify potential challenges and discuss intervention implementation strategies.

Productivity. As reflected in my updated bio-sketch, I have published three first-authored manuscripts and currently have multiple manuscripts under review since the last submission, which demonstrates my productivity in research and addresses reviewers' concerns in training potentials. I plan to maintain and further cultivate the momentum of my research by setting milestones for training goals. For example, I propose to conduct a systematic review of papers studying the underlying mechanisms of the association between social isolation and cognitive decline. I will lead the planning, writing, and publishing manuscripts while working with sponsors and collaborators. The proposed dissertation research will produce at least three publishable manuscripts.

Integration of the Research and Training Plans. I further developed the research and training plans so that the two essential components are closely related. I now propose specific research activities using data from Drs. Dodge and Kaye's existing projects that will enable me to utilize the skills developed in the training plan. I provide justifications for proposing to use a mixed-methods approach in the Research Strategy section. The mixed-methods approach is uniquely advantaged for the design and refinement of a behavioral health intervention.

BACKGROUND AND GOALS FOR FELLOWSHIP TRAINING

A. Doctoral Dissertation and Research Experience

My research interests in understanding the aging process and improving older adults' quality of life were first developed in the three-year research assistant experience in the lab of aging and cognition at Zhejiang University. I was awarded the Zhejiang Province Undergraduate Innovation and Entrepreneurship Fellowship to test the reliability and feasibility of using an internet administered 9-item Patient Health Questionnaire (PHQ-9) as a depression screening tool. We developed a mobile application, integrating the function of self-diagnosis, mood monitoring, and online intervention materials. I was drawn to explore the potential of employing information and communication technology (ICT) through the first experience of using mobile technology to deliver mental health services. This research experience produced a paper published in *Asia-Pacific Psychiatry*.[33]

As a first-generation college student at Zhejiang University, I seized as many opportunities as I could to receive research training. I found gratification in building knowledge and was fascinated by the possibilities of making positive changes through research. I visited the University of Surabaya to collaborate with investigators in Indonesia to conduct cross-culture research. I worked with Dr. Christ Mayhorn at North Carolina State University on the topic of evaluating the social engineering of phishing attacks. I was selected as one of the three undergraduates to work as a research intern at the University of Alberta School of Nursing. During the three-month internship, I participated in a systematic review of the barriers and facilitators of oral health care for nursing home residents, which was published in the *International Journal of Nursing Studies*.[34] **My experiences at multiple cities in China, Indonesia, Canada, and the United States exposed me to the rich cultures in various regions of the world and inspired me to work with populations with diverse backgrounds.** By working on aging-related research topics, I realized the vast unmet needs in the older adult population globally and decided to commit my research to this vulnerable group with complex challenges.

By the end of my undergraduate training, I had a strong aspiration of becoming an investigator in applied science. Therefore, I applied for the MSW/PhD dual degree program at the University of Southern California Suzanne Dworak-Peck School of Social Work (USC SSW). In the first two years of the MSW/PhD dual degree program, I took both MSW and PhD level courses and completed social work clinical internship requirements. I interned at ABC Adult Day Health Care Center for 20+ hours per week. In my clinical practice, I was captivated by the gap between the prevalent experience of loneliness and social isolation and older adults' strong wishes in getting connected with family and friends. Social isolation and loneliness have become the primary outcomes in my research since then. Many older adults came to consult me on how to use their mobile phones to contact friends and share meaningful memories online with their families. **My clinical experience with older adults affirmed the value of addressing social isolation by harnessing the potential of ICT.**

As my training focus shifted to research in the third year, I pursued my interest in studying social isolation and loneliness in directed research and my qualifying exam project. Using data from the Health and Retirement Study (HRS), I exanimated the longitudinal association between internet use and loneliness for older people and tested the hypothesized mediation effect of social contact. My first-authored manuscript prepared with this work was published in *the Journal of Gerontology: Series B Social Science*. To gain knowledge and skills in statistical methods, I sought training in structural equation modeling (SEM) and longitudinal data modeling at the Inter-university Consortium for Political and Social Research (ICPSR) summer training at the University of Michigan. Learning advanced statistical methodology enabled me to model the relationships of key variables of interests in a sophisticated manner. I honed my data management and modeling skills by conducting independent research with the qualifying exam. The qualifying exam project examined the longitudinal bidirectional relationship between geriatric conditions and loneliness. The study results show a longitudinal vicious circle relationship between the number of falls and loneliness. My first-authored manuscript is in press at the Journal of the American Medical Directors Association (JAMDA).

Driven by my interest in employing technology to improve older adults' health outcomes, I participated in the Intergenerational Mobile Technology Opportunities Program (IMTOP) led by Drs. Shinyi Wu and Iris Chi. IMTOP is a mHealth intervention that recruited college-student tutors to help older patients using mobile technology to facilitate diabetes self-management. Using the data collected in the IMTOP intervention study, my first-authored paper reporting the longitudinal IMTOP intervention effects in rural Taiwan **won the 2018 APHA Nobuo Maeda International Research Award**. The manuscript was published at *The Diabetes Educator*.[35] The hands-on research activities with IMTOP not only helped me **learn about designing and implementing mobile health (mHealth) interventions but also gained exceptional training in theoretical thinking and scientific writing.**

IMTOP study leads to two more research opportunities to which I significantly contributed – IMTOP pilot study with immigrant populations in Los Angeles and a feasibility study of an Intergenerational mHealth Program for Affordable Housing Communities (ImPAHC) in Sacramento. I led the IMTOP pilot study to test the feasibility of using the technology-facilitated diabetes self-management intervention to address the prevalent problem of type 2 diabetes (T2D) among first-generation Chinese and Hispanic immigrants in LA. The research project adopts a mixed-method approach, using both survey and individual semi-structured interviews to collect data. We collaborated with four community-based organizations (CBOs) serving Chinese and Hispanic immigrants. I paid weekly visits to all four CBOs to meet with managers and staff members. I supervised three MSW students and one undergraduate student and provided training in qualitative research methods and questionnaire administration. Working together, we have collected 140 questionnaires with immigrants with type 2 diabetes and conducted 24 in-depth individual interviews in Cantonese, Mandarin, and Spanish. Because of this pilot study, **I had the opportunity to work closely with community-based organizations and gained experience in data collection and research program management.** My first-authored manuscript reporting the IMTOP pilot study findings is currently under review at the *Journal of Cultural and Ethnic Diversity in Social Work*. The experience of working with CBOs helped me to apply for and receive a **travel award** from the Stanford Longevity Center to participate in the ***Forming Science-Industry Research Collaboration Workshop*** at the 2019 GSA annual science meeting. The workshop provided training on developing, adapting, and evaluating interventions in community settings and collaboration with Industry. Working with CBOs prepared me to achieve the long term goal of becoming an impactful intervention researcher. The results of the IMTOP pilot study highlighted the importance of social and cultural environments on health outcomes. Additionally, I **served as a co-investigator on the ImPAHC project**. We paid visits to three retirement communities in Sacramento and conducted interviews and focus groups to assess the feasibility of delivering mobile technology facilitated T2D management intervention in affordable senior housing. Working with retirement housing communities inspires me to investigate the aging process from an ecological perspective. I decided to examine the impacts of physical and social environments on social isolation and cognitive health in my dissertation project.

Older people often are not included as part of the intervention/technological tool development team, which adds barriers for older people to learn to use technology. To address this issue, I serve as a co-investigator on a research project, led by Drs. Wu and Chi, employing **a participatory co-design approach to develop a caregiver self-management program (CSMP) app** for Chinese immigrant caregivers. The co-design CSMP app proposal was funded by the USC Clinical and Translational Science Institute (CTSI). We have completed the conceptual design, interactive prototype testing, and internal real-life condition testing phases of the co-design process. I lead the writing of a manuscript reporting the unmet needs of the Chinese immigrant caregiver population and their acceptance of technology-facilitated interventions. The manuscript is currently under review at *the Journal of Health Care for the Poor and Underserved (JHCPU)*. I gained first-hand experience using the co-design process for intervention tool development. I am currently engaged in writing an R21 grant proposal to support the further development of the CSMP app.

Socially isolated older adults are at high risk of cognitive decline and the onset of Alzheimer's disease and related dementias (ADRD). My past research experiences paved the path for me to study the longitudinal association between environmental factors, social isolation, and cognitive health. The fieldwork at affordable senior housing communities made me realize the research gap on contextual risk factors impacting older adults' cognitive health and social well-being. Therefore, I devoted my **doctoral dissertation** to examining the effects of physical and social environments on older adults' social isolation and cognitive health. Recognizing cognitive health in later life is a new research field for me, I sought the highly individualized training opportunities that could be made possible with the F99/K00 award. Having proposed the project with the dissertation committee, I have been working on conducting the proposed dissertation research. <u>By the starting time of the award, I will have time to participate in the rich training activities planned</u>. If awarded, this funding will enable me to receive exceptional mentored research and training opportunities, successfully transition to the new research direction, and prepare me to grow into an impactful independent investigator.

B. Training Goals and Objectives

My research is guided by the overarching goal of promoting long, healthy, and meaningful lives. My long-term career development objective is to build substantial knowledge that will prepare me to develop behavioral interventions to reduce social isolation and prevent cognitive decline. It is my aspiration to become a leading scientist in a renowned research-oriented university and make impacts with my research. To achieve this long-term career goal, for the F99/K00 award tenure, I propose to increase knowledge in cognitive health and ADRD research, cultivate methodological expertise, and gain experience in intervention development, implementation, and assessment. These training goals will prepare me to achieve the Specific Aims of my research plan. Working with my sponsors, I developed the following specific training goals and objectives and describe how we plan to collaborate to achieve the goals.

Training Goal 1: increase knowledge in prevention, clinical practice, intervention programs, and policy pertaining to cognitive decline in normal aging and the development of ADRD. Achieving training goal 1 will prepare me to accomplish specific aims of the proposed research by enhancing my ability to interpret cognitive health results and design intervention research projects. I will take a course on cognitive decline provided at USC Davis School of Gerontology. I have built connections with scholars at USC Alzheimer's Disease Research Center (ADRC) through attending weekly neurobehavior research meetings where I presented my dissertation and future research plans. The USC ADRC research meetings address diverse ADRD related topics, including protective and risk factors of cognitive decline, current clinical screening and treatment practice, and intervention program design for preventing the onset and progress of ADRD. I will learn about recent ADRD research and clinical practice by attending USC ADRC research meetings and training sessions provided by the ADRC Research and Education Core (REC). I will work with co-sponsors Drs. Chi, Dodge, Silbert, and Wu to develop a database of literature and conduct a systematic review of literature on the underlying mechanisms of the association between social isolation and cognitive decline. I will lead the writing of the systematic review manuscript as the first-author. To gain understanding of the current screening, treatment, and health services for older adults at risk of ADRD, I will complete clinical internships in memory care clinics mentored by neurologists. Dr. John Ringman from USC ADRC will mentor me in the clinical internship in F99, and the mentorship for the K00 stage is to be provided by Dr. Jeffrey Kaye at OHSU ADC. I will get involved with the research activities at the Layton Aging and Alzheimer's Disease Center at Oregon Health and Science University (OADC). Dr. Dodge (Data Core Director at OADC) will be the liaison for the resources at OADC, directing me to training opportunities at research cores according to my learning needs. Dr. Dodge and Dr. Silbert (MRI Core Director) will provide directed reading to facilitate my learning for substantial topics related to cognitive decline and ADRD. Additionally, I will attend the monthly OADC investigator meeting held at OHSU where researchers present their ongoing research and proposals.

Training Goal 2: Gain methodological skills for investigating the underlying mechanism of how social isolation influences cognitive health in diverse environments. I am trained in analyzing longitudinal data with Structural Equation Modeling and Multilevel Modeling methods. All three of my first-authored publications employed longitudinal statistical methods.[35–37] Working with Drs. Chi & Wu, I will further hone my longitudinal data analysis skills by conducting dissertation research using five waves of longitudinal data from the NHATS study, which will prepare me to conduct casual inference analysis for the association between social isolation and cognitive decline. I will further cultivate efficiency in statistical analysis by taking an R programming course, which will allow me to better collaborate with research teams. I will familiarize myself with examining neuroimaging outcomes in the F99 stage by working with collaborator Dr. Judy Pa (see letter of support) and drafting publishable manuscripts with her study datasets. My postdoctoral research training involves investigating the underlying mechanisms for the association between social isolation and cognitive health. I will further investigate and expand the underlying mechanisms that will be discovered in the systematic review, including both social behavioral and biological pathways. I will receive training in volumetric and functional MRI that allows me to link the variation observed in behavioral testing of cognition with brain activity. I plan to receive training in interpreting Neuroimaging results working with Dr. Silbert. With the guidance of Dr. Dodge, I will work with neuroimaging experts in the I-CONECT research teams and use the data from the existing studies to gain hands-on experience that will prepare me for future independent research. The study findings will be disseminated via publications and conference presentations.

Training Goal 3: **Exploring future directions for harnessing technology to promote a positive aging experience.** Through participating in IMTOP and CSMP research projects, I had gained experience in employing mobile technology to promote health in the older adult population. Technology develops rapidly, yet individuals most in need are less likely to benefit from technological progress. In fact, older adults of disadvantaged socio-economical backgrounds experience the most significant disparity in technology adoption. I plan to explore future directions for using technology to promote positive aging. Through participating in future CSMP research activities, I will contribute to developing health behavior change theories guided mobile health apps, receive training on assessing user engagement, and examine the effectiveness of such interventions. I will observe Dr. Pa's research team implementing an intervention study using virtual reality technology for ADRD prevention. Getting exposure to different formats of technology-facilitated interventions will broaden my prospects for developing behavioral health interventions.

The OADC is specialized in collecting digital biomarker data. Digital biomarkers are collected using wearable devices in a non-intrusive manner, such as walking speed assessed using in-home sensors. In the K00 stage, I plan to enhance my understanding of collecting and analyzing high-frequency digital biomarker data by participating in the Collaborative Aging Research Using Technology Project (CART) lead by co-sponsor Dr.

Jeffrey Kaye. For example, CART uses ambient technology devices to measure the daily activities of older participants and enables researchers to link health outcomes to their social connectedness. I-CONECT project uses electronic pillboxes that record older adults' medication intake. Gaining skills in collecting and analyzing digital biomarker data could support my employment of detailed information on an older person's daily living in ADRD related research. Wearable devices are one of the research directions for the area of aging and technology. Learning about digital biomarker collection methods could enable me to design wearable device-based intervention studies for preventing ADRD.

Training Goal 4: learn to design, evaluate, and implement community-based interventions. The F99/K00 award will provide me the opportunities to gain intervention research experience by observing fieldwork of I-CONECT. I will work with Dr. Dodge and the I-CONECT team to conduct conversational intervention sessions and provide assistance after participants completed the trials. This series of observation/shadowing will give me first-hand experience of conducting non-pharmacological trials in addressing social isolation and cognitive decline using ICT. Dr. Chi has excessive experience working with community-based organizations that mainly serve the older adult population. With the support of this award, I will work with Dr. Chi to receive training on conducting community-based research. I have been working closely with Dr. Wu to harness the capacity of mobile technology to promote health among T2D patients. With the support of the F99/K00 grant, I will continue to conduct research in collaboration with Dr. Wu to examine the role of ICT in enhancing the health and social connectedness of older adults. These community-based research experience will prepare me to become a future faculty member in a reputable academic institution.

The mixed-methods research approach is uniquely advantaged in obtaining a holistic perspective of the problem targeted by the intervention within the context. I have taken a course in mixed-methods taught by Dr. Lawrence Palinkas. In the K00 stage, I proposed to use a mixed-methods approach to solicit older adults' and community stakeholders' perspectives of the community-based intervention that will be developed. Gaining mentorship and guidance in conducting mixed-methods research will greatly benefit the development of the social isolation and cognitive health intervention. I will work with Dr. Lawrence Palinkas to formulate the purposeful sampling plan and design the interview guides. I will also receive direct qualitative data analysis training from Dr. Palinkas in Biweekly meetings in the F99 stage, Monthly meetings in the K00 stage. The purpose of the F stage consultation is to develop interview guides. The K stage consultation is focused on the analysis of qualitative data and considerations of intervention implementation. Dr. Palinkas is an expert in implementation science. Interventions developed without considerations of the implementation process are at a high risk of ineffectiveness. Therefore, I will work with Dr. Palinkas on the design and the evaluation of the intervention implementation.

If awarded, the F99/K00 award will provide me the exceptional opportunities working with leaders and experts in the fields of Gerontology, Engineering, Neurology, Mixed-methods research, and Implementation Science. Applying to the F99/K00 award encouraged me to think beyond the dissertation stage of research and identified a postdoctoral mentor who matches my career development and training needs. If awarded, I will be able to work with both my doctoral mentors (Chi & Wu) as well as my postdoctoral mentor (Dodge) simultaneously and receive individualized training by working with Drs. Palinkas, Silbert, and Kaye. Having a smooth transition from the predoctoral training to the postdoctoral research will benefit the realization of my short and long-term career development goals by increasing the efficiency and continuity of the transitional stage and enhancing academic collaboration with faculty members at both USC and OHSU.

C. Activities Planned Under This Award

The F99/K00 award will afford me the resource and training opportunities that are essential for my professional development. During the proposed five years of F99 and K00 phases of training, 75% of the effort will be dedicated to research activities, including conducting literature reviews, data collection, analysis, and preparing manuscripts and submitting them to leading peer-reviewed journals in the fields of Gerontology and Neurology. To prepare for applying to faculty positions in top-ranked universities, I will seek solo-teaching opportunities in the F99 phase and the last year of the K00 phase. About 5 to 20% of the effort will be paid to develop grantsmanship and writing proposals. Specifically, I plan to spend about 20% of the time developing a K99/R00 grant in the third year of the postdoctoral stage. I will spend about 5% of the time to disseminate my research findings and interact with scholars by attending major academic conferences in the fields of Gerontology and Neuroscience. The following table shows the activities planned for achieving each training goal.

Table 1. Training Activities and Timeline

Training activities	Sponsor/Professor	F99	K00			
		Year 1	Year 1	Year 2	Year 3	Year 4

Training Goal 1: Increase knowledge in prevention, intervention and policy pertain cognitive decline in normal aging and the development of ADRD.						
Course: PM-605 Systematic Review and Meta-Analysis (Spring)	Cortessis, Victoria	X				
Conduct a systematic literature review on the underlying mechanisms of the association between social isolation and cognitive decline.	Chi, Dodge, Silbert, Wu	X	X			
Course: GERO-566 Cognitive Decline: Alzheimer's Disease and Dementia (Spring)	De Martino, Massimo Ulderico	X				
Attend training sessions provided by USC ADRC Research and Education Core (REC)	Yassine, Hussein	X				
Clinical internship with neurologists at memory clinics	Ringman (at USC) & Kaye (at OHSU)	X	X			
Join monthly OADC investigator meetings (Online at F99 phase, in-person at K00 phase)	Dodge	X	X	X	X	X
Training Goal 2: Gain methodological knowledge and skills for investigating the underlying mechanism of how social isolation influences cognitive health in diverse environments.						
Course: PM-560 Statistical Programming with R (Summer)	Li, Arthur	X				
Attend OADC imaging core weekly lab meeting	Silbert		X	X	X	X
Attend AAIC Neuroimaging pre-conference workshop	Lowe, Val		X	X	X	X
Get familiar with analyzing brain biomarker outcomes with collaborator Pa's dataset	Pa	X				
Collaborate with neuroimaging experts and examining neuroimaging outcomes using I-CONECT data	Dodge & Silbert		X	X	X	X
Training Goal 3: Exploring future directions for harnessing technology to promote positive aging experience.						
Observe virtual reality intervention sessions for preventing ADRD	Pa	X				
CSMP project - Develop health behavior theories guided mHealth intervention	Chi & Wu	X	X			
Learn to incorporate digital biomarker outcomes with hands-on research activities using I-CONECT and CART data	Dodge & Kaye		X	X	X	X
Training Goal 4: learn to design, implement, and evaluate community-based interventions.						
Collect pilot data to inform intervention design	Dodge			X		
Attend training at the Clinical Trial and Methods Professional Interest Area (PIA) supported by ISTAART and participate in the IMPACT-AD course provided by ACTC	Dodge		X			
Immersion training with I-CONECT fieldwork	Dodge	X	X			
Regular consultation on mixed-methods research and intervention implementation	Palinkas		X	X	X	
CSMP project - conducting collaborative research with community-based organizations	Chi & Wu	X	X			
Online course: Study Designs for Intervention Research in Real-World Settings (UCSF)	Handley, Margaret & Shade, Starley		X			
NINDS-sponsored Clinical Trial Methodology Course (R25NS088248)	Meurer, William		X	X		
Professional Development Activities						
Submit conference abstracts, attend research conferences and present works	All sponsors	X	X	X	X	X

Preparing manuscripts and submit to high impact peer-review journals	All sponsors	X	X	X	X	X
Training on conduct of research	All sponsors	X	X	X	X	X
"Design Studio" and Advanced Grant Writing Course at OHSU	Dodge		X	X	X	
Gain further grant writing experiences and prepare to submit K99/R00 application	All sponsors			X	X	X
Formulate future research agenda and plan for R21/mini R01 applications	All sponsors				X	X

In the F99 phase, I plan to take a course at the USC School of Gerontology dedicated to cognitive decline in later life in the Spring of 2022 (year 1 of F99). This course covers cognitive decline for people with Alzheimer's Disease and other forms of dementia, and imaging technology to identify treatment options. To enhance efficiency in analyzing longitudinal data and prepare for research collaboration in postdoctoral training, I will complete an R programming course from USC Keck School of Medicine in the summer of 2021 (year 1 of F99 stage). Coursework will make up about 10% of my total professional efforts in the F99 phase.

In the K00 phase, I will move to Portland and receiving training at OHSU. I will develop the capacity to examine neuroimaging outcomes in intervention research by attending the weekly OADC imaging core lab meetings led by Dr. Silbert. Furthermore, I will receive training at the annual Alzheimer's Association International Conference (AAIC) Neuroimaging pre-conference workshops. I will complete a formal online course on Study Designs for Intervention Research in Real-World Settings at UCSF. This course will help me to gain knowledge in determining which design is most suited to a range of 'real-world' implementation settings and circumstances, and on choosing between designs to maximize overall study quality. I will apply for the NINDS-sponsored Clinical Trial Methodology Course (R25NS088248), a two-year intensive, engaging program with online and in-person training sessions designed to help junior investigators develop scientifically rigorous, yet practical clinical trial protocols, and to focus funding mechanisms as a key trial planning activity.

Table 2. Percentage of Efforts

	F99-Year 1	K00-Year 1	K00- Year 2	K00- Year 3	K00- Year 4
Research	75%	75%	75%	75%	75%
Teaching	5%	0%	0%	0%	5%
Coursework	10%	5%	5%	0%	0%
Grant writing	5%	15%	15%	20%	15%
Conferences	5%	5%	5%	5%	5%

I will work with my co-sponsors to continuously apply to funding opportunities from NIH and other prestigious funders to support the development and pilot testing of social isolation and cognitive decline interventions. I will engage in peer-support writing groups, complete seminars in responsible conduct of research, and attend the "Design Studio" and Advanced Grant Writing Course at OHSU to refine my research proposals to be submitted during the K00 award period. Grant writing will make up for about 5% to 20% of my total efforts. I plan to submit to pilot funding opportunities at OHSU, which will allow me to collect preliminary data for intervention development and prepare future NIH R21 or mini R01 applications. More time is allocated to grant writing in year 3 of the K00 stage as I will submit an application to the K99/R00 mechanism.

Knowledge developed from the proposed dissertation will be presented at the annual meetings of the GSA, APHA, and SSWR. I am a member of all three scientific organizations. I will continue to make scholarly contributions to the organization and groups with new knowledge and skills gained from my proposed research projects. In the K00 phase, I will expand my network to ADRD related research groups. I will attend and submit abstracts to present at the Alzheimer's Association International Conference (AAIC), GSA and Clinical Trials in Alzheimer's Disease Conference (CTAD). I will have conversations with distinguished researchers in my research area and foster long-term collaborative relationships in the academic community. Conferences take about 5% of my efforts.

My short-term career milestone is to graduate with distinction from USC SSW. Within four years after finishing the dissertation, I envision myself successfully attaining another career development award and hold a tenure-track assistant professor position at a research-intensive university. Nine years after I complete my dissertation, I envision myself achieving tenure and will have developed an R01 funded intervention project on social isolation and cognitive health in later life, utilizing technology for trial outcome monitoring and assessments and as intervention tools.

SPECIFIC AIMS

Two essential domains fundamentally affecting the aging process are meaningful social engagement and physical health.[1] Three decades of research have consistently shown that social isolation is associated with the onset of cardiovascular disease,[2–4] declines in cognitive function,[5–7] and lower overall life satisfaction.[8–10] Perceived social isolation was associated with a higher cortical β-amyloid burden in older adults, supporting the crucial role social isolation plays in the development of Alzheimer's disease and related dementia.[11,12] The potential causes of social isolation are multi-faceted. While the existing literature studied individual-level factors, such as age,[13,14] personality,[15–17] and health status ,[18–21] only a few studies have examined the extent to which the living environment influences the experience of social isolation among older adults. Furthermore, environments serve as one of the most pervasive and complex stimuli to the brain and contribute to shaping cognitive functionality.[22]

The Ecological Theory of Aging (ETA) posits that the living environment profoundly influences the aging process.[23,24] The process of personal and environmental resource exchange determines the aging outcomes.[23] Socially isolated individuals are at a higher risk of cognitive decline.[25,26] Social isolation might mediate the association between environments and cognitive health. The proposed dissertation work aspires to investigate the longitudinal influence of living environments on social isolation and cognitive health of older adults, including physical environment as measured by community and in-home disorder and social environment as measured by neighborhood cohesion.

Information and Communication Technology (ICT) has brought a revolution to the way people communicate and access resources.[27,28] With increased popularity among older people,[29,30] ICT could serve as an intervention delivery tool to alleviate social isolation and promote cognitive health.[31] An increasing amount of research investigating the association between ICT use and social isolation has been published. However, the interactive effect of ICT use and environmental factors on the risk of social isolation has not yet been empirically examined. As ICT increases individuals' access to information and resources, adopting ICT possesses the potential to mitigate the effect of a disadvantaged living environment.

Cognitive health is the foundation for independent living and quality of life. However, the effect of ICT use on cognitive health has not been extensively researched. Providing social engagement through technology-based approaches can alleviate social isolation among older people and simultaneously prevent cognitive decline. Nevertheless, only a few randomized controlled intervention studies have been conducted to examine the effectiveness of ICT-enhanced tools on cognitive functioning and social isolation.[32] The existing trial showed that ICT-enhanced intervention had beneficial effects on executive functions.

In the F99 phase, I will examine the longitudinal influence of living environments on social isolation and cognitive health with five waves of data from the National Health & Aging Trend Study (NHATS). The moderating effect of using technology for medical, social, and instrumental purposes will be scrutinized within the living contexts. **In the K00 phase**, I will use the longitudinal data from the ongoing randomized controlled trial "Internet-based Conversational Engagement Clinical Trial (I-CONECT)" (NIA R01AG056102, R01AG056628 & R56AG051628) to investigate the underlying pathways in the association between social engagement and cognitive functioning. The I-CONECT trial collects digital bio-marker and neuroimaging data, including region-specific structural and functional pre-post changes in the brain. Additionally, I will explore how technology can effectively alleviate isolation and improve cognitive functions with data collected in the Collaborative Aging Research Using Technology (CART) initiative (NIA U2C AG054397). I will recruit I-CONECT participants to co-design a behavioral health intervention using a mixed-methods approach. The K00 stage research will produce a technology-facilitated behavioral health intervention protocol to address social isolation and cognitive decline. In summary, the specific aims are as follows.

Aim 1: **My dissertation project** will examine the effect of physical and social environments of living on social isolation and cognitive health. The effects of ICT use will be investigated within living environments.

Aim 2: **My postdoctoral research** will build knowledge on the behavioral and contextual pathways that could explain the association between social isolation and cognitive decline in later life. Employing a community participatory co-design approach, I will develop a technology-facilitated behavioral health intervention to reduce social isolation and prevent cognitive decline in the older adult population.

Successfully achieving the aims of this proposal will generate knowledge on the environmental factors on the risk of social isolation, contribute to the existing literature on the mechanisms of social engagement on cognitive health, and lead to the development of a technology-facilitated behavioral intervention program for alleviating social isolation and preventing cognitive decline. Conducting the research activities in the current proposal will help the PI to develop into an independent intervention researcher focused on promoting social engagement and preventing ADRD among older adults.

RESEARCH STRATEGY

1. Significance

1.1. Environment and Social Isolation

Social isolation is the absence of contact with others, lack of relationships and social integration.[38] A meta-analysis of 70 studies published over three decades showed that socially isolated individuals had a 29% increased likelihood of mortality.[4] AARP estimated that about 4 million older adults enrolled in Medicare are socially isolated, and Medicare spends an additional $6.7 billion on isolated individuals than otherwise if they are socially connected.[39] Eradicating social isolation requires a social approach.[40–42] While the current literature has studied the individual-level risk and protective factors of social isolation, the impacts of physical and social environments have not been thoroughly researched.

The Ecological Theory of Aging (ETA) posits that environmental factors significantly influence health and social wellbeing.[23,40,43] Living environment provides access to resources, social interaction settings, and an essential source of cognitive stimulation.[22] Living environments are often assessed through diverse dimensions, including physical and social aspects of environments.[23,40] Physical environments refer to the objective aspects of the neighborhood and the facilities, for example, access to amenities, green open spaces and aesthetics.[44] While the social environment has been operationalized as the social participation and social integration perceived by people.[45]

Previous research reported that older residents of a high-crime neighborhood are more likely to be socially isolated despite their strong desire for social integration.[42,46] The community disorders, which can present itself as litters on the streets, deserted buildings, and vandalism, could lead to the perception of lack of safety, thus causes an increased likelihood of social isolation. Outdoor space & public buildings, and common areas have a critical role in keeping the residents living an active social life and contributing to their general wellbeing.[44] Although existing literature sheds light on the role of the physical environment on social isolation. The evidence is built mainly on cross-sectional data. Longitudinal association between the two warrants further research. The proposed dissertation work aspires to build evidence with longitudinal data, and the following hypothesis was derived based on cross-sectional results and theoretical reasoning of ETA.

H1: Negative physical characteristics of the environment, i.e., disorders in the community and at home, are associated with an increase in social isolation longitudinally.

Social environment characteristics, such as a sense of belongingness, togetherness, and trust, were associated with a less sedentary lifestyle and more social participation and interpersonal interaction, thus predicting a lower extent of social isolation.[47] Residential satisfaction and a sense of belonging were found negatively associated with perceived social isolation.[48] Middle-age to older adults who perceive the community as trustworthy experienced less subjective social isolation (i.e., loneliness) over four years.[49] Existing literature cohesively suggests that a positive social environment is protective of being socially isolated. Based on the literature review, the following hypothesis was derived regarding the longitudinal relationship between social environment and social isolation.

H2: Positive social environment factors are associated with less social isolation longitudinally.

One's perception of the social environment is likely to be influenced by the physical characteristics of the environment. Sampson and Raudenbush (2004) reported that physical deterioration and disorder has a social meaning.[50] Physical characteristics of a living environment could change an individual's opinions of the social environments, resulting in less perceived opportunities in the neighborhood. For example, visible vandalism on the street could prevent residents from walking on the streets, increasing the likelihood of social isolation.[51]

H3: Social environment buffers the influence of physical environment on social isolation over time.

1.2. Living Environment, Social Isolation, and Cognitive Health

Physical and social environments may affect the social wellbeing of older people and influence their cognitive health. Environments serve as one of the most pervasive and complex stimuli to the brain and contribute to shaping cognitive functionality through the mechanism of neuroplasticity.[22] Environmental challenges, as simple as daily tasks like shopping, was hypothesized to simulate the brain because it requires an older person to remember the route and shopping list despite the distractors in the environments. Environmental stimuli, just like education and cognitively-demanding jobs, were theorized to be associated with an increased cognitive reserve and prevent cognitive decline.[22] Considerable amount of research/practice interests and efforts have been paid to building aging-friendly communities.[51,52] However, **longitudinal evidence regarding the environmental influences on cognitive health is yet to be built.**[22]

Previous conceptual work pointed out the vital influence of the living environment on older adults' cognitive health. Community disorder, such as cluttered and unsafe environment, increases the cognitive burden and the environmental stressors on the residents. Older adults living in communities with a higher number of adverse physical environmental characteristics are more likely to be deprived of health care resources and cognitively stimulating activities.[54] Positive social environments provide the resources and opportunities that are necessary for cognitive simulating engagements.[45,55] Social environments were found to be significantly associated with cognitive functioning with cross-sectional data.[55] Although the theoretical reasons supporting the association between specific physical and social environments and cognitive decline is substantial, empirical examinations of the relationships are scarce.[22,56] **The proposed research project aspires to contribute to the knowledge by building empirical evidence on physical environmental factors and cognitive health using longitudinal data.** The following hypotheses were derived.

H4: Disadvantaged physical environments, i.e., the disorder in the community and at homes, predict deterioration in cognitive health over time.

H5: Positive social environment has a direct longitudinal effect on cognitive health among older adults.

Besides the lack of longitudinal evidence on environment and cognition, the current literature rarely addresses how the living environment "gets under the skin." The proposed research project aspires to contribute to the literature by empirically examining one possible mechanism through which living environments impact cognitive health. Social isolation could mediate the association between physical and social environments and cognitive functioning. The previous section's literature review established the reasoning for the associations between physical and social environments and social isolation. Social isolation was found to predict cognitive decline longitudinally.[25,57] Perceived social isolation was associated with a higher cortical Amyloid burden in older adults, supporting the critical role of social isolation in preclinical Alzheimer's disease and related dementia.[11,58] Social isolation may not fully explain the effects of the living environment on cognitive health. Other factors related to living context, such as mobility, could also serve as explanatory mechanisms of the link between living environments and cognitive health. The proposed dissertation focuses on the longitudinal relationships between social isolation and cognitive health within the environment.

H6: Social isolation partially mediates the association between physical and social environments and cognitive health over time.

Although ETA highlights the importance of the environment in the aging process in general, the theory did not specifically discuss the effect of environments on cognitive health.[23,24] The knowledge that will be built in the proposed study could enrich the content and expanding the scope of ETA.

1.3. ICT, Social Isolation and Cognitive Health

ICT use has been found associated with reduced social isolation.[59,60] A systematic review reported that ICT alleviates the elderly's social isolation through four mechanisms: connecting to the outside world, gaining social support, engaging in activities of interests, and boosting self-confidence.[60] The ETA hypothesized ICT use promotes the wellbeing of older adults through enhancing their agency.[40] Nevertheless, the effect of ICT interventions on social isolation could not last for more than 6 months after termination.[60] Older adults, defined by their age, form a diverse group with various personal characteristics and life experiences. Previous research noted that ICT intervention might not work for everyone.[60] Future research are needed to identify how to implement and contextualize ICT interventions in the older adult's daily life so that the population would be more likely to benefit from ICT use.[60] Additionally, the nature and content of ICT use might result in different effects. Internet use for social purposes was reported to be associated with decreased perceived social isolation in the following year, while the association between social isolation with instrumental use of the internet was insignificant.[61] **Contextualizing the effect of ICT use within older adults' ecological living environment could contribute to the literature with potential mechanisms and intervention delivering format for reducing social isolation and preventing cognitive decline**.

The impact of ICT use within the living environment will be explored in the proposed dissertation work and postdoctoral research. ICT adoption has the potential to reduce the negative influence of disadvantaged physical and social environments by providing a new avenue for older adults who suffer health issues to interact with family, peers, and society.[27,28,36] ICT users reported that internet made it easier for them to reach people and increased the quantity of communication with others, thus supporting them in feeling less isolated.[59] ICT corresponds to the information & communication aspect of the community environment. We hypothesize the effect of physical and social environment on social isolation is different for ICT users and non-ICT users. In the proposed dissertation project, I will investigate the interacted influence of physical and social environments and using ICT for different purposes (social, instrumental, and medical) on social isolation.

H7: ICT use for social purposes alleviate social isolation over time.

H8: ICT use for non-social purposes is not related to social isolation longitudinally.

H9: ICT use buffers the longitudinal association between physical social environments and social isolation.

1.4. The Significance of Developing an ICT Facilitated Behavior Health Intervention

The effects of ICT use on cognitive health are not well-understood. I-CONECT intervention employs a Randomized Controlled Trial (RCT) study design to investigate the impact of a specific type of ICT use, i.e., Internet-facilitated conversational engagement via video-chat, on cognitive functioning among older adults who are cognitively intact or with Mild Cognitive Impairment (MCI). The task of conversation is highly cognitively simulating, which requires control of attention, working memory, executive functions, and mental interpretation of others' emotions and intentions.[62,63] Linguistic ability is known as highly related to cognitive health among older adults.[64] In the K00 Phase, I will utilize data collected in the I-CONECT project to examine the longitudinal intervention effect of ICT facilitated conversation on the social wellbeing and cognitive health of older adults.

Despite the empirically supported benefits of ICT for older adults' wellbeing, most of the currently available technology tools, such as social media sites and mobile health apps, are not designed with older adults' unique needs in mind. Older people often are not included as part of the intervention/technological tool development team. It was not until recent years the researchers in the health service and related fields start to employ a co-design approach to develop intervention programs from the target populations' perspective.[65–67] On the other hand, anxiety and resistance remains among the older adult population when it comes to learn, trust and safely adopt new technologies.[68–70]

Qualitative research is well suited to understand stakeholders' perspectives, considerations of feasibility, and the culture and contexts. Quantitative studies are better suited for examining associations and understanding mechanisms. Combining the advantages of the two schools of research design, a mixed-methods study is uniquely advantaged for the design and refinement of a behavioral health intervention.[71–73] I will employ a mixed-methods approach to **co-design a technology-facilitated community-based intervention to reduce social isolation and prevent cognitive decline** at the K00 research phase, leading to future research agenda on pilot testing the potential effects and feasibility of the intervention program.

The current I-CONECT intervention did not consider the impact of physical and social environments, and very few existing intervention projects have addressed the issues of social isolation and cognitive decline from an ecological perspective. With the evidence will be built in my dissertation work, I will explore the possibility of designing a sustainable community-based program to offer a technology-facilitated solution to alleviate social isolation and prevent cognitive decline.[74] Studying the effectiveness of technology in the living environment and exploring new approaches of technology-facilitated care could transform the older adults' home to become more than a living space, but also a platform that integrates social and health services.[75]

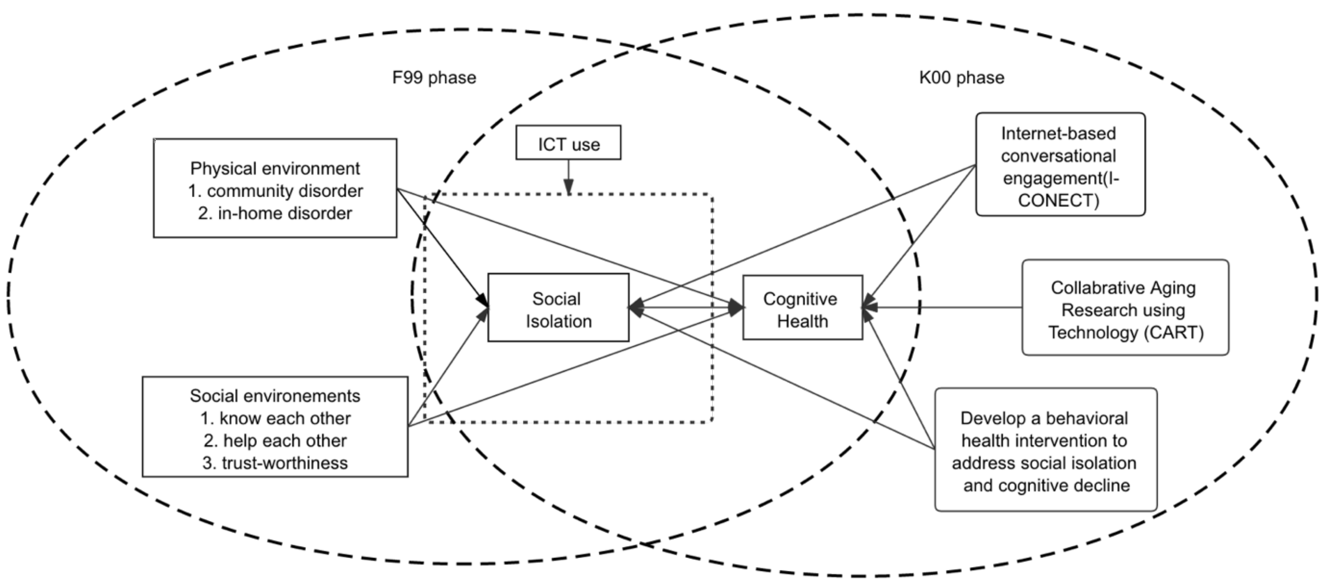

Figure 1. Conceptual framework of the F99 and K00 phases of research

Figure 1 illustrates the conceptual framework for both F99 and K00 phases. **The proposed research can produce an ecological theory informed solution to the challenge of social isolation and the risk of cognitive decline.** By contextualizing the role of technology in older adults' living environments, knowledge will be built in the proposed research will address the challenges of independent living and aging in place with technology tools that fit the circumstances. The proposed work will **contribute to the literature and expand**

ETA's scope by studying both the physical and social environments and their relationships with social isolation and cognitive health.

2. Approach

2.1. F99- predoctoral stage (Specific Aim 1)

The proposed dissertation will use data from NHATS wave 5 (collected in 2015) to 9 (collected in 2019). NHATS is a longitudinal study that annually surveys Medicare beneficiaries aged 65 and above in the contiguous United States. The waves 5 to 9 were selected because the physical environment measures were included since wave 4, and the NHATS sample was replenished in the 5th wave (2015). Starting the analysis from wave 5 prevents the problem of having a high percentage of missing in the baseline of longitudinal analysis. In total, there were 8,334 complete cases in wave 5 data collection.[76]

2.1.1. Measurements

Physical environments were reported by the interviewers who visited the respondents at their home. Standing **in front of the respondent's home/building and looking around**, the interviewer rated on a four-point scale (1=none, 4=a lot) regarding 1) the amount of litter on the street, 2) graffiti on buildings and walls, 3) vacant or deserted houses or stores. The physical condition **inside the responder's home/apartment** was assessed by observing the existence of 1) peeling paint, 2) evidence of pests, 3) broken furniture, 4) flooring need of repair, 5) tripping hazards, and 6) cluttered room. The scale score of the physical conditions of the community, home/building, and inside home/apartments will be derived from the sum score of the corresponding items. The use of interviewers' observation could provide more accurate information regarding each participant's physical environment than using the census data by zip code, which tends to reduce the complexity of the real-life condition.

Social environments. This dissertation work operationalizes social environment as community cohesion perceived by older adults. Respondents reported whether the people in the community know each other very well, willing to help each other, and people in the community can be trusted on a three-point scale (1=agree a lot, 2=agree a little, 3=do not agree). A sum score of the three items was used as the scale score for the social environment. Social environment scale scores ranged from 3 to 9, with a higher score indicates more neighborhood cohesion.

ICT use. NHATS assessed three types of ICT use: instrumental internet use, social internet use, and internet use for medical purposes. **Instrumental** internet was measured by asking the respondents whether they shop online and pay bills or banking with the internet. **Social** use was measured with one item asking in the last month if the respondent went online to visit social network sites (such as Facebook or Linkedin). Use the internet for **medical** purposes had four items: order or refill prescriptions, contact medical providers, handle health insurance matters, and get information about one's health conditions. The three aspects of ICT use will be used as moderators that are independent of each other.

Social isolation was constructed with six dichotomously coded indicators: no marriage or partnership, no family identified to talk with about important things, no friend identified to talk with about important things, never meeting in person with family or friends in the past month, no participation in religious service, and no club, class or organization activities participation.[38] All items of social isolation will be coded as a higher indicates a more isolated situation. The scale score is the sum of six dichotomous indicators, ranging from 0-6. Scores less than 1 represent least isolated; scale score 2-3 indicates somewhat isolated, and scores equal or above 4 indicate social isolation.

Cognitive Health. For sampled persons self-responded cases, the NHATS study assessed 1) **memory**: using immediate and delayed words recall; 2) **orientation**: date, month, year and day of the week, naming the President and Vice President of the United States; and 3) **executive function** measured by clock drawing test.[77]

2.1.2. Quantitative data analysis strategies

Data management and variable re-coding will be conducted using Stata 15 SE. Descriptive univariate analyses will be run before model building. Correlation of key variables will be run to estimate the effect size and detect any potential multicollinearity issues. The proposed dissertation project will produce three publishable manuscripts. Paper 1 will employ longitudinal path analysis methods to model the longitudinal relationships between physical environments, social environment, and social isolation (H1-H3). NHATS wave 5 physical environment and social environment variables will be included in the model to predict social isolation in wave 9, controlling for baseline social isolation. Paper 2 will study the longitudinal association between wave 5 physical and social environments and cognitive health in wave 9 (H4-H5). The social isolation score in wave 6 is hypothesized to mediate the relationship between environmental variables and cognitive health (H6). Paper 3 will use exanimate the direct effects of ICT use for different purposes in wave 5 on social isolation in wave 9 (H7-H8). Meanwhile, the moderating effect of different purposes of ICT use in baseline on the longitudinal association between baseline physical and social environments and wave 9 social isolation will be modeled (H9). Sampling weight will be applied to the models to match the representativeness of the racial/ethnic groups as designed in the sampling strategies.

2.2. K00- postdoctoral stage (Specific Aim 2)

The postdoctoral stage is guided by the overarching aim of building knowledge on the mechanisms that could explain the association between social isolation and cognitive decline and developing a community-based behavioral health intervention. Previous research reported six steps for quality intervention development: 1) defining and analyzing the problem; 2) identifying modifiable factors; 3) deciding on the intervention mechanism; 4) clarify the intervention delivery process; 5) testing and revising the intervention; 6) a more rigorous effectiveness evaluation.[78] The proposed dissertation research addresses step 1 and 2 of intervention development, i.e., understanding the causes of the problem and modifiable factors. The existing evidence that could explain the mechanism through which social isolation might affect cognitive health will be summarized in a **systematic review** (step 1-3). I will further explore the mechanisms of change and clarify the intervention delivery format in my postdoctoral research using data from the **I-CONECT and CART** projects (step 3-4). Focus groups will be held with older adults at risk of social isolation to design desirable approaches for intervention delivery (step 4). In the later years of postdoctoral training, I will apply for pilot funding at OHSU or externally **to collect preliminary data** to evaluate the feasibility and determine the effect size of the behavioral health intervention (step 5). The timeline of the proposed K00 stage research activates is outlined in Table 1.

2.2.1. Systematic review of social isolation and cognitive health

In the first year of K00 stage, I will lead a systematic review of existing evidence on the modifiable pathways of the association between social isolation and cognitive health. Research with human subjects will be included in the review. Modifiable pathways considered will include behavioral and clinical mechanisms, technology adoption, and contextual factors. The systematic review methods will follow the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The review procedure will be registered. With the assistance of a librarian and in consultation with the sponsor team, I will formulate search teams, identify electronic databases, and further specify the review's inclusion and exclusion criteria. At least two reviewers will independently evaluate studies, extract data, and meet to resolve disagreements. We will assess the research quality using GRADE guidelines and GRADEpro software as recommended by Cochrane Collaborative. **This research activity will produce a systematic review published in a high impact journal and help me better define the scope of the proposed behavioral health intervention, identify mechanisms of the intervention, and clarify the delivery process.**

2.2.2. Internet-based Conversational Engagement Clinical Trial (I-CONECT)

I-CONECT study is conducted in collaboration by Oregon Health and Science University (OHSU) and the University of Michigan (UM) (NIA R01AG056102, R01AG056628 & R56AG051628; ClinicalTrials.gov Identifier: NCT02871921). One hundred sixty socially isolated adults aged 75 plus will be recruited at each site (OHSU & UM) and randomized into either the video-chat intervention group and the control group. The inclusion criteria are aged 75 or older, consent to magnetic resonance imaging (MRI) if physically able to receive one, socially isolated, adequate vision and hearing, understand English, and normal cognition or MCI (based on consensus diagnosis). The intervention group participants received computers and internet service during the intervention, which ensures the participants can video chat with study staff for 30 minutes/day, 4 times/week for 6 months (high does), and 2 times/week for an additional 6 months (maintenance dose). Study staff briefly check-in by phone call (approximately 10 minutes) with both intervention and control group participants once per week. In-home neuropsychological assessments will be administered at baseline, 6 months, and 12 months. All participants would have MRI scans at baseline and 6 months if MRI were safe for them. Participants at OHSU will have their medication compliance tracked using electronic pillboxes. The estimated primary completion date for I-CONECT study is Oct 2022.

The primary intervention outcomes are executive and memory functioning assessed with cognitive tests using National Alzheimer's Coordinating Center Uniform Data Set neuropsychological battery (UDS V3: https://www.alz.washington.edu/WEB/dataforms_main.html), cross-validated by NIH Toolbox.[79] Region-specific structural and functional pre-post changes in the brain are evaluated with magnetic resonance imaging (MRI). **Medical adherence** is measured with electronic pillboxes. Compared to traditional self-reported medical adherence, the digital biomarker collected with electronic pillboxes can eliminate the social desirability bias and record the exact time when the participants opened the pillboxes.[80] **Social Isolation** was defined as: i. Score ≤12 on the 6-item Lubben Social Network Scale (LSNS-6);[81] or ii. Engages in conversations lasting 30 minutes or longer no more than twice per week, per subject self-report.

Among the outcomes measured in I-CONECT, my interests focus on cognitive health and social isolation. I will collaborate with I-CONECT team to examine the intervention effects. Data will be analyzed to evaluate the differences in cognitive function change and extent of social isolation from baseline to 6 months between the intervention and the control groups. Similar analyses will be conducted for the change between 6 to 12 months. I will also employ I-CONECT data to examine the association between social isolation and change in neuropsychological outcomes, such as volumes and structural MRI variables. **I-CONECT will serve as an example for me to learn about the design and implementation of behavioral intervention for cognitive health.**

2.2.3. Collaborative Aging Research Using Technology (CART)

The CART initiative uses ambient and wearable technology devices to assess health and activity in the home and community to help older adults aging in place. It built an open research platform for researchers to detect the longitudinal change in social engagement, health, and wellbeing. The CART facilities equipped homes include unobtrusive sensors blended in the environment to evaluate older adults' daily activities, walking speed, sleep patterns, mobility, and computer use. Such health behavior and physiological data collected with digital deceives are known as digital biomarkers.[82] Using digital biomarker outcomes could minimize self-report bias and allow researchers to address questions with high-frequency data. I will utilize the longitudinal data collected using CART platform to investigate the relationships between daily activities, social engagement, and the cognitive health of older participants. The pathways identified in the systematic review will be further validated using the longitudinal data from the CART initiative. Working with co-sponsor Dr. Kaye, I will explore the possibility of employing technological devices in the CART initiative to deliver ETA informed behavioral health intervention and assess intervention effectiveness.

2.2.4. Co-Design a Behavioral Health intervention and Develop a Pilot Study Proposal

A mixed-methods approach will be employed to co-design a behavioral health intervention protocol for addressing social isolation and prevent cognitive decline. I will build collaborative relationships with community service organizations serving the older adult population in Portland to elicit their perspectives on the intervention design, recruitment strategies, study procedure, and outcome evaluation. Experts panels consist of all sponsors will be convened quarterly to consult the development of the intervention.

Evidence on social isolation and cognitive health built with I-CONECT and CART data will serve as the quantitative part of the mixed-method approach. Focus groups will be held with socially older adults to collect information on the participant's opinion towards the intervention design and clarify the intervention delivery process. Focus group participants will be recruited purposefully among the current I-CONECT participants. Based on the maximum variation strategy,[73] individuals with diverse living environment conditions will be selected to participate in the focus group discussion. Focus group members will be selected from both study sites of I-CONECT (OHSU & UM). Two focus groups will be held at each site: one with intervention group participants, the other with control group participants. It is targeted for each focus group to have 8 participants (total N=32). The interview guide will be developed informed by ETA concepts and in consultation with experts with qualitative/mixed-methods research specialties (Palinkas) and with the best knowledge of I-CONECT participants (Dodge). The PD/PI (Yu) will facilitate the focus group discussion at OHSU and UM. Conversations in focus group sessions will be audio-recorded and then transcribe verbatim for analysis. The transcripts will be analyzed using inductive content analysis method by at least two trained independent coders.[83]

A behavioral health intervention protocol for social isolation and cognitive health will be developed by synthesizing study results from the qualitative and quantitative components. A pilot study proposal will be drafted for examining the implementation feasibility and estimate the intervention's effect size. OADC pilot award and Extramural funding opportunities will be sought to support the pilot study.

RESPECTIVE CONTRIBUTIONS

This research and training proposal was informed by the applicant's previous research, clinical experience, coursework, and theoretical thinking toward the research questions. The research and training plans were developed, reviewed, and edited in collaboration with all sponsors and co-sponsors. Preparing for this application, I had weekly consultation meetings with Drs. Chi and Wu who were involved in the conceptualization of the specific aims, selection of theoretical framework, and training plan development. As the PI of the parent study and designated the postdoctoral stage primary mentor, Dr. Dodge worked with me to formulate the postdoctoral research and training plan. She oriented me with resources and personals available at OHSU and OADC, helped me develop the specific aim, training and research plans. She reviewed and provided feedback on several proposal drafts. Dr. Silbert guided me to solidify the training plan, especially for planning activities related to neuroimaging methodology. Dr. Kaye worked with me to refine the training goals, planning technology-facilitated ADRD research activities in the K00 stage. Dr. Palinkas guided the planning of the mixed-methods research design, helped develop the training plan, and provided invaluable input on revising the proposal. Several drafts of the proposal and resubmission were carefully reviewed and edited by the sponsor and co-sponsors to provide feedback on the training plan, research study aims, conceptual framework, analysis plans, and feasibility of study aims. All sponsors reviewed and agreed on the finalized sponsor & co-sponsor statement.

To accomplish the proposed research, Dr. Chi will guide me on developing and conducting community-based intervention research. She will offer expertise in building collaborative relationships with community service organizations and ethical considerations of conducting research with older adults. Dr. Wu will ensure my continued research engagement with existing technology-facilitated health intervention studies. Dr. Dodge is the primary mentor for the postdoctoral training. She will work with to design more detailed K00 research plans and monitor the progress. Dr. Dodge will also provide her extensive experience in biostatistics and conducting intervention research with socially isolated older adults. Dr. Kaye will serve as a clinical AD mentor and guide me to explore future directions in technology-facilitated ADRD intervention research. Dr. Silbert will be the neuroimaging methodology expert in the sponsor team and work with me to develop knowledge and skills for examining brain biomarker outcomes in intervention research. Dr. Palinkas will guide the mixed-methods design planning and work with me to formulate an analytic plan for focus group discussions. All six sponsors will formulate an expert panel to consult on the scientific rigor, contribution to theory and practice, and innovation of the planned secondary data analysis and intervention development research.

SELECTION OF SPONSOR & INSTITUTION

Dr. Iris Chi is the Frances Wu Chair Professor at USC SSW. She has over 30 years of research experience in gerontology and social work. Dr. Chi has led and consulted seven cross-national studies in aging research. Dr. Chi has mentored Ms. Yu for the past four years and serves as a co-chair for Ms. Yu's dissertation committee. Dr. Chi is selected as the primary sponsor of this application because of her extensive research experience with the older adult population. Dr. Chi has been working closely with community-based organizations to design and implement interventions. Dr. Chi will continue to contribute to Ms. Yu's professional development through weekly meetings, providing guidance in conducting responsible research, theoretical interpretations of empirical findings, and intervention development and adaptation.

Dr. Hiroko Dodge is the Professor of Neurology at OHSU. She has been successfully directing the UM and OHSU Data Management and Biostatistics Core over the past ten years. She has previously mentored 17 PhD and postdoctoral fellows and four career development awardees (K01/K23/K25 mechanisms). Dr. Dodge and Ms. Yu have established mentoring relationships in the past two years by developing and resubmitting this F99/K00 application. Dr. Dodge will be the primary mentor in the postdoctoral stage, providing training in developing clinical trial designs, recruiting underserved populations, selecting clinical trial outcomes, clinical trial enrichment, and analyses of high-frequency digital biomarker longitudinal data. She will assist Ms. Yu in familiarizing herself with dementia research, especially in the area of dementia and social isolation.

Dr. Jeffrey Kaye is the Layton Endowed Professor of Neurology and Biomedical Engineering at OHSU. He serves as the PI for the Collaborative Aging Research Using Technology (CART) initiative and directs the Oregon Layton Aging and Alzheimer's Disease Center (OADC). Dr. Kaye's history as a mentor spans 25 years, including 27 previous PhD students and postdoctoral fellows, and seven previous Mentored Career Development Awardees (KL2, K01, K23, K25m, and VA career development mechanisms). Dr. Kaye will guide Ms. Yu to build empirical knowledge with data from the CART initiative and use technological infrastructure built in the CART homes to design a behavioral health intervention. As an experienced physician, Dr. Kaye will orient Ms. Yu to identify and empirically examine the clinical challenges an older adult at risk of ADRD encounters and address them in the proposed behavioral intervention.

Dr. Lawrence Palinkas is the Albert G. and Frances Lomas Feldman Professor of Social Policy and Health at USC. He has received continued support from NIDA, NIMH, and NICHD. Dr. Palinkas has published broadly on innovative mixed-methods and qualitative research design to address the health needs of older adults. Dr. Palinkas's recent research focuses on the implementation process of evidence-based interventions. He will assist Ms. Yu to conduct quality mixed-methods research that could inform the development of a community-based intervention for social isolation and cognitive health. He will also provide consultation on the implementation considerations in the intervention program design.

Dr. Lisa Silbert is a Professor in the Department of Neurology at OHSU and the Neuroimaging Core Director at the OADC. She serves as the director of the dementia clinic at the VA Portland Health Care System. Dr. Silbert has unique expertise in the neurological evaluation and cognitive assessment of older individuals with Alzheimer’s Disease and Related Dementias (ADRD). As an expert in neuroimaging methodologies, Dr. Silbert will work with Ms. Yu to gain knowledge and skills to collaborate with neuroimaging specialists to examine biological outcomes in epidemiological and intervention research projects.

Dr. Shinyi Wu is an Associate Professor of Social Work and Industrial & System Engineering at USC. Her research has focused on improving system efficiency in delivering health care and reducing health disparities. Dr. Wu has served as Ms. Yu's co-mentor and the co-chair of her dissertation. In the past four years, Dr. Wu has worked with Ms. Yu to examine the feasibility and effectiveness of harnessing mobile technology to promote chronic disease self-management. Dr. Wu is selected as a co-sponsor because of her research expertise in technology delivered health interventions. In the F99 phase, Dr. Wu will directly work with Ms. Yu to continue to examine the effects of ICT use on social well-being and cognitive health.

USC is the primary site for Ms. Yu's F99 phase research and training plans. OHSU will be the prospect training sites in the K00 phase. USC is a prominent academic and research university that has been training leading researchers. The resources at USC SSW will fully support Ms. Yu's training. Ms. Yu is a student member of the USC Edward R. Roybal Institute on Aging and the USC ADRC, where she can access frontier research and network with investigators of similar interests. Ms. Yu pursues training opportunities at the university level, including coursework at the Keck School of Medicine and Davis School of Gerontology. OHSU will support the applicant's research and training agenda in the postdoctoral stage. The OHSU School of Neurology specializes in using technology-facilitated approaches to address cognitive decline in older people. OADC provides research methods training, grant writing workshops, and pilot funding opportunities to help Ms. Yu formulate future research agendas. Ms. Yu's research interests match well with the research emphasis at OADC.

TRAINING IN RESPONSIBLE CONDUCT OF RESEARCH

The USC Suzanne Dworak-Peck School of Social Work (USC SSW) provides instruction for the responsible conduct of research in several venues: (a) research methods courses; (b) professoriate workshops and scholarly lectures; (c) the Hamovitch Center for Science in the Human Services; and (d) IRB research training courses.

Research methods courses. Format: Courses are designed for small in-person group discussions (7 or 8 students per course). Subject matter: Courses cover a wide range of topics, including ethical research conduct, conflict of interest, protecting human subjects, community-based and collaborative research practices, peer review, publication ethics and authorship, responsibilities to humanity, avoiding and handling misconduct, and ethical and unethical study designs. Faculty participation: Tenure-track faculty members instruct all courses. Duration of instruction: Courses meet weekly for 3 to 4 hours for the duration of the semester. Frequency of instruction: The PI successfully completed three research methods courses in fall 2017, spring 2018, and fall 2018 (see biosketch for reference).

Professional development workshops and scholarly lectures. Format: Workshops and lectures are designed for face-to-face discussion among doctoral students. Subject matter: Professional development workshops and scholarly lectures cover topics relevant to becoming a successful researcher: responsible and ethical conduct in publishing, authorship, peer reviews, data collection and analysis, study design, grant proposals, use of financial awards, and managing collaborative research teams. Faculty participation: Senior and tenure-track research doctoral faculty members, in partnership with research professionals at the Hamovitch Center, conduct the workshops and lectures. Duration of instruction: Workshops and lectures range from 1 to 2 hours and are scheduled monthly throughout the academic year. Attendance is mandatory for all doctoral students. Frequency of instruction: The PI has successfully completed 24 professional development workshops and scholarly lectures (2017 to present) and plans to continue to attend the workshops and scholarly lectures.

Hamovitch Center for Science in the Human Services. (see Facilities and Other Resources document for more information). Format: scholarly workshops, lectures, and individual consultation. Subject matter: The Hamovitch Center specializes in all aspects of conducting research and provides workshops, lectures, and individual consultation from full-time professionals with expertise in conducting responsible and quality research. Faculty participation: Full-time senior research faculty members and research professionals are available for USC faculty members and doctoral students engaged in research. Duration of instruction: Workshops and lectures are scheduled semiquarterly throughout the academic year and individual consultation is available as needed. Frequency of instruction: The PI has a working relationship with several full-time professionals at the Hamovitch Center, engaging in activities that include data analysis consultation, manuscript writing, study design, and grant proposals.

IRB courses. Format: Online training courses. Subject matter: Courses completed are (December 2016): Human Subjects; Good clinical practice; Conflicts of Interest, and Health Insurance Portability and Accountability Act (HIPAA) Privacy Education. Faculty participation: All sponsor/cosponsors have completed these trainings. Duration of instruction: These IRB course trainings are required for all proposed studies at USC, as are periodic refresher courses. Frequency of instruction: Since initial completion of the IRB courses, the PI has taken refresher training of Human Subjects and Good clinical practice courses in December 2019. The upcoming refresher training is scheduled in December 2022.

Besides the trainings made available at USC SSW, the PI has engaged in the Digital Scholar Training Program provided by USC Clinical and Translational Science Institute (CTSI), where a series of responsible conduct of research topics are discussed.

CTSI Digital Scholar Training Program. Format: Online training courses. Subject matter: This program helps researchers who are involved in the health sciences to develop competency in digital research practices. Subjects covered including selecting right journal for publication, collaboration with industry, and participant recruitment using non-traditional approaches etc. Faculty participation: tenured faculty members and senior scientists are invited to give webinar presentations. Duration of instruction: Online courses typically last for an hour with Q&A sessions at the end. Frequency of instruction: The CTSI digital scholar training program schedules online courses every month. The PI has signed up for the notifications for webinars. She has attended about 9 webinars and plan to continue to attend trainings that suit her learning needs.

SPONSOR AND CO-SPONSOR STATEMENTS

A. Research Support Available

Chi	
Dodge	
NIA R01 AG051628	Conversational Engagement as a Means to Delay Alzheimer's Disease Onset: Phase II (Dodge, PI), 09/01/16 – 05/31/21
NIA R01 AG056102	Web-enabled social interaction to delay cognitive decline among seniors with MCI: Phase I (Dodge, PI), 07/01/17 – 03/31/22
NIA U2C AG054397	ORCATECH Collaborative Aging (in Place) Research Using Technology (CART) (Kaye, PI), 09/01/16 - 04/31/21
Kaye	
NIA U2C AG054397	ORCATECH Collaborative Aging (in Place) Research Using Technology (CART) (Kaye, PI), 09/01/16 - 04/31/21
NIA P30 AG024978	Oregon Roybal Center for Care Support Translational Research Advantaged by Integrating Technology (Kaye, PI) 09/04 - 07/24
NIA P30 AG008017	Oregon Alzheimer Disease Research Center (Kaye, PI) 07/90 - 03/25
NIA R01 AG051628	Conversational Engagement as a Means to Delay Alzheimer's Disease Onset: Phase II (Dodge, PI), 09/16 – 05/21
NIA R01 AG056102	Web-enabled social interaction to delay cognitive decline among seniors with MCI: Phase I (Dodge, PI), 07/17 – 03/22
NIA U01 AG10483	Alzheimer's Disease Cooperative Study (UCSD subcontract) (Feldman, PI) 09/91 - 06/21
NIA U01 AG016976	National Alzheimer's Coordinating Center (University of Washington subcontract) (Kukull, PI) 07/99 - 06/21
Palinkas	
NIDA P30 DA027828	Center for Prevention Implementation Methods for Drug Abuse and Sex Risk Behavior (Brown, PI), 9/1/16 –6/30/21
NIMH P50 MH113662-01A1	Accelerator Strategies for States to Improve System Transformations Affecting Children, Youth and Families (Hoagwood, PI), 9/21/19 – 4/30/23
NICHD R01 HD092489-01A1	Dissemination and implementation of the Safe Environment for Every Kid (SEEK) Model for Preventing Child Maltreatment. (Dubowitz, PI), 4/1/19 – 8/31/23
Silbert	
Department of Veterans Affairs	HSR&D COVID-19 Supplement (Silbert, PI) 06/01/20 – 02/28/21
NIA R01 AG056712	White Hyperintensity-Associated Astrocytopathy In Alzheimers Disease And Vascular Cognitive Impairment A Targeted Histopathologic Study Using Postmortem 7T MRI (Silbert/Woltjer, PI), 04/01/18 – 01/31/23
NIA P30 AG066518	Oregon Alzheimer Disease Research Center Neuroimaging Core (Silbert, PI), 9/1/16 –6/30/21
NIA U2C AG054397	ORCATECH Collaborative Aging (in Place) Research Using Technology (CART) (Kaye, PI), 09/01/16 - 04/31/21
NIA R01 AG051628	Conversational Engagement as a Means to Delay Alzheimer's Disease Onset: Phase II (Dodge, PI), 09/16 – 05/21
NIA R01 AG056102	Web-enabled social interaction to delay cognitive decline among seniors with MCI: Phase I (Dodge, PI), 07/17 – 03/22
Wu	

[redacted]

[redacted]	[redacted]
[redacted]	[redacted]

B. Sponsor's/co-sponsor's Previous Fellows/Trainees

Chi – has mentored 32 doctoral students and postdoc fellows in total		
Xinming Song	1991-1995	Professor, Peking University
Vivian Lou	1993-1999	Associate Professor, University of Hong Kong
Angela Leung	2000-2007	Associate Professor, Hong Kong Polytechnic University
Man Guo	2005-2011	Associate Professor, University of Iowa
Ling Xu	2007-2012	Assistant Professor, University of Texas, Arlington
Dodge – has mentored 17 doctoral students and postdoc fellows in total		
Adriana Seelye	2014-2017	Assistant Professor, Minnesota VA hospital
Meysam Asgari	2014-2016	Assistant Professor, Oregon Health and Science University
Junko Nishihira	2014-2016	Assistant Professor, Ryukyu University
Jiayu Zhou	2016-2018	Assistant Professor, Michigan State University
Ming Li	2017-2018	Researcher, Alibaba Seattle
Kaye – has mentored 27 previous PhD students and postdoctoral fellows in total		
David Salat	1998-2000	Associate Professor, Radiology, Harvard Medical School
Neil Thomas	2017-2019	Assistant Professor, University of Ottawa, Canada
David Bissig	2018-2020	Assistant Professor, UC Davis
Silbert – has mentored 6 previous PhD students and postdoctoral fellows in total		
Michelle Cameron	2005-2008	Assistant Professor, OHSU
Erin Boespflug	2013-2018	Assistant Professor, OHSU
Juan Piantino	2013-2018	Post-doc in fMRI and Research Associate, OADC
Palinkas – has mentored 70 doctoral students and postdoc fellows in total		
Maxwell Davis	2007-2008	Director of the CALSWEC Mental Health Program, University of California, Berkeley
Ian Holloway	2010–2012	Associate Professor, University of California, Los Angeles
Kim Brimhall	2016-2017	Assistant Professor, Binghamton University
Wu – has mentored 15 doctoral students and postdoc fellows in total		
Irene Vidyanti	2009-2014	Data Scientist, Los Angeles County Chief Information Office
Magaly Ramirez	2011-2016	Assistant Professor, University of Washington
Haomiao Jin	2012-2016	Research Assistant Professor, University of Southern California

C. Training Plan, Environment, Research Facilities

In the F99 stage, Ms. Yu proposes to examine the longitudinal influence of physical and social living environments on social isolation and cognitive health among older adults with National Health & Aging Trends (NHATS) data. The proposed dissertation will contextualize the role of ICT use by examining the interaction effects of the physical social environments with ICT use. An ecological perspective has rarely been adopted to studying the intertwined relationships between social wellbeing and cognitive health. With the population aging rate increasing, Ms. Yu's research will enhance the research field's understanding of the ecological support system on social wellbeing and cognitive health. In the K00 stage, Ms. Yu proposes developing and pilot testing a community-based intervention that can provide a sustainable solution to keep the older adults socially connected and prevent the onset of dementia. Her work will be meaningful to address the challenge of social isolation and the burden of care brought by the increasingly prevalent ADRD. Ms. Yu's training plan incorporates coursework, seminars, and mentored clinical internship opportunities provided by sponsors with diverse and complementary skillsets. The proposed research and training agenda works together to advance Ms. Yu's research skills, scholarship, and help her become an independent investigator.

With the receipt of this award, Ms. Yu will have the opportunity to receive training from an outstanding mentoring team made up of leaders in the fields of Neurology, Biostatistics, Engineering, Anthropology, and Social Work. Each of the sponsors has influential research works committed to improving the quality of lives for the older adult population. The sponsor team will collaborate to help Ms. Yu achieve her long-term career goal of becoming an independent social scientist and an impactful intervention researcher dedicated to improving social connectedness and preventing cognitive decline for the older adult population.

Depending on her over 30 years of experience in conducting aging research, **Dr. Chi** will provide content and professional development expertise on developing or adapting interventions in collaboration with the target age group and community-based organizations, scientific writing, and navigating the peer-review process. **Dr. Wu** has expertise in employing technology as an intervention tool for the older adult population. In the F99 phase, Dr. Wu will directly work with Ms. Yu to examine the effects of ICT use on social wellbeing and cognitive health. After Ms. Yu graduates from USC and start the K00 phase of research, Dr. Wu will consult with Ms. Yu in using technology-facilitated tools to improve health outcomes and increase access to health care for the populations in need. Ms. Yu will work with **Dr. Dodge** to increase her ADRD research knowledge and using I-CONECT program as an example to gain hands-on research experience in conducting and evaluating clinical trials. **Dr. Silbert** is an expert in MRI methodologies and the director of the neuroimaging core at OADC. She will help Ms. Yu to grow her capacity of neuroimaging methodology and using digital biomarkers in her research. **Dr. Kaye** will collaborate with Ms. Yu to explore the potential to incorporate ambient technology and wearable devices to design an ecological perspective informed behavioral health intervention. As an experienced clinician, Dr. Kaye will serve as Ms. Yu's K00 stage clinical internship at a memory clinic. Ms. Yu will work with **Dr. Palinkas** to refine her mixed-methods research skills and receive consultation on designing and implementing community-based health interventions. Ms. Yu will apply the skills developed from the individualized training plan to write her dissertation, seek to publish in high-impact journals (e.g., The Gerontologist, Journal of Gerontology, Journal of Alzheimer's Disease), and develop a research agenda beyond the postdoctoral training stage than leads to her career success.

Dr. Chi serves as the primary sponsor of Ms. Yu's F99/K00 application and faculty advisor at predoctoral stage. She will be the coordinator for communication among the co-sponsors at USC (Chi, Wu, Palinkas). Dr. Dodge will be the primary advisor in the postdoctoral stage, she will help communicate and coordinator mentoring effort with sponsors at OHSU (Dodge, Silbert, Kaye). Ms. Yu will also take initiatives to solicit quarterly meetings with all sponsors to consult on the development and implementation of the proposed behavioral health intervention and check in to achieve the following listed training goals.

<u>Training Goal 1</u>: **Ms. Yu will increase knowledge in prevention, clinical practice, intervention, and policy pertain cognitive decline in normal aging and the development of ADRD.** Dr. Wu is currently conducting research with older adults with ADRD and their family caregivers. She has connected Ms. Yu with researchers with similar interests at the USC Alzheimer's Disease Research Center (ADRC). Ms. Yu will work with Dr. Wu, Dr. Chi, Dr. Dodge, and Dr. Silbert to develop a literature review database on cognitive health and ADRD research. We will collaborate with Ms. Yu to conduct a systematic review of the pathways that could shed light on the association between social isolation and cognitive decline. Since Ms. Yu is new to the field of cognitive health and dementia research, the Layton Aging and Alzheimer's Disease Center at Oregon Health and Science University will be an outstanding training resource for her. Dr. Dodge will direct Ms. Yu to different cores of the Layton center as her research projects require, such as the cores of outreach, recruitment and education, neuroimaging, and data management and analysis. The systematic review and involvement at both USC and Oregon ADRC will formulate an intellectual community and academic support system that will benefit Ms. Yu's professional development and help her achieve training goal 1.

Dr. Dodge is an expert in pharmacological and non-pharmacological intervention strategies urgently needed to delay the onset of dementia and reduce its societal burden. She has contributed to the knowledge of distinguishing the normal cognitive aging and pathological cognitive decline. Dr. Dodge is the Principal Investigator of Internet-based Conversational Engagement Clinical Trial (I-CONECT, NIA R01AG056102, R01AG056628 & R56AG051628, ClinicalTrials.gov Identifier: NCT02871921), examining the effect of technology-facilitated conversation engagement on delaying cognitive decline. Dr. Dodge has primary responsibility for leading and overseeing the I-CONECT trial. She is the ideal senior scientist in Neurology to work with Ms. Yu to develop her knowledge base in cognitive health and ADRD through hands-on research experiences and directed learning. Dr. Dodge and Ms. Yu will collaborate on drafting and publishing manuscripts using I-CONECT data. I-CONECT will serve as an example trial study for Ms. Yu to learn about intervention study design, implementation, and evaluation. Dr. Dodge will work with Ms. Yu to prepare to apply for a pilot grant at OHSU to conduct qualitative focus groups to understand the older adults' perspective on the design of a behavioral health intervention that addresses social isolation and cognitive decline. At the F99 phase, Ms. Yu will meet with Dr. Dodge biweekly. Ms. Yu will move to Portland in the K00 stage and meet with Dr. Dodge weekly. Ms. Yu will have access to Dr. Dodge with any substantial and methodological questions relevant to ADRD prevention, intervention, and policy research. As the data core director at OADC, Dr. Dodge will connect Ms. Yu with scholars at Layton Aging and Alzheimer's Disease Center at Oregon Health and Science University (OADC). Ms. Yu will attend the monthly OADC investigator meetings using Zoom at her F99 phase, and in-person at the K00 phase. At these investigator meetings, Ms. Yu will attend the research grant presentation by leading researchers in ADRD related domains.

Ms. Yu will gain experience in the clinical world of ADRC by having two mentored internship experiences at USC and OHSU separately. Dr. John Ringman (collaborator) will serve as the internship supervisor in the F99 phase (see letter of support), and co-sponsor Dr. Kaye will serve as the clinical internship mentor in the first year of the K00 phase. Ms. Yu has selected courses that suit her learning needs. In the first year of F99, She will take a cognitive decline and ADRD special topic course at USC Leonard Davis School of Gerontology. The course is designed to cover the normal and abnormal brain and cognitive function changes in the aging process. Ms. Yu will learn from the course approaches of prevention, early detection, and management of ADRD. She will also take a course on the methodology of systematic review and meta-analysis, which will equip her to conduct rigorous systematic reviews that could inform intervention development. Ms. Yu will expand her academic network by submitting abstracts and presenting in cognitive health-related conferences, such as the Alzheimer's Association International Conference (AAIC), Gerontological Association of America annual meeting (GSA), and Clinical Trials in Alzheimer's Disease Conference (CTAD).

Training Goal 2: Ms. Yu will gain methodological skills for investigating the underlying pathways of how social isolation influences cognitive health in diverse environments. Conducting the longitudinal analysis with NHATS data in the proposed dissertation work will exert Ms. Yu's existing strength in SEM and MLM modeling and use complex datasets to contribute to the literature. Ms. Yu will further cultivate efficiency in statistical analysis by taking a course on R programming in the F99 phase. Gaining R programming skills will assist Ms. Yu to better collaborate with large scale datasets and the I-CONECT and CART research teams. Prior to the postdoctoral phase, Ms. Yu will start to explore Neuroimaging outcomes and biomarkers of brain health working with collaborator Dr. Judy Pa at USC Keck School of Medicine (See letter of support). Getting started with cognitive health research using biological outcomes will help Ms. Yu transition from a researcher trained in social work to a versatile ADRD researcher with unique expertise. Dr. Dodge has a strong background in statistical methods. She will supervise Ms. Yu to conduct research with I-CONECT data to evaluate the intervention effects and to explore behavioral pathways that could provide causal implications on the association between social isolation and cognitive declines. Ms. Yu currently does not have experience in using biomarker outcomes in study projects with social isolation and cognitive health-related research. She will attend OADC imaging core weekly lab meeting lead by Dr. Silbert, where studies using biomarkers and neuroimaging methods will be discussed. She will attend Ms. Yu will collaborate with Dr. Dodge and Dr. Silbert to gain hands-on experience in analyzing biomarker outcomes with the I-CONECT data. Dr. Dodge will also connect Ms. Yu with other researchers at OADC with specialties in neuroimaging methods and analyzing digital biomarkers as clinical intervention outcomes.

Training Goal 3: Ms. Yu will explore future directions in technology and aging research. Technology is an inclusive term that describes various tools that can potentially be utilized to improve older adults' quality of life, including mobile apps, ambient technology, wearable devices, and virtual reality techniques, etc. With the sponsor team and collaborators' support, Ms. Yu will explore future directions in technology and aging research and uses these tools in her future intervention program. She has been a critical contributor to Dr. Chi and Dr. Wu's intervention studies using mobile technology as intervention delivery tools to promote self-care among T2D patients and immigrant caregivers of older adults. With the support of the F99/K00 grant, she will continue to work with Drs. Chi and Wu to develop health behavior and ecological perspective informed mHealth interventions. Ms. Yu will observe collaborator Dr. Pa's intervention sessions using virtual reality technology and collaborate on manuscripts. In the K00 phase, Dr. Kaye will meet with Ms. Yu biweekly to discuss investigation plan with the CART initiative, using digital biomarker data collected with ambient sensors and wearable devices. Working on the I-CONECT project will also serve as an avenue for Ms. Yu to explore future directions in employing technology in intervention research. These collaborations will build knowledge on innovative technology use in aging research and produce publications on high impact journals.

Training Goal 4: **Ms. Yu will receive advanced training on design, implementation, and evaluation of community-based interventions.** Ms. Yu has a clear vision of reducing social isolation and delay the onset of dementia with community-based intervention research. Ms. Yu has learned how to build collaborative relationships with community partners, participated in recruitment, and data collection by serving as a co-investigator in the pilot study effort employing a co-design approach to develop a caregiver self-management program (CSMP) app for Chinese caregivers. She has further advanced her experience in community-based research by working with Dr. Wu and Dr. Chi on pilot studies examining the acceptability and feasibility of IMTOP intervention among recent immigrants in the United States and for low-income older residents in affordable retirement communities. With Dr. Chi's experiences in intervention and behavioral science research, she will continue to guide Ms. Yu to conceptualize her intervention study and provide constructive feedback to her theoretical reasoning and design of the intervention. Dr. Wu and Dr. Chi will discuss with Ms. Yu in the weekly mentoring meetings the practical, intervention, and policy interventions for her findings of the longitudinal analysis using NHATS data. Dr. Wu has served as Principle Investigator for several intervention studies examining the effectiveness of employing mobile technology as enabling tools for improving the health outcomes of older adults. Dr. Wu is uniquely qualified to guide Ms. Yu to make sense of the analytical results of survey data related to ICT use in the older adult population and help her to translate the empirical findings to intervention mechanisms. Dr. Chi will guide Kexin to interpret the hypothesized longitudinal association between physical and social environments and social isolation with her extensive research experiences on researching with the older adult population and their support networks.

Dr. Palinkas is a leader in the mixed-methods research methodology. His seminal paper on the new development in purposeful sampling for mixed-methods implementation research has been cited for over 2,500 times. Ms. Yu proposes to adopt a mixed-methods approach for the development of a behavioral health intervention to address social isolation and cognitive decline. Dr. Palinkas is the ideal mentor to work with Ms. Yu to formulate a purposeful sampling, data collection, and analysis plan and work with Ms. Yu to prepare manuscripts reporting mixed-methods study findings. Dr. Palinkas is also an expert in implementation and dissemination science. Ms. Yu will consult Dr. Palinkas on the process of intervention implementation and develop a draft of the intervention research proposal with his guidance. Dr. Palinkas will have biweekly consultation meetings with Ms. Yu. She will be able to receive individualized training on methodological and substantive topics.

With the recipient of the award, Ms. Yu will have the opportunity to receive immersion training with I-CONECT fieldwork in the K00 phase. She will take an online course provided by UCSF on study design for intervention research in real-world settings. The course fits Ms. Yu's training goal particularly well, given her interests in conducting community-based intervention research from the social ecology perspective. Ms. Yu will also apply for the National Institute of Neurological Disorders and Stroke (NINDS) supported clinical trial methodology course (R25NS088248) to receive two years of continuous training with both online and in-person formats. This course is designed to help junior investigators develop scientifically rigorous, yet practical clinical trial protocols, and focus on applying for funding as a critical trial planning activity. Taking this course will aid Ms. Yu to grow both scholarly in knowledge and professional in grant preparation.

The six sponsors formulate an experienced and committed sponsor/co-sponsor team to contribute to Ms. Yu's professional development and help her achieve her long-term career goal of becoming an independent scientist and impactful intervention researcher. As the primary sponsor of this F99/K00 proposal, Dr. Chi will be the coordinator for the communication between co-sponsors. Ms. Yu will take the initiative to request a meeting with all sponsors and co-sponsors in the process of conducting the proposed research activities. All sponsors and co-sponsors will meet in-person once in a year at major scientific meetings, such as the GSA annual conference.

Environment and Research Facilities

Ms. Yu has been a student of Dr. Palinkas, a mentee of Dr. Wu and Dr. Chi for the past four years. Ms. Yu has established a collaborative relationship with Dr. Dodge through the development of this F99/K00 application. This recently established mentoring relationship has already been proven extremely valuable to Ms. Yu. Dr. Dodge has been very instrumental in helping Ms. Yu refining her thinking in methodology and training plans related to her research at both dissertation and postdoctoral stages. Drs. Silbert adds to the clinical and neurological research methodology expertise to the sponsor team. She will supervise Kexin to access training and research resources at the OADC neuroimaging core. Dr. Kaye is the director of the Layton Aging and Alzheimer's Disease Center at OHSU. He will help Ms. Yu to gain

The Suzanne Dworak Peck School of Social Work is among the top schools of social work in the nation according to the U.S. News & World Report and has a vast amount of available resources. Ms. Yu also has access to the school's research center, the Hamovitch Center for Science in the Human Services; the USC Edward R. Roybal Institute on Aging; the USC ADRC. The Layton Aging and Alzheimer's Disease Center at OHSU will be an outstanding training resource for Ms. Yu. Dr. Dodge will direct Ms. Yu to different cores of OADC as her research projects require, such as the cores of outreach, recruitment and education, neuroimaging, and data management and analysis. Details regarding these resources are thoroughly described in the Facilities and Other Resources and Description of Institutional Environment and Commitment to Training sections of the proposal.

D. Number of Fellows/Trainees to be Supervised During the Fellowship

Besides working with Ms. Yu, Dr. Chi will be co-mentoring Ruotong Liu and Yi-Hsien Tung with Dr. Wu during the period of the award. Dr. Wu will also be independently supervising Anthony Nguyen, a PhD student in

Industrial and Systematic Engineering. Because Dr. Wu and Dr. Chi work closely on research projects and share responsibilities of mentoring PhD students, they will have sufficient time and resources to support Ms. Yu's dissertation and postdoctoral research. Dr. Dodge will be mentoring three other postdoctoral fellows, Patrick Pruitt, Chao-Yi Wu, and WanTai (Michael) AuYeung, during the period of the F99/K00 award. Dr. Palinkas will be supervising Abigail Palmer- Molina, a recent recipient of F31. Dr. Silbert will not supervising doctoral and postdoctoral level trainees, she will serve as a mentor to two Assistant Professors at OHSU. Besides supporting Ms. Yu, Dr. Kaye will supervise two postdoctoral fellows, Clara Berridge, and Lyndsey Miller. The three co-sponsors based at OHSU collaborate on grant projects (e.g., CART and I-CONECT), which forms a natural foundation for them to co-mentor Ms. Yu in her postdoctoral research and training.

E. Applicant's qualifications and Potential for a Research Career

Since she joined the MSW/ PhD program at USC, Ms. Yu has been an integral part of the research team and has developed to become an emerging scholar in the scientific community. Ms. Yu's critical thinking, understanding of the literature, methodological skills are impressive. In the first two years of her training, Ms. Yu took MSW courses, worked as a social work intern while receiving doctoral training. Her class performance was excellent, as reflected in her grade report. Furthermore, Ms. Yu lead the primary data collection of the IMTOP pilot study. She trained and supervised student workers to recruit participants at community-based organizations. The pilot study provides essential data for adapting the IMTOP intervention to the recent and first-generation immigrants with Type 2 diabetes in the greater Los Angeles. For the past four years of intensive training in the dual degree program, Ms. Yu not only gained her MSW degree but also had 17 conference presentations and published three first-authored manuscripts in high impact journals. Ms. Yu has demonstrated the ability to thrive despite the stress. Her publishing record shows an increase in research productivity over time. The sponsors believe the existing training and research experiences provide a foundation for the unique, novel, and highly valuable research she proposes in her dissertation and onward.

Ms. Yu has been actively engaged in grant proposal preparation, survey and qualitative interview guide design, and data collection with caregivers and older adults with disadvantaged backgrounds. Ms. Yu's first-authored work using the IMTOP study data was recognized as the winner of the 2018 APHA Nobuo Maeda International Research Award. Ms. Yu worked with low-income retirement communities to evaluate the needs of a mHealth intervention within this specific living environment. This research experience sparked Ms. Yu's passion for studying the physical and social environments on social isolation and cognitive health as she observed the impacts of living environments on social and health outcomes. Furthermore, Ms. Yu has been actively engaging in the co-design CSMP app pilot study with Chinese immigrant caregivers. She contributed significantly to the study design and data collection using semi-structured interviews. Ms. Yu is currently leading the qualitative data analysis and manuscript preparation for the co-design CSMP app study.

Ms. Yu has pursued her passion in addressing social isolation among older people using large scale, nationally representative longitudinal datasets, such as Health and Retirement Study and National Health & Aging Trends Study (NHATS). Her first-authored paper examining the longitudinal association between internet use and perceived loneliness in older adults was published in the Journal of Gerontology: Series B. Ms. Yu passed her qualifying exam and advanced to candidacy in September of 2019. In the process of completing the proposed project, Ms. Yu demonstrated outstanding theoretical thinking, independence in learning, and advanced analytical skills. Her qualifying exam project produced a quality manuscript published in the Journal of the American Medical Directors Association (JAMDA).

We strongly attest to Ms. Yu's promise and preparedness to undertake the course of training and research proposed in her application. We will continue to mentor and collaborate with her throughout the period of the award and will use our experiences and expertise to help her in developing her long-term career goal of addressing the largely unstudied issue of living conditions on social isolation and the role of use of technology among older adults. In our specialized training, Ms. Yu will use findings from her project to develop technology assisted interventions to defend the negative impact of social isolation on cognitive impairments for older persons. In summary, Kexin is an enthusiastic social scientist who is capable and self-motivated. She produces high-quality research output. Her good interpersonal skills will path the way for her future success. The support and training she would receive through this F99/K00 award will, we firmly believe, ensure her trail toward becoming an accomplished independent investigator and scientist. We enthusiastically offer our strongest possible recommendation for granting the fellowship award to Ms. Kexin Yu. Ms. Yu proactively seeks grant opportunities at a very early stage of her career. She will be continuously working with an extraordinary sponsor team to refine her thinking, develop thoughtful and innovative research proposals, and secure support for her future independent research. We have every confidence that Ms. Yu is going to succeed as an independent researcher.

Keck School
of Medicine
of USC

DEPARTMENT OF NEUROLOGY

October 13, 2020

Dear F99/K00 Award Review Committee,

It's my pleasure to write this letter in support of Ms. Yu's F99/K00 grant application. I am John Ringman, Professor of Clinical Neurology at the Memory and Aging Center at the Keck School of Medicine of USC. I regularly see patients with cognitive or behavioral problems referable to neurological and, in particular, neurodegenerative disease. Over the years I have engaged in diverse research activities devoted to Alzheimer's disease (AD) and other neurobehavioral disorders and I have an active NIH-funded research agenda regarding highly genetic forms of AD and dementia among Latinos. I am happy to serve as a collaborator to Ms. Yu's research during the tenure of the F99/K00 award.

I got to know Ms. Yu in the past year initially from her talk on her F99/K00 grant proposal to the USC Alzheimer's Disease Research Center (ADRC) research meeting. Ms. Yu presented herself as a promising young scholar who is dedicated to addressing ADRD by alleviating social isolation and improving older adults' living environment. She brings a fresh perspective to ADRD research with her social work training background, emphasis on social justice, and research experience with older adults with diverse backgrounds.

I am delighted to support Ms. Yu's professional development. Ms. Yu will shadow my clinical practice with older adults in biweekly clinical internship sessions in our Memory Disorders clinic. This internship experience will help Ms. Yu understand the current screening, diagnosis, and clinical guidelines pertaining to ADRD. As a social work Ph.D. candidate, she is uniquely qualified to examine the patients' unmet needs and challenges within the social contexts. I firmly believe that gaining clinical internship experience will help Ms. Yu identify intervention opportunities and develop programs with significant practical implications.

I have reviewed Ms. Yu's research and training plans. The proposed dissertation and postdoctoral research agenda will provide an excellent opportunity to prepare her to become an independent ADRD researcher. I look forward to collaborating with her and witness her professional achievements with time.

Sincerely,

[redacted]

John M. Ringman, M.D., M.S.
Helene and Lou Galen Professor of Clinical Neurology
Department of Neurology
Keck School of Medicine at USC
[redacted]

[redacted] Academic office
[redacted] Clinical matters
[redacted]

USC Suzanne Dworak-Peck School of Social Work

Ph.D. Program

Michael Hurlburt, Ph.D.
Associate Professor
Director of Doctoral Programs and Chair of the PhD Program
Suzanne Dworak-Peck School of Social Work
University of Southern California (USC)

October 16, 2020

M E M O R A N D U M

From: Michael Hurlburt, Ph.D. [redacted]
Associate Professor
Director of Doctoral Programs and Chair of the Ph.D. Program
Suzanne Dworak-Peck School of Social Work
University of Southern California (USC)

RE: **Kexin Yu, PhD Candidate at the University of Southern California, Suzanne Dworak-Peck School of Social Work**

DESCRIPTION OF INSTITUTIONAL ENVIRONMENT & COMMITMENT TO TRAINING

Dear Sir/Madame,

Social Work Ph.D. Program. The USC Suzanne Dworak-Peck School of Social Work (USCSSW) established the first social work doctoral program in the western United States in 1953. Today we continue the tradition of admitting highly motivated, self-directed individuals interested in university research and teaching. Students pursue an in-depth, customized course of study in an atmosphere of careful mentoring and respect for scholarship. Our doctoral curriculum is highly interdisciplinary with the intent of producing graduates who are capable of original research and passionate about advancing the profession's knowledge base. Our program fosters early engagement in research and publication, together with a structured and sequential experience in classroom teaching. Unique developmental opportunities are offered to improve students' presentation skills, networking with other universities, and leadership potential. Our goal is to make students competitive for the best available positions in this country and elsewhere in this world.

Program Structure. Doctoral students at USCSSW are required to complete a minimum of 45 course units beyond the master's degree (exclusive of the Doctoral Dissertation). During the first year of the PhD program, students complete 8 courses (16 units) in social work policy practice and research and participate in faculty-directed research projects. Core Substantive Courses include: Theories of Human Behavior in the Context of the Social Environment; Explanatory Theories for Larger Social Systems; Theories for Practice with Small and Large Systems. Core Research/Statistics Courses include: Introduction to Social Work Statistics; Multiple Regression in Social Work Research; Advanced Social Work Research Methods. During the second year of the PhD Program, students are required to enroll in advanced multivariate statistics and at least three courses (per semester) in other departments or schools within the university. At least eight of these 12 units must be in courses with a substantive rather than a research methodology or statistics focus. Additional second year requirements include three student-directed tutorials (2-unit) under the mentorship of a social work faculty member. Tutorials provide greater understanding of the student's chosen specialty through a closer examination of relevant practice theories, explanatory theories, and research methodologies. Each tutorial is designed to produce a study of publishable quality. In the third year of the PhD Program, students must pass a qualifying examination in their research area, which consists of a written paper of publishable quality and includes an oral defense. After passing the qualifying exam, students are advanced to candidacy. Year Four and Beyond- Dissertation: Before proceeding with the dissertation, students must establish a dissertation committee approved by the Graduate School and submit a dissertation proposal for approval. Students submit their completed dissertation to their dissertation committee and orally defend it.

University of Southern California
[redacted]

Teaching. During their third and fourth year, doctoral students are required to teach one course per year. Teaching requirements may be fulfilled by co-teaching, teaching as an assistant or solo teaching. Before beginning these experiences, students must take a teaching course approved by the doctoral committee.

Average time to degree. The average time to degree over the past 10 years is 5.5 years. More than 95% of enrolled doctoral students in the school complete their doctoral studies and, during the last 5 years, 92% of students have accepted academic research positions immediate upon completion of doctoral studies.

Monitoring and evaluation. The USCSSW doctoral program requires an End of Semester Report from PhD students at the end of each semester for the entire duration of their doctoral studies. This report is completed by the student's faculty mentor and includes information on milestones and other deliverables (e.g., proposals, publications, presentations), as well as a description of the student's dissertation progress. This report is jointly signed by the student and mentor, and is submitted to the director of the doctoral program. The doctoral program also requires a Progress Report Timeline from PhD students at the end of each year of their doctoral studies. This Progress Report Timeline is used as a tool to guide and monitor each student's individual progress and ensures the PhD Program that each student is on track with tasks and milestones specific to each year of doctoral studies. This yearly report is completed by the PhD student's faculty mentor and is jointly signed by the student and faculty; this report is then submitted to the director of the doctoral program and to USC's graduate school department. These semester and yearly progress reports help doctoral students to navigate through the program productively and expeditiously.

Additional Program Opportunities and Commitment to Training. USCSSW and the University of Southern California are collectively committed to outstanding training, mentorship, and professional development for all of our doctoral students, including the applicant, Kexin Yu. Within USCSSW, the research clusters of excellence provide an environmental structure in which students have opportunities for discourse, collaboration, and professional training in the context of faculty, peers, and external collaborators who have broadly shared research interests. The research clusters regularly hold research seminars, which involve presentations and discussions of research by cluster members as well as by external speakers. Kexin will have opportunities through the research centers, particularly the USC Edward R. Roybal Institute on Aging, to present and discuss her own work. Students also have strong collaborative connections with one another, which is facilitated by availability of dedicated office space for doctoral students at the Hamovitch Research Center office in downtown Los Angeles, and in a large, computer-equipped doctoral training lab in the School of Social Work on the main university campus. The Graduate School at USC provides additional resources for doctoral students, including specialized training in grant writing skills and other areas of professional development, which Kexin will take advantage of as part of her training plan. Students also have opportunities to engage in intellectual activities around the university, related to social science disciplines that Kexin will participate in. Finally, faculty within the USCSSW have strong connections with external colleagues and collaborators. With these strong and enduring ties, the applicant will have an easy ability to work with faculty outside of the SSW and to participate in the intellectual opportunities present at USC's Davis School of Gerontology, Keck School of Medicine and USC Alzheimer's Disease Research Center, such as seminars, courses, and ongoing lab involvement, as described extensively in the applicant's training plan. At all levels (institutional, school, and individual faculty), there is a strong commitment to the applicant's proposed training.

Kexin's dissertation chair and primary sponsor is Dr. Iris Chi, Dr. Frances Wu Chair for Chinese Elderly Professor, and distinguished scholar who has extensive experience conducting research focused on addressing health and mental health needs of older adults. Dr. Chi has more than 30 years of experience mentoring PhD students. She has been elected as a Fellow of the Gerontological Society of America, and the American Academy of Social Work and Social Welfare. All co-sponsors on the proposed project also possess strong records of being NIH grant recipients and are enthusiastic about the intellectual contributions Kexin brings to their ongoing research activities.

Progress/status of the F99/K00 applicant in relation to the program's timeline. Kexin is in excellent academic standing with respect to the timeline set forth by the USCSSW PhD program and anticipates completing her dissertation on time within standard degree completion. She has already completed all coursework, passed her qualifying exam, and advanced to PhD Candidacy. Kexin has defended her dissertation proposal. At the time of award notification, which will coincide with her 5th year in the program, Kexin will also continue her work with her lead sponsor, Dr. Iris Chi, as well as other faculty members on projects that are designed to complement her research agenda and strengthen manuscript development, grant-writing, and grant management abilities.

As chair of the doctoral program, I am pleased to offer my full support for Kexin's F99/K00 application. Kexin has remarkable promise for a career trajectory in academic research and teaching, specifically in the area of dementia and social isolation intervention among older adults.

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001

Expiration Date: 02/28/2023

Use of Human Specimens and/or Data

Does any of the proposed research in the application involve human specimens and/or data *	❍ Yes ● No
Provide an explanation for any use of human specimens and/or data not considered to be human subjects research.	
Are Human Subjects Involved	● Yes ❍ No
Is the Project Exempt from Federal regulations?	● Yes ❍ No
Exemption Number	❑ 1 ☑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ☑ 8
Other Requested Information	

Human Subject Studies

Study#	Study Title	Clinical Trial?
1	Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology	No

Section 1 - Basic Information (Study 1)

OMB Number: 0925-0001

Expiration Date: 02/28/2023

1.1. Study Title *

Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology

1.2. Is this study exempt from Federal Regulations *	● Yes	❍ No

1.3. Exemption Number: ❑ 1 ☑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ☑ 8

1.4. Clinical Trial Questionnaire *

1.4.a. Does the study involve human participants?	● Yes	❍ No
1.4.b. Are the participants prospectively assigned to an intervention?	❍ Yes	● No
1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?	❍ Yes	● No
1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?	❍ Yes	● No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics (Study 1)

2.1. Conditions or Focus of Study

- Social isolation
- Cognitive functioning

2.2. Eligibility Criteria

Participants of the National Health & Aging Trends Study or pariticpants of the Internet-based Conversational Engagement Clinical Trial (I-CONECT)

2.3. Age Limits	Min Age: 65 Years	Max Age: N/A (No limit)
2.3.a. Inclusion of Individuals Across the Lifespan	INCLUSION_OF_INDIV_ACROSS_LIFE_SPAN1019212515.pdf	
2.4. Inclusion of Women and Minorities	Inclusion_of_Women___Minorities1019212477.pdf	
2.5. Recruitment and Retention Plan	Recruitment_and_Retention_Plan1019212478.pdf	
2.6. Recruitment Status	Not yet recruiting	
2.7. Study Timeline	Study_Timeline1019212479.pdf	
2.8. Enrollment of First Participant	05/01/2023	Anticipated

INCLUSION OF INDIVIDUALS ACROSS LIFE SPAN

Rational for Exclusion Based on Age

The proposed research project is focused on social isolation and cognitive health of older adults defined as age 65 or above. The Social Security Administration defines age 65 as the standard retirement age. Existing aging research has widely used the age of 65 and above to define older adulthood. ADRD is one of the most prominent risk factors for later life mortality and causes a significant burden of health care on individuals, families, and society. It is of clinical and policy significance to focus the proposed project on the older adult population.

Description of Investigation Team Expertise and Facilities

The applicant has dedicated the past four years of doctoral training to aging research and proposed continuing to contribute to the gerontological social work knowledge in the dissertation and postdoctoral research. She will transition from a predoctoral scholar trained in social work to conduct interdisciplinary research for ADRD prevention as a postdoctoral fellow at OHSU. The sponsor team has extensive experience in working with older adults. All sponsors have been conducting aging research for over 20 years. The USC USC Alzheimer Disease Research Center (ADRC), Oregon Layton Aging and Alzheimer's Disease Center (OADC), and the USC Edward R. Roybal Institute on Aging will provide facility support for conducting research with older adults.

Existing Datasets or Resources

The proposed research will utilize existing data from the National Health and Aging Trends Study (NHATS), the ongoing Internet-based Conversational Engagement Clinical Trial (I-CONECT) led by co-sponsor Dr. Hiroko Dodge, and the Collaborative Aging (in Place) Research using Technology (CART) initiative led by co-sponsor Dr. Jeffrey Kaye. The NHATS study over samples the oldest old and non-Hispanic Black older adults, offering a large scale longitudinal dataset to achieve the proposed research aims. Both I-CONECT and CART projects are still enrolling older adult participants. Gender and racial break-down information are not available at this moment. The two studies focused on studying the health and wellbeing of older adults and do not exclude any individuals on the basis of gender and race.

INCLUSION OF WOMEN & MINORITIES

National Health and Aging Trends Study (NHATS) follows a nationally representative cohort of individuals aged 65 and above and enrolled in Medicare, with oversamples of the oldest-old age groups and non-Hispanic Black persons. At the baseline of the secondary data analysis with NHATS data (N=8,334), 59.12% were female, 20.52% self-identified as non-Hispanic Black, and 5.6% identified as Hispanic. These racial and gender statistics are largely representative of older adults across the nation.

The Internet-based Conversational Engagement Clinical Trial (I-CONECT) and Collaborative Aging (in Place) Research using Technology (CART) initiative are still enrolling participants. Gender and racial break-down information are not available at this moment. The two studies do not exclude any individuals on the basis of race and gender. The participants of the focus group discussion will be sampled from the I-CONECT participants. A purposeful sampling method will be used to recruit focus group participants to increase the probability of elicit diverse and rich information. The anticipated enrollment for the focus groups has slightly more female participants than male participants, given women usually composite higher than 50% of the persons in older age groups. The purposeful sampling aims to recruit participants from racially diverse backgrounds.

RECRUITMENT AND RETENTION PLAN

Co-Design focus group participants will be recruited from the participants of the I-CONECT study. Working with the I-CONECT research team, the PI will contact individuals who meet the purposeful sampling criteria via phone calls and invite them to participate in the focus group discussion. The study purpose, participants' rights and options will be carefully explained. If the individual agrees to participate, his/her availability will be asked and recorded in order to schedule the focus group sessions. Five days before the planned focus group session, the PI or a research assistant will seek confirmation of attendance from the participants. Because the qualitative focus group study does not have a longitudinal design, a retention plan is inapplicable in this case.

STUDY TIMELINE

Research activities	F99		K00			
	Year 0	Year 1	Year 1	Year 2	Year 3	Year 4
Submit IRB application	x					
Dissertation Research • Secondary data analysis using longitudinal NHATS data • Prepare and submit manuscripts to peer-review journals	x	x				
Systematic review of social isolation and cognitive health • Develop a systematic review protocol • Conduct literature search, screening, and information extraction • Prepare and submit a manuscript to peer-review journals		x	x			
I-COENCT: • Conduct hands-on research with I-CONECT data to Examine underlying pathways of the association between cognitive health and social isolation • Prepare and submit manuscripts to peer-review journals			x	x	x	
CART: • Investigate the longitudinal relationships between social engagement and cognitive health • Explore the possibility of employing technological infrastructure in the CART initiative to deliver ETA informed behavioral health intervention and assess outcomes. • Prepare and submit manuscripts to peer-review journals			x	x	x	
Co-design of a behavioral health intervention with older adults and potential community partners • Building collaborative relationships with community service organizations. • Consult with sponsors and cosponsors to prepare the interview guide for the qualitative study • Recruit I-CONECT participants and conduct co-design focus groups. • Transcribe and analyze the focus group data • Prepare and submit manuscripts to peer-review journals				x	x	
Develop an intervention pilot testing protocol and apply for pilot funding (OADC pilot award/ NIH R21) • Convene quarterly expert panel meetings with all sponsors to consult on the intervention design and implementation. • Prepare and submit the pilot study grant proposal				x	x	x
Prepare and submit K99/R00 award application				x	x	

2.9. Inclusion Enrollment Reports

IER ID#	Enrollment Location Type	Enrollment Location
Study 1, IER 1	Domestic	

Inclusion Enrollment Report 1

1. Inclusion Enrollment Report Title* : Inclusion Report for F99/K00 Project 1 F99 AG068492-01

2. Using an Existing Dataset or Resource* : ● Yes ❍ No

3. Enrollment Location Type* : ● Domestic ❍ Foreign

4. Enrollment Country(ies): USA: UNITED STATES

5. Enrollment Location(s):

6. Comments:

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	1	1	0	0	2
Asian	1	1	0	0	2
Native Hawaiian or Other Pacific Islander	1	1	1	1	4
Black or African American	3	3	1	1	8
White	4	4	1	1	10
More than One Race	1	1	2	2	6
Total	11	11	5	5	32

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	1064	646	0	0	0	0	0	0	0	1710
White	3351	2352	0	0	0	0	0	0	0	5703
More than One Race	3	5	0	0	0	0	0	0	0	8
Unknown or Not Reported	128	116	0	266	201	0	115	87	0	913
Total	4546	3119	0	266	201	0	115	87	0	8334

Section 3 - Protection and Monitoring Plans (Study 1)

3.1. Protection of Human Subjects	Protection_of_Human_Subjects1019212480.pdf
3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site?	❍ Yes ● No ❍ N/A
If yes, describe the single IRB plan	
3.3. Data and Safety Monitoring Plan	
3.4. Will a Data and Safety Monitoring Board be appointed for this study?	❍ Yes ● No
3.5. Overall structure of the study team	

PROTECTION OF HUMAN SUBJECTS

Risk to Human Subjects

a. Human Subjects Involvement, Characteristics, and Design

The proposed study involves both primary qualitative data analysis and secondary quantitative data analysis to examine the influence of the physical and social living environment, ICT use within the context of the environment, on social isolation and cognitive health. Secondary data analysis for the dissertation work will use wave 5 to 9 data form the National Health & Aging Trends Study (NHATS) (U01AG032947). A cohort of 8,334 individuals aged 65 and above responded to the wave 5 NHATS survey and received follow-up survey interviews annually after.

The proposed postdoctoral stage study will use data from the Internet-based Conversational Engagement Clinical Trial (I-CONECT) study and Collaborative Aging (in Place) Research using Technology (CART) initiative. The I-CONECT project is currently recruiting socially isolated adults, aged 75 and above, and with normal cognition or mild cognitive impairment (MCI), to participate in the trial. I-CONECT is a multi-site study conducted at both the University of Michigan (UM) and Oregon Health and Science University (OHSU). The recruiting goal is 160 participants at each site, 320 participants in total. As of October of 2020, 180 participants have enrolled in the study. The CART Initiative is a multi-year nationwide study that seeks to facilitate the health and independence of older adults who are members of diverse communities. CART is a four year non-randomized longitudinal infrastructure development and validation study, made up of participants recruited from 5 different sites (OHSU, Portland VA, Rush University Medical Center, University of Miami, and Weill Cornell Medicine). This study does not include an intervention. As of the end of October 2020, 225 homes have been enrolled to participate in the CART initiative nationwide.

The participants of the behavioral health intervention co-design focus group will be purposefully sampled from the I-CONECT participants with an intention to recruit individuals residing in varied living environments and with diverse demographic backgrounds. Two focus groups will be held at each site with a target of 8 participants in each focus group, resulting in 32 participants for the focus groups anticipated.

The secondary data analysis procedures will be submitted to the University of Southern California's (USC) Institutional Review Board (IRB). The primary data collection and analysis procedure will be submitted to the IRB at USC, OHSU, and UM. I-CONECT research team members at OHSU and UM will help with the participant recruitment for the focus groups and coordinate to book spaces for the focus groups.

b. Study Procedures and Materials

The secondary data analysis for the proposed dissertation study will use longitudinal data from the NHATS study. The proposed postdoctoral research will use data from I-CONECT and CART. NHATS data survey data were collected with in-person interviews using surveys with detailed data collection instructions designed by NHATS team researchers.

I-CONECT is currently recruiting participants from the communities around UM and OHSU, and randomizing enrolled participants into the video-chat intervention group and control group. The intervention group will have a computer installed in their homes with internet service. During 6-months long high dose duration, the participants assigned to the intervention group will have a video chat with study staff for 30 minutes/day, 4 times/week. Afterward, the participants will have 2 times/week video chats, 30 minutes/day, for an additional 6-months maintenance dose period. Both intervention and control group participants will have a 10-minutes telephone check-in with the study staff, answering brief questions to monitor their social activities and health conditions. At baseline, 6-month and 12 months, research staff will visit the participants at their homes to administer assessments in-person. Participants at both study sites will have MRIs at baseline and 6 months if receiving MRIs were safe for them. All video chat and in-home assessment sessions will be recorded for speech and language analysis. Only participants at OHSU will have their medication compliance tracked with electronic pillboxes.

CART study includes two components that last for three years: clinical assessments and in-home monitoring. The participants will be asked to have a baseline study visit with research staff, including the consent process, filling out demographic and health information, and baseline assessments with research staff. Each year, the participants will have a follow-up assessment visit and will update demographic and health information at that time. In-home sensors will be set up to monitor physical and computer activity patterns and how those patterns relate to changes in medication and cognition. The system detects activity in homes, and wirelessly sends the information to the research project staff. All data is kept strictly confidential.

Co-design focus groups participants will be recruited from both intervention and control groups of the I-CONECT participants. At each study site, two focus groups will be held – one with intervention group participants, the other with control group participants. The focus group sessions will be led by the PI and coordinated by a research assistant who will be taking notes for the discussion sessions. Participants will be notified that the focus groups will be audio recorded when scheduling the focus group sessions with them via phone calls. The PI/research assistant will explain to the participants the audio recording and notes will be used for research purpose only. The purposes of the focus group study, discussion topics, time frame, and incentive for data collection will be explicit in the informed consent form. Each focus group session will last for about an hour, at a selected meeting room that allows or privacy where the participants will not be seen or overheard by someone not participating in the focus groups. All participants will be advised to keep the discussion to the group and respect each other's opinions. The recordings of the focus groups will be transcribed verbatim for the use of data analysis.

c. Potential Risks

The risk of participating in the quantitative and qualitative components of the proposed study will be minimal. All primary and secondary data is going to be analyzed will be de-identified. The PI has no access to the names or any identifying data for the NHATS and CART datasets. In order to purposefully sample I-CONECT participants for the focus group discussion, the group assignment, age, gender, race, and location will be used. Participants' contact information will be obtained to contact participants via phone calls. The contact information will be stored separately in a password-protected file. Under no circumstances, the participant ID will be put together with the name and participant ID in the same file. Transportations from and back to their homes will be arranged for the participants to attend the focus group sessions. During the focus group discussion, participants might experience mild discomfort due to the discussion of topics related to social isolation. It will be explained to the participants they have the right not to answer any questions they do not wish to respond. Participants will be informed they can withdraw from the study at any time and be compensated for their time and effort even if they cannot complete the focus group discussion. All data will be analyzed and presented in manuscripts anonymously.

Adequacy of Protection Against Risks

a. Informed Consent and Assent

Both I-CONECT and CART participants signed informed consent for the parent studies. Additional informed consent will be sought for the qualitative focus groups. To avoid contamination with the intervention effect, only participants who have completed the I-CONECT study will be invited for the focus groups. Participants will be first contacted via phone calls. The PI/research assistant will describe the focus group procedures, including the plan of audio-recording, to the participants. If the individual agrees to participate, the PI/research assistant will schedule the time of the focus group with him/her. At the time of the focus groups, before any data would be collected, the PI or a research assistant will present written informed consent to the participants, answer any questions they might have, and then have the participants sign the informed consent. Individuals with MCI can provide their informed consent. The informed consent forms are going to be stored in a locked file cabinet for three years after completion of the study unless IRB deemed the requirement of the informed consent could be waived.

b. Protection Against Risk

All secondary data will be cleaned and recoded prior to being accessed by the PI, which reduces the risk of confidentiality. Primary qualitative focus group transcripts and notes will be stored in password-protected files. In the data collection process, participants will be informed that their participation in both the I-CONECT and the qualitative focus groups are voluntary. They have the right to withdraw at any point in the study without any penalty. The data collected are confidential and will only be used for research-related purposes. In the focus group discussions, participants will be addressed with their first name. Their last name will not be asked or recorded. Only the research team and the Human Subjects Protection Program will have access to the primary raw data. Any research assistant who will help to contact participants, take note in the focus group discussion, to help with transcribing the audio recording, and/or to help with data analysis will receive IRB training and been added to the IRB record as research personnel prior to any research-related activities. In an appropriate time after the completion of data analysis, the original focus group summaries will be destroyed. The results from the analysis of the primary and secondary data will be published or presented at conferences in an anonymous/aggregated form. No identifiable information will be used.

Potential Benefits of Proposed Research to Human Subjects and Others

Secondary analysis using the NHATS data will contribute to the literature with longitudinal empirical evidence on the influences of physical and social environments on social isolation and cognitive health. Participants of the

NHATS study might benefit from participating in the study as the evidence built from the data will be used to inform the development of interventions and policy related to aging friendly communities and aging in place. The I-CONECT study is one of the first clinical trials examining the effectiveness of internet facilitated video chat intervention on reducing social isolation and prevent cognitive decline. Participants could directly benefit from the potential positive effects of the intervention. Knowledge will be built using the data collected in the I-CONECT could benefit the participants indirectly by explaining one of the mechanisms through which internet-enhanced social interaction promote older adults' wellbeing and cognitive health. Participants of CART study could benefit from using the computer and in-home sensors provided by the research project. The PI will explore the possibility of designing a behavioral health intervention supported by the CART home structure. The development of such an intervention program could directly benefit the participants' health and well-being in the future. Participating in the focus group discussion could alleviate the experience of social isolation among the participants. Participants of the focus group will be engaged in a conversation exploring the design for a sustainable community-based program for isolation and cognitive health, which could indirectly benefit the participants by informing the development of an intervention.

Importance of the Knowledge to be Gained

Knowledge built in the proposed dissertation and postdoctoral research could lead to the development of a sustainable community-based ICT-facilitated program with the aim to address social isolation and cognitive decline. Risks to participation will be minimal. Knowledge to be gained is of extreme importance as the accelerated population aging poses significant challenges in providing health care and social services that meet the needs of the older adult population. The proposed research has the potential to inform the policy and practice regarding aging in place, to promote cognitive health, and to aid older adults to live longer independent lives.

Section 4 - Protocol Synopsis (Study 1)

4.1. Study Design

4.1.a. Detailed Description

4.1.b. Primary Purpose

4.1.c. Interventions

Type	Name	Description

4.1.d. Study Phase

Is this an NIH-defined Phase III Clinical Trial? ❍ Yes ❍ No

4.1.e. Intervention Model

4.1.f. Masking ❍ Yes ❍ No

❑ Participant ❑ Care Provider ❑ Investigator ❑ Outcomes Assessor

4.1.g. Allocation

4.2. Outcome Measures

Type	Name	Time Frame	Brief Description

4.3. Statistical Design and Power

4.4. Subject Participation Duration

4.5. Will the study use an FDA-regulated intervention? ❍ Yes ❍ No

4.5.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/ Investigational Device Exemption (IDE) status

4.6. Is this an applicable clinical trial under FDAAA? ❍ Yes ❍ No

4.7. Dissemination Plan

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

AUTHENTICATION OF KEY BIOLOGICALS AND CHEMICAL RESOURCES

No key biological and/or chemical resources will be used in the activities proposed in this application.